ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Marine Technician Machinery Maintainer (00379)

USQ-S2-MAR TECH Machinery Maintainer (ALMI)

MITE Course IDs:
Residential Phase – 121388

Approved: 06 2024
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RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NTG) and the Commandant of the Curriculum Control Authority CCA (NFS(P) The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J “Course History”.

Revision	Approval Date	Authority
1.0		
1.1		
2.0	14 June 2024	CNTG

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FOREWORD

1. This training document is developed under the authority of the Commander Naval Training Group (CNTG) and in accordance with the following references:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. Annex C – 5555-00379-MAR-TECH dated 15 Dec 16; and

   c. NAVORD 4500-0 - Structured Delegation, and NPTORD 4500-22 Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met, and the standards listed in the QSP(s) for the associated co-qualifications have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA, who will forward to NTG for approval, as appropriate.

5. The designated TE(s) for this training is indicated below:

TE	Phase/Module	MITE Course ID	NQual
NFS(P)	All	121388	ALMI
NFS(A)	All	121388	ALMI

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL TRAINING GROUP (CNTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Marine Technician Machinery Maintainer. This QSP is designed to assist training establishments in developing, conducting, evaluating, and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22 Qualification Standard and Plan, para 4.5. Training day calculations will be based off a 400-minute training day.

a. Minimum course length:

	Minutes
Actual Training Time (ATT) (Sum of all EOs/POs excluding PC&ECs)	38950
Enabling Checks (ECs)	3760
Performance Checks (PCs)	1475
Sub-total (@400 minutes per day)	44185 **110.5 trg days**
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	1445.55
Training Administrative Time (TAT)	1445.55
Divisions (parades)	662.775
Physical Fitness (PT)	3976.65
Sub-total (@400 minutes per day)	7530.5 **18.8 trg days**
Co-qualifications (to be scheduled with/within the training) (BAT,TAT,TTT,PT already calculated within the respective QSP)	
Naval Technical Communication – Module A (MITE ID: 118128)	2150
Attack Team Leader (MITE ID: 118787)	800
Live Fire Training	400
Sub-total (@400 minutes per day)	3350 **8.4 trg days**
Course Duration Calculation (ATT+PCs&ECs+BAT+TTT+TAT+PT) * **Partial days must be rounded up (400 minutes per day)**	51715.5 **130 trg days** + 3350

	9 co-qualification trg days = **139 trg days**

b. Maximum course length:

	Minutes
Actual Training Time (ATT) (Sum of all EOs/POs excluding PC&ECs)	38950
Enabling Checks (ECs)	14870
Performance Checks (PCs)	3125
Sub-total **(@400 minutes per day)**	56945 **142.4 trg days**
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	1828.35
Training Administrative Time (TAT)	1828.35
Divisions (parades)	854.175
Physical Fitness (PT)	5125.05
Sub-total **(@400 minutes per day)**	9635.925 **24.1 trg days**
Co-qualifications (to be scheduled with/within the training) (BAT,TAT,TTT,PT already calculated within the respective QSP)	
Naval Technical Communication – Module A (MITE ID: 118128)	2150
Attack Team Leader (MITE ID: 118787)	800
Live Fire Training	400
Sub-total **(@400 minutes per day)**	3350 **8.4 trg days**
Course Duration Calculation (ATT+PCs&ECs+BAT+TTT+TAT+PT) * **Partial days must be rounded up (400 minutes per day)**	66580.925 **167 trg days** + 3350 **9 co-qualification trg days** = **176 trg days**

*BAT, TAT, TTT, and PT are calculated using (ATT+PCs & ECs) = number of weeks.
(5 days per week)

1. These timings may fluctuate based on instructor/equipment to trainee ratios. Schedulers must consider these ratios.

2. To accurately schedule for each course load:

a. Confirm course size; and

b. Confirm all EO/EC/PC timings and equipment requirements.

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional training establishment resources can be listed in Annex H if required.

Required Course Staffing						
MOD	EO/PO	EO Name	Occupation	Min Rank	Qty for Instruction	Qty for Assessment
1	006.01	Perform Mathematics Calculations	Civ	Civ	1	1
	006.02	Perform General Physics Calculations			1	1
	007.02	Apply General Chemistry			1	1
	008.02	Apply General Thermodynamics			1	1
2	011.01	Perform Electrical Circuit Analysis	Civ	Civ	1 (2 During Labs)	1
	012.01	Perform Rotating Machinery Analysis			1 (2 During Labs)	1
3	003.01	Maintain Twin Agent Unit System	HT ET (Locker contents/ submersible pump).	MS or above	1	0
	003.02	Maintain Portable Damage Control Equipment			1	0
	003.03	Maintain Portable Rescue Equipment			1	0
4	005.01	Attend the Attack Team Leader Course	DC School	NA	As per ATL QSP	
	005.02	Attack Live Fire	Civ		As per civilian training requirements	
	005.03	Lead a Flood Control Team	DC School		1	3
	005.04	Lead a Casualty Power Team			1	3
5	004.01	Inspect Ship Structures and Hull	HT	MS or above	1	1
	004.02	Maintain Materials			2	2

6	001.01	Perform Maintenance Administration	MAR TECH Supply Tech	MS or above	1	1
	001.02	Perform Maintenance Logistics			1	1
	001.03	Naval Technical Communication Module A	Civ	NA	1	1
6	009.01	Maintain Gasoline Engines	MAR ENG	MS or above	1	6
	009.02	Perform first line maintenance on small boat diesels	MAR ENG		1	1
	009.03	Troubleshoot small boat diesels	MAR ENG		1	1
	009.04	Maintain small boat stern drive			1	1
	009.05	Maintain small boat components			1	1
7	011.02	Maintain Electrical Components	ET (Bearing Maintence and Technology course)	MS or above	1	1
	011.03	Maintain Field Devices			1	1
	011.04	Maintain Sensors			1	1
	012.02	Maintain Electrical Motors			1	2
	012.03	Maintain Motor Control Devices			1	EC 1 PC 1 for every 2 trainees
8	002.03	Maintain Centrifuge	MAR ENG	MS or above	1	1
	002.04	Maintain Oil Pollution Abatement			1	1
	010.01	Maintain Gas Turbine Water Wash System			1	1
	010.02	Maintain Main Shaftline			1	5
9	002.01	Maintain Pumps	MAR ENG	MS or above	1	2
	002.02	Maintain RAS Equipment			1	1
	007.01	Maintain Auxiliary Steam Generator			1	EC 6 PC 6
	008.01	Maintain ROD Units			1	0
	008.03	Maintain AC Refrigeration	Advance Refrigeration	MS or above	1	4
10	006.03	Maintain Compressors	MAR ENG	MS or above	1	6
	008.04	Maintain HVAC	HT		1	4

Critical Training Resources	No of Days

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40 Quantity Control Process for the NTS. The course capacity is based on the civilian standard for safety practices for the practical phases (ratio 1:12). Resource implications need to be considered if a course size is above the minimum load.

a. Maximum:

(1) Practical - 12; and
(2) Theoretical - 24.

b. Minimum – 1; and

c. Optimum – 12 (any ratio above the optimum requires additional instructors).

MITE

5. Loading. The authority to load members onto this course lies with the Training Establishment (TE). NTG is to be included as an INFO addressee on all loading messages.

6. Scheduling. Each session of this course shall be scheduled IAW NPTORD 4500-1 Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System; in MITE by the Training Establishment conducting the training, and NTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

7. Prerequisites. To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

a. One of the following residential courses:

(1) RQ S3 Marine Technician Basic Maintainer (NQUAL - ALJS); or
(2) CSQ S2 Marine Systems Roundsperson (NQUAL - AMBP); and

b. At least one of the following on-board phases completed:

(1) USQ Roundsperson HFX (NQAUL – ALMJ); or
(2) USQ Roundsperson KIN (NQUAL - ALMK); or
(3) USQ Roundsperson VIC (NQUAL - ALVD); or

(4) USQ Roundsperson ORCA (NQUAL - ALML); or
(5) USQ Roundsperson HDW (NQUAL - AMBT); or
(6) USQ Roundsperson JSS (NQUAL – TBD).

c. Candidates must be physically fit IAW DAOD 5023-1 and CANFORGEN 047/03, free from any medical restrictions, which would preclude full participation in all performance objectives IAW ADM (HRMil) Instruction 11/04 identification 6600-2-10.

8. Qualification. Successful completion of this course and specified co-qualifications constitutes the primary means of attaining the Marine Technician S2 Unique Speciality Qualification (ALMI). Additionally, the following MITE Course IDs will be granted:

a. Marine Technician S2 Unique Speciality Qualification - MITE Course ID 121388;

b. Naval Technical Communication Module A - MITE ID 118128;

c. Attack Team Leader – MITE ID 118787;

d. Fire Fighter Refresher – MITE ID 109369; and

e. Flood Refresher – MITE ID 109370.

9. Co-Qualification. In addition to the performance objectives specific to this training, the following additional qualifications are included in this training program to meet the requirements of the job. The MITE and NQuals associated with these qualifications shall be granted upon completion of the training program and shall be listed within the course report and will be delivered IAW the following:

a. Naval Technical Communication – Module A (MITE ID: 118128);

b. Attack Team Leader (MITE ID: 118787); and

c. Live Fire Training.

COURSE REPORTING

10. Upon passing this course, trainees shall receive a Course Report (CF 377) and/or Certificate of Military Achievement (CF 289) completed IAW CMP Instruction 01/06 - Course Reporting and Certificates and NPTORD 4500-51 Course Reports and Certification.

EVALUATION

11. Program evaluation shall be conducted IAW the guidance provided in A-P9-050-000 CFITES Vol. 11 Evaluation and Validation Techniques.

VALIDATION

12. This training will be validated every five years IAW CFITES Vol. 8 Validation of Instructional Programs and based on the task list in Annex B.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

13. PLARs shall be conducted in accordance with NPTORD 4500-27 PLAR, DAOD 5031-1 Canadian Forces Military Equivalencies Program and CFITES Vol 12 Canadian Forces Military Equivalencies Programme: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

14. All Candidates are required to be in possession of a valid Fit Testing card for Half Mask and Firefighting Mask prior to commencement of PO 004 MAINTAIN SHIPBOARD COMPONENTS, PO 005 LEAD A DAMAGE CONTROL TEAM and PO 010 MAINTAIN ANCILLARY SYSTEMS.

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL TRAINING GROUP (CNTG)

PO 001 PERFORM TECHNICAL ADMINISTRATION

1. Performance: Perform Technical Administration.

2. Conditions:

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Machinery Maintainer shall perform technical administration IAW refs by:

   a. Performing Maintenance Administration;

   b. Performing Logistical Administration; and

   c. Naval Technical Communication Module A.

4. References: Specified in applicable EOs.

5. Task Numbers: AFT0340, AFT 0350, and AFT 0310.

6. Training Limitations: NFS may not have all required equipment/ trainer.

PO 002 MAINTAIN FLUID SYSTEM COMPONENTS

1. Performance: Maintain Fluid System Components.

2. Conditions:

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Machinery Maintainer Shall Maintain Fluid System Components IAW Ref by:

   a. Maintaining Pumps;

   b. Maintaining RAS Equipment;

   c. Maintaining Centrifuges; and

   d. Maintaining oil Pollution and Abatement.

4. References: Specified in applicable EOs.

5. Task Numbers: AZT0360, AZT 0070, AKT2180, AKT 2320, AKT 2310, AZT0180, CT0190, AHT0760, AKT0130, AKT0140, AKT1610, AHT0510, AHT0520, AKT2820, AIT0510, AYT0250, AYT0220, AYT0030, AYT0040, and UT0250.

6. Training Limitations: NFS may not have all required equipment/ trainer.

PO 003 MAINTAIN DAMAGE CONTROL SYSTEMS

1. Performance. Maintain Damage Control Systems.

2. Conditions.

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Marine Technician Machinery Maintainer Shall Maintain Damage Control Systems IAW refs by:

   a. Maintaining Twin Agent Unit;

   b. Maintaining Portable Damage Control Equipment; and

   c. Maintaining Portable Rescue Equipment.

4. References. Specified in applicable EOs.

5. Task Numbers. ANT0590, ANT1180, ATT0570, ANT1460, FT0440, ANT1110, ANT1740, ANT0110, ANT0620, ANT1450, ANT1100, ANT1000, ANT0840, ANT0390, AJT0520, AYT0430, ANT1730, AJT0620, AJT0010, AJT0370, and AJT0180.

6. Training Limitations. Nil.

PO 004 MAINTAIN SHIPBOARD COMPONENTS

1. Performance: Maintain Shipboard Components.

2. Conditions:

   a. Given:

      (1) Applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Machinery Maintainer Shall Maintain Shipboard Components IAW refs by:

   a. Manufacturing Materials; and

   b. Inspecting Ship Structures and Hull.

4. References: Specified in applicable EOs.

5. Task Numbers: AMT1010, AMT0370, AMT0720, AMT0250, AMT0290, AMT0300, AMT0070, AMT0670, AMT0100, AMT0400, BBT0350, BBT0340, BBT0320, BBT0330, and BBT0430.

6. Training Limitations: Nil.

PO 005 LEAD A DAMAGE CONTROL TEAM

1. Performance: Lead a Damage Control Team.

2. Conditions:

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Machinery Maintainer Shall Lead a Damage Control Team, IAW refs by:

   a. Completing the ATL Course MITE ID 118787;

   b. Attacking Live Fires;

   c. Leading a Flood Control Team; and

   d. Leading a Casualty Power Team.

4. References: Specified in applicable EOs.

5. Task Numbers: ANT0200, and ANT0340. (As per Attack Team Leader – MITE ID 118787 QSP)

6. Training Limitations: NFS may not have all required equipment/ trainer.

PO 006 MAINTAIN COMPRESSED AIR SYSTEMS

1. Performance: Maintain Compressed Air Systems.

2. Conditions:

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Machinery Maintainer Shall Maintain Compressed Air Systems IAW refs by:

   a. Applying General Mathematics;

   b. Performing General Applied Physics; and

   c. Maintaining Compressed Air System Components.

4. References: Specified in applicable EOs.

5. Task Numbers: AIT0530, AIT0550, AIT0590, AHT0690, AIT0540, AIT0560, AIT0810, ATT0140, AUT0740, AUT0820, AUT1120, AUT0830, AUT2940, AUT2950, AUT3990, AUT0310, AYT0160, AYT0280, AYT0150, AKT1550, AKT0480, AMT0160, AMT0160, AT0170, AT0070, OT0050, and AST0040.

6. Training Limitations: NFS may not have all required equipment/ trainer.

PO 007 MAINTAIN AUXILIARY STEAM SYSYEMS

1. Performance: Maintain Auxiliary Steam Systems.

2. Conditions:

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Machinery Maintainer Shall Maintain Auxiliary Steam Systems IAW refs by:

   a. Maintaining Auxiliary Steam Generator; and

   b. Performing General Chemistry.

4. References: Specified in applicable EOs.

5. Task Numbers: AUT0740, AUT0820, AUT1120, AUT0830, AUT2940, AUT2950, AUT3990, AUT0310, AYT0160, AYT0280, AYT0150, AKT1550, AKT0480, AMT0160, AMT0160, AT0170, AT0070, OT0050, AST0040, AHT0550, AHT0780, AIT0670, AHT0700, AIT0650, AIT0660, AT0090, ET0120, AHT0800, and AHT0810.

6. Training Limitations: NFS may not have all required equipment/ trainer.

PO 008 MAINTAIN DOMESTIC SERVICE SYSTEMS

1. Performance: Maintain Domestic Service Systems.

2. Conditions:

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Machinery Maintainer Shall Maintain Domestic Service Systems IAW refs by:

   a. Maintaining the Shipboard Reverse Osmosis Desalination System (SROD);

   b. Applying General Thermodynamic Principles to Engineering Applications;

   c. Maintaining Refrigeration Components; and

   d. Maintaining Heating, Ventilation and Air Conditioning Systems (HVAC).

4. References: Specified in applicable EOs.

5. Task Numbers: ATT1410, AOT0190, AOT0310, JT0480, AOT1010, AOT0040, AWT0120, AT0060, AT0160, AT0280, ATO270, AOT0770, AOT0780, AKT1820, and AOT0030.

6. Training Limitations: Nil.

PO 009 PERFORM SMALL BOAT MAINTENANCE

1. Performance: Perform Small Boat Maintenance.

2. Conditions:

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Machinery Maintainer shall Perform Small Boat Maintenance IAW Refs by:

   a. Maintaining Small Boat Gasoline Engines;

   b. Performing First Line Maintenance on Small Boat Diesel Engines;

   c. Troubleshooting Small Boat Diesel Engines;

   d. Maintaining Small Boat Stern Drives; and

   e. Maintaining Small Boat Components.

4. References: Specified in applicable EOs.

5. Task Numbers: AZT0100, AZT0120, AZT0020, AZT0200, AZT0390, AZT0290, AZT0240, VT0020, AZT0190, AZT0370, AZT0400, AZT0250, AZT0300, AZT0210, AZT0260, and AZT0270.

6. Training Limitations: Nil.

PO 010 MAINTAIN ANCILLARY SYSTEMS

1. Performance: Maintain Ancillary Systems.

2. Conditions:

   a. Given:

      (1) Applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Machinery Maintainer Shall Maintain Ancillary Systems IAW refs by:

   a. Maintaining the Gas Turbine Water Wash System; and

   b. Maintaining the Main Shaftline.

4. References: Specified in applicable EOs.

5. Task Numbers: AVT1430, AVT1120, AHT0580, AVT0390, AUT1150, XT0020, BT0190, BT0010, BAT0010, BAT0250, BAT0300, AIT0530, AIT0550, and AKT0700.

6. Training Limitations: If the right amount of equipment per student is not available for this training, PO 010 will be required to be extended to accommodate the shortfall.

PO 011 MAINTAIN SHIPBOARD ELECTRICAL SYSTEMS

1. Performance: Maintain Shipboard Electrical Systems.

2. Conditions:

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Machinery Maintainer Shall Maintain Shipboard Electrical Systems IAW Refs by:

   a. Performing Electrical Circuit Analysis;

   b. Maintaining Electrical Components;

   c. Maintaining Field Devices; and

   d. Maintaining Sensors.

4. References: Specified in applicable EOs.

5. Task Numbers: AUT2330, AUT1540, AUT1450, AUT0960, AUT1420, AUT2210, AUT2870, AMT2070. ALT1230, AUT3080, AUT1390, AUT1530, AUT2830, AKT2530, AMT0270, AUT1590, AUT2960, AKT2420, AUT2640, AUT2470, AUT2620, AUT2670, AUT2770, ATT0260, and AUT2440.

6. Training Limitations: Nil.

PO 012 MAINTAIN ELECTRICAL ROTATING MACHINERY AND CONTROLS

1. Performance: Maintain Electrical Rotating Machinery and Controls.

2. Conditions:

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) Access to DRMIS;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Machinery Maintainer Shall Maintain Electrical Rotating Machinery and Controls IAW Refs by:

   a. Performing Rotating Machinery Analysis;

   b. Maintaining Electric Motors; and

   c. Maintaining Electrical Control Devices.

4. References: Specified in applicable EOs.

5. Task Numbers: AUT0090, AKT1380, AU2470, AKT0690, AKT0530, AUT1290, AUT1280, AUT1100, AUT1660, AKT1410, AUT2320, AUT1600, AUT1010, AUT1350, AUT1340, AUT1710, AUT0980, AUT2230, AUT2550, AUT2540, AUT2390, AUT0350, AUT0660, AUT2190, AUT0270, AUT3290, AUT3170, AUT0670, AUT2450, AZT0350, AUT3240, AUT3230, AKT0470, AUT3200, AKT2910, AKT0030, AUT2260, and AUT3470.

6. Training Limitations: Nil.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDANT OF THE CCA

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification. PCs will be administered in the form of (PRACTICAL or WRITTEN) examinations and the trainee must obtain a pass in all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative tests that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and trainees. Marks for ECs are not calculated in the trainee's final mark.

3. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be done IAW CFITES Vol. 7 Evaluation.

5. Senior instructor or delegated staff provides quality control by administering and scoring all Performance Checks/Critical Enabling Checks. Any PC/CEC not administered or invigilated by staff will be administered by a non-involved instructor designated by the senior instructor. Enabling Checks are the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 Conduct of Instruction Programmes and TE SOs or SOPs.

7. Continuous monitoring of candidate progress is required on this course in order to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for the TRB or PRB (if required); and

d. Information to assist in completion of the CF 377 Course Report.

8. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment SOs or SOPs.

9. PC/CEC failures and supplemental PC/CEC failures are to be handled IAW Training Establishment SOs or SOPs. Supplemental CECs and PCs shall not be given prior to 24 hours having elapsed since trainee was informed of their CEC or PC failure.

TRAINING REVIEW BOARD

10. TRBs shall be convened and conducted IAW with NPTORD 4500-54 Training Review Board Process.

COURSE REPORTS

11. Course reports (CF 377) shall be prepared and distributed IAW CFAO 26-12, 4500-51 Course Reporting and Certificates.

12. Trainee will be graded pass or fail. IAW NPTORD 4500-51 Course Reporting and Certificates, students will not be ranked.

13. Where necessary for Rank Qualifications, the narrative will indicate trainee's strengths and weaknesses based on PC/CEC results to provide information to the Career Manager. Training limitations and/or incomplete training must also be included in the course report. Record of trainee counselling is to be maintained by the training establishment.

CHAPTER 4: LESSON SPECIFICATION

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDANT OF THE CCA

PO 001 PERFORM TECHNICAL ADMINISTRATION

EO 001.01 Perform Maintenance Administration

1. Performance: Perform Maintenance Administration.

2. Conditions:

   a. Given:

      (1) References;
      (2) Computer and software;
      (3) DRMIS; and
      (4) Scenario.

   b. Denied: Nil.

   c. Environmental: Classroom with DWAN access computers.

3. Standard: The Marine Technician Machinery Maintainer shall independently perform maintenance administration IAW refs, by:

   a. Preparing reports and returns, to include:

      (1) Unsatisfactory Condition Reports (UCR) (AFT0340);
      (2) Pre-installation Failure (PIF) reports (AFT0350); and
      (3) Operational Deficiencies (OPDEF).

4. Time Required:

   a. EO – 175 minutes.

5. Instructional Methods:

   a. Demonstration - Performance

6. Training Aids:

   a. Completed examples of required forms;

   b. Training scenarios; and

   c. Computers with DRMIS access, one per trainee.

7. Test Details:

a. EO 001.01 will be assessed in PC 001.

8. References: A93, A72, and A24.

9. Remarks: Nil

10. Teaching Points:

Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Prepare Reports and Returns, by:						
1.1 Preparing UCR (CF 777), to include: a. Purpose and use; b. Who can raise; c. Use of Unsatisfactory Condition Reporting System; and d. Staffing process.		25	50	DP	A93 A72 A24	Instructor to describe purpose of and demo of how to draft a UCR, PIF, and OPDEF. Trainees to interpret requisite data for each document.
1.2 Preparing PIF, to include: a. Purpose and use; b. Who can raise; c. How to populate; and d. Staffing process.		25	50	DP		
1.3 Preparing OPDEF, to include: a. Purpose and use; b. When to use; c. Categories and timings; d. How to draft; and e. Staffing process.		25		DP		

Enabling Check – EO 001.01

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Trainees will be provided feedback throughout the EO. No formal EC is included in EO 001.01.

EO 001.02 Perform Logistical Administration

1. Performance: Perform Logistical Administration.

2. Conditions:

   a. Given:

      (1) References;
      (2) Computer and software;
      (3) DRMIS; and
      (4) Scenarios.

   b. Denied: Nil.

   c. Environmental: Classroom with access to DWAN computers.

3. Standard: The Marine Technician Machinery Maintainer shall perform logistical administration independently IAW refs, by:

   a. Preparing marine engineering equipment for shipment or storage (AFT0310), to include:

      (1) Preparing a CFSS Supply document (DND 2227);
      (2) Utilizing Canadian Government Cataloguing System CGCS;
      (3) Preparing a Material Condition Tag (CF 942);
      (4) Preparing a Local Cash Procurement Account (LCPA) and Local Purchase Order (LPO);
      (5) Preparing a High Priority Request (HPR); and
      (6) Preparing a Transfer Request (TRANREQ).

4. Time Required:

   a. EO - 340 minutes.

5. Instructional Methods:

   a. Interactive lecture;

   b. Demonstration – performance; and

   c. Field trip.

6. Training Aids:

   a. Completed examples of required forms;

   b. Training scenarios;

   c. Access to return stores; and

   d. Computers with DRMIS access, one per trainee.

7. Test Details:

   a. EO 001.02 will be assessed in PC 001.

8. References: A24, A25, A26, A47, A93 and A104.

9. Remarks: Snr instructor to co-ordinate field trip to return stores, ideally scheduled upon completion of training module of this EO. Field trip is beneficial to trainees, however not essential to the completion of the EO.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP 1 Prepare Marine Engineering equipment for shipment or storage, by:						
1.1 Preparing DND 2227, to include: a. Purpose and use; and b. Procedure for populating.	10		10	IL DP	A24 A25 A26 A47 A93 A104	Instructor to review DRMIS logistics requirements and describe purpose of supply documents. Trainees to fill out appropriate paperwork.
1.2 Utilizing CGCS, to include: a. Purpose and use.	20		20			
1.3 Preparing CF 942, to include: a. Purpose and use; b. Procedure for populating; and c. Tracking of.	10		10			
1.4 Preparing LCPA and LPO, to include: a. Purpose and use; and b. Signing authorities.	10		10			
1.5 Preparing HPR, to include: a. Purpose and use; b. How to populate; c. Timings; and d. Staffing process.	25		50			
1.6 Preparing TRANREQ, to include: a. Purpose and use; b. How to populate; and c. Staffing process.	25		50			
1.7 Field Trip to Return Stores.			90	Tour		30-minute tour includes 60 minutes of travel time.

Enabling Check – EO 001.02

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Trainees will be provided feedback throughout the EO. No formal EC is included in EO 001.02.

EO 001.03 Naval Technical Communication Module A

1. Performance: The Enabling Objective and associated tasks shall be trained and assessed IAW the QSP for Naval Technical Communication Module A, MITE ID 118128.

2. Remarks: Any considerations or special instructions (ie: Instructor requirements, etc) are IAW Naval Technical Communication Module (A) QSP.

Enabling Check – EO 001.03

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Trainees will be provided feedback throughout the EO. All testing will be completed IAW Naval Technical Communication Module A – QSP.

PERFORMANCE CHECK – PO 001

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom with computer access.

Arrangements and Procedures

2. PC 001 will be a practical assessment based on content from PO 001.

3. Pass/Fail.

4. Time required: 75 minutes.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001 will be a practical assessment based on content from PO 001.

2. During this practical assessment, you will be required to perform technical administration. You must pass this test to pass PO 001. To pass the test, you must perform technical administration in accordance with the standard.

3. References will be provided.

4. Pass/Fail IAW PC checklist.

5. You will have 75 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Trainees will be given a scenario in which they are tasked to order a part (NSN to be provided).

2. When the part arrives, it will be u/s and the trainee will have to utilize applicable CF/DND forms/messages to bring awareness to the defect and return the part to the system.

3. In advance of this PC, trainees will have completed homework assignments similar to the PC.

PART IV – PERFORMANCE CHECK LIST

Checklist	SAT	UNSAT	Comments
Did trainee utilize appropriate paperwork?			
1. DND 2227			a. Trainee completed DND 2227 to order part requested with no more than two errors. b. Trainee completed DND 2227 to a company return item (a & b) results in a pass for this item.
2. HPR			Trainee to indicate whether this requires HPR. Yes or no. Resulting in a pass.
3. UCR			Trainee to draft UCR message with no more than three errors to result in a pass.
4. PIF			Trainee to report failure with no more than 3 errors resulting in a pass for this item.
5. OPDEF required? Yes/No			Student to indicate whether OPDEF required, no need to draft one.
6. CF 942			Trainee to complete CF 942 tag with no more than two errors resulting in a pass for this item.
Instructions to Evaluator: **1.** The trainee must receive 5 SATs to pass this EC. **2.** In order to receive SATs the trainee must demonstrate that they are aware of the technical/logistical administrative requirements. **3.** All forms shall be filled out to the standard required in the homework assignments.			

PO 002 MAINTAIN FLUID SYSTEM COMPONENTS

EO 002.01 Maintain Pumps

1. Performance: Maintain Pumps.

2. Conditions:

   a. Given:

      (1) References;
      (2) Access to pumps;
      (3) Appropriate tools;
      (4) PPE; and
      (5) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom; and engineering workshop.

3. Standard: The Marine Technician Machinery Maintainer shall maintain pumps under supervision, IAW refs by:

   a. Inspecting pumps, to include:

      (1) Inspecting rotary piston pumps;
      (2) Inspecting rotary vane pumps;
      (3) Inspecting fuel pump; and
      (4) Inspecting Centrifugal pumps.

   b. Repairing pumps, to include:

      (1) Repairing positive displacement pumps (AKT2180);
      (2) Repairing fuel pumps (AZT0180);
      (3) Repairing rotary piston pumps (AKT2310);
      (4) Repairing rotary vane pumps (AKT2320); and
      (5) Repairing Centrifugal pumps.

   c. Testing pumps, to include:

      (1) Testing fuel pump (AZT0360);
      (2) Testing rotary piston pumps;
      (3) Testing rotary vane pumps; and
      (4) Testing centrifugal pumps.

4. Time Required:

   a. EO – 980 minutes; and

   b. EC – 30 minutes per pair of trainees.

5. Instructional Methods:

   a. Interactive lecture;

   b. Demonstration – performance; and

   c. Self-study assignment.

6. Training Aids:

   a. Multimedia projector;

   b. Handouts; and

   c. Pump trainer.

7. Test Details:

   a. EO 002.01 will be tested in EC 002.01.

8. References: C28.

9. Remarks: Trainee is eligible for testing after they have disassembled/assembled 1 type of pump. Tested pump may or may not be the same as practiced.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspect pumps, to include:						
1.1 Inspecting positive displacement pumps by: a. Naming pump types and general uses for each; b. Explaining the basic action of a displacement pump; c. Explaining the necessity for a relief valve to be fitted in the discharge of an positive displacement pump; d. Explaining that a relief valve on the discharge of a pump handling oil or other hazardous material must relieve within the system; e. Describing with the aid of diagrams how a reciprocating displacement pump works; f. Explaining the purpose of an air vessel fitted to the discharge; g. Describing the characteristics of a reciprocating pump, referring to: (1) Suction lift; (2) Priming;	200			IL	C28	Using Text book of Engineering Theory knowledge, Power point or handouts to teach pump fundamentals. Using Pictures and/or passing around defective components, demonstrate common defects in common pump components.

	(3) Discharge pressure; and (4) Vapour, or gas, in the fluid being pumped. h. Explaining the principle of rotary displacement pumps; i. Identifying the principle parts of: (1) Gear pump; (2) Rotary-vane pump; (3) Screw-displacement pump; and (4) Rotary piston pump. j. Describing the principles of operation of an axial flow pump; k. Describing the type of duty best suited to an axial-flow pump; l. Identifying Reciprocating pump defects of: (1) Cylinders; (2) Piston/buckets; (3) Rings; (4) Valves; (5) Joins; (6) Glands; and (7) Relief valves. m. Identifying screw and gear pump defects of: (1) Rotors and gears; and (2) Seals. n. Identifying rotary pump defects of: (1) Vanes; (2) Rotor; (3) Casing; and (4) Springs.						
1.2	Inspecting centrifugal pumps by: a. Explaining the principles of a centrifugal pump, referring to the purpose of: (1) The impeller; and (2) The Diffuser or volute. b. Explaining what is meant by a single and double-entry pumps; c. Identifying the arrangement of a centrifugal multi-stage single entry centrifugal pump; d. Describing the characteristics of a centrifugal pump, referring to: (1) Suction lift; (2) Priming; (3) Discharge pressure; and (4) Vapour or gas in the fluid being pumped. e. Explaining the necessity of priming and/or air extraction is necessary; f. Explaining the principle of an ejector; g. Identifying Indicators of Excessive Pump Wear; h. Identifying centrifugal pump component defects of: (1) Casing; (2) Impeller; (3) Wear rings;	100			IL	C28	Using Text book of Engineering Theory knowledge, Power point or handouts to teach pump fundamentals. Using Pictures and/or passing around defective components, demonstrate common defects in common pump components.

(4) Shaft; (5) Gland/seal; (6) Air pump; and (7) Float chamber.						
TP2 Repair pumps, to include:						
2.1 Repairing positive displacement pumps, to include: a. Screw and gear pump; b. Reciprocating pumps; and c. Rotary pumps.		100	300	DP		Instructor to demonstrate how to disassemble different types of pumps and allow the trainee time to work in groups gaining practical experience using the applicable OEM manual or CFTO.
2.2 Repairing centrifugal pumps, to include: a. Centripedal pump; b. Axial flow centrifugal pump; c. Single and double volute; and d. Multistage centrifugal pump.		50	150	DP		Instructor to demonstrate how to disassemble different types of pumps and allow the trainee time to work in groups gaining practical experience using the applicable OEM manual or CFTO.
TP3 Test pumps, to include:						
3.1 Testing positive displacement pumps by: a. Running pump after inspection; b. Ensuring proper operating parameters are met: (1) Pressure; (2) Flow; (3) Noise; (4) Vibration; (5) Heat; (6) Freedom of movement; and (7) Current draw.		10	30	DP		
3.2 Testing centrifugal pumps by: a. Running pump after inspection; b. Ensuring proper operating parameters are met: (1) Pressure; (2) Flow; (3) Noise; (4) Vibration; (5) Heat; (6) Freedom of movement; and (7) Current draw.		10	30	DP		

Enabling Check – EO 002.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop.

Arrangements and Procedures

2. EC 002.01 will be a practical assessment based on content from EO 002.01.

3. Pass/Fail.

4. Time required: 30 minutes per pair of trainees.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.01 will be a practical assessment based on content from EO 002.01.

2. During this practical assessment, you will be required to maintain pumps. You must pass this test to pass EO 002.01. To pass the test, you must complete the pump repair in accordance with the standard.

3. References will be provided.

4. Pass/Fail.

5. You will have 30 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Student is required to disassemble 1 of the pump types covered in the EO. Student will identify defects when disassembled. Student will reassemble the pump afterwards.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
1. Proper PPE worn.			
2. Ensured equipment is safe to work on.			
3. Did not cause damage to pump components during disassembly.			
4. Identified all defective components.			
5. Identified the type of defect.			
6. Identified possible cause of defect.			
7. Determined properly whether part needs to be replaced.			
8. Reassembled pump.			
9. Pump was Reassembled with no damage caused.			
10. Pump was in a serviceable state upon completion.			
11. Safe working practices were conducted throughout the test.			

Instructions to Evaluator:

1. The instructor is to have the trainees, in pairs, disassemble and then reassemble a pump. There should be defective components in the pump;
2. The defect can be anything; as long as it was taught on the EO;
3. Only one defect needs to be given to the trainee, if the instructor has enough time and needs to validate the trainee further, a second fault can be added;
4. Trainees begin with 100 points; each fault subtracts 8 from the total;
5. The trainee must receive more than 75 points in total to pass this EC/PC; and
6. The only grades awarded for this EC/PC are PASS (for totals 75 points or greater) or FAIL (for totals less than 75 points).

EO 002.02 Maintain Replenishment at Sea (RAS) Equipment

1. Performance: Maintain RAS Equipment

2. Conditions:

   a. Given:

      (1) References;
      (2) RAS equipment;
      (3) Appropriate tools;
      (4) PPE; and
      (5) Supervision.

   b. Denied: Nil.

   c. Environmental: Engineering workshop; and classroom.

3. Standard: The Marine Technician Machinery Maintainer shall maintain RAS equipment under supervision, IAW refs by:

   a. Investigating the operation of replenishment at sea systems (CT0190), to include:

      (1) Inspecting sliding pad eyes;
      (2) Inspecting fuel probe;
      (3) Inspecting emergency breakaway fuel coupling;
      (4) Inspecting RAS fuel receiver; and
      (5) Inspecting fuel hoses and couplings.

   b. Repairing RAS equipment, to include:

      (1) Assembling and disassembling fuel probe (AKT0140);
      (2) Assembling and disassembling emergency breakaway fuel coupling (AKT0130);
      (3) Repairing emergency breakaway fuel coupling (AKT1610); and
      (4) Repairing replenishment at sea fuel receiver (AHT0760).

   c. Testing of RAS equipment.

4. Time Required:

   a. EO – 500 minutes; and

   b. CEC – 20 minutes per trainee.

5. Instructional Methods:

   a. Interactive lecture;

   b. Demonstration – performance; and

   c. Guided discussion.

6. Training Aids:

   a. Fuel Probe;

   b. Emergency breakaway coupling;

   c. Fuel hose couplings;

   d. Fuel hoses; and

   e. Fuel receiving bell.

7. Test Details:

   a. EO 002.02 will be tested in a critical EC 002.02.

8. References: A42, A101, and A98.

9. Remarks: 2 fuel probes, 2 fuel receivers, 2 emergency breakaway couplings and 6, 7” section fuel hoses with clamps will be required to perform the maintenance routines. Trainee can be tested once they have completed the fuel receiver maintenance.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Investigate the operation of replenishment at sea systems, to include:						
1.1 Inspecting sliding padeye by: a. Checking for physical wear; b. Checking for corrosion; c. Ensuring it’s properly greased; and d. Ensuring freedom of movement.	10	10		IL	A42	Using pictures and/or props show all components and visual defects.
1.2 Inspecting fuel probe by: a. Checking for physical damage; b. Checking for corrosion; and c. Checking visually for defects.	5	5		IL	A98	
1.3 Inspecting Emergency breakaway fuel coupling by: a. Checking physical condition of coupling; and b. Checking physical condition of bolts and flanges.	5	5		IL	A98	

1.4	Inspecting RAS fuel receiver by: a. Inspecting for physical wear; b. Checking for freedom of movement; c. Inspecting visually condition of o - ring; and d. Ensuring bell is properly greased inside prior to a RAS.	10	10		IL	A98	
1.5	Inspecting fuel hoses and couplings by: a. Checking physical integrity of hoses for: (1) Cracking of the rubber; (2) Splitting; and (3) UV damage. b. Checking couplings for: (1) Leaks; (2) Properly seated; (3) Alignment; (4) Physical damage; and (5) Corrosion and/or wear.	10	10		IL	A98	
TP2 Repair RAS equipment, to include:							
2.1	Assembling / Disassembling Fuel probe by: a. Disassembling; b. Replacing seal; and c. Reassembling.		25	150	DP	A98	
2.2	Repairing emergency breakaway fuel coupling by: a. Disassembling; b. Checking internal components; and c. Reassembling.		25	150	DP	A98	
2.3	Repairing RAS fuel receiver by: a. Replacing O-rings; and b. Greasing linkages.		5	50	DP	A98, A101	Conduct PM routine for fuel receiver.
TP3 Test RAS equipment, to include:							
3.1	Discussing conditions for testing RAS equipment by: a. Explaining how most equipment doesn't have the ability to be bench tested; b. Explaining how testing the padeye occurs with assistance from FMF; and c. Explaining how each piece of equipment is tested during a RAS and the importance of equipment is assembled properly.	15			GD	A98	

Critical Enabling Check – EO 002.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop

Arrangements and Procedures

2. EC 002.02 will be a practical assessment based on content from EO 002.02.

3. Pass/Fail.

4. Time required: 20 minutes per trainee.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. CEC 002.02 will be a practical assessment based on content from EO 002.02.

2. During this practical assessment, you will be required to repair RAS equipment. You must pass this test to pass EO 002.02. To pass the test, you must complete the repair of RAS equipment in accordance with the standard.

3. References will be provided.

4. Pass/Fail.

5. You will have 20 mins to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Instructions to Evaluator:

1. The instructor is to have the trainees conduct 6 monthly PM routine as well as looking for faults. The instructor is to tell the trainee what the fault is and have them fix it.

2. The fault can be anything; as long as it was taught on the EO.

3. Only one fault needs to be given to the trainee, if the instructor has enough time and needs to validate the trainee further, a second fault can be added.

4. Trainees begin with 100 points; each fault subtracts 8 from the total.

5. The trainee must receive more than 75 points in total to pass this CEC.

6. The only grades awarded for this CEC are PASS (for totals 75 points or greater) or FAIL (for totals less than 75 points).

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
1. Proper PPE is worn by the trainees			Normally just nitrile gloves for using grease.
2. Checks quick release pin for freedom of movement.			
3. Ensures receiver can swivel freely on mount			
4. Applies grease to all grease fittings.			
5. Checks operation of manual release mechanism and visual latch indicator flags are functioning properly			
6. Checks receiver for corrosion.			
7. Wipes receiver clean.			
8. Inspect seal.			
9. Applies light coat of grease inside the bell mouth			
10. Identifies visually defects.			

Instructions to Evaluator:

1. The instructor is to have the trainees, in pairs, disassemble and then reassemble a pump. There should be defective components in the pump;
2. The defect can be anything; as long as it was taught on the EO;
3. Only one defect needs to be given to the trainee, if the instructor has enough time and needs to validate the trainee further, a second fault can be added;
4. Trainees begin with 100 points; each fault subtracts 8 from the total;
5. The trainee must receive more than 75 points in total to pass this EC/PC; and
6. The only grades awarded for this EC/PC are PASS (for totals 75 points or greater) or FAIL (for totals less than 75 points).

EO 002.03 Maintain Centrifuges

1. Performance: Maintain Centrifuges

2. Conditions:

   a. Given:

      (1) References;
      (2) Functional centrifuge;
      (3) Appropriate tools;
      (4) PPE; and
      (5) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom; and Workshop.

3. Standard: The Marine Technician Machinery Maintainer shall maintain centrifuges independently, IAW refs by:

   a. Inspecting centrifuges, to include:

      (1) Inspecting fuel oil system components;
      (2) Inspecting fuel oil filling and transfer system components; and
      (3) Inspecting lube oil filling and transfer system components.

   b. Repairing centrifuges, to include:

      (1) Assembling and disassembling fuel oil system components (AHT0510);
      (2) Assembling and disassembling fuel oil filling and transfer components (AIT0510);
      (3) Assembling and disassembling main lube oil filling and transfer system components (AHT0520); and
      (4) Troubleshooting centrifuges (AKT2820).

   c. Testing centrifuges, to include:

      (1) Operating centrifuges.

4. Time Required:

   a. EO – 1425 minutes; and

   b. CEC – 50 minutes per 2 trainees.

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids:

a. Centrifuge;

b. Centrifuge maintenance tools;

c. Fuel or oil tank; and

d. F-75 or F-76 Marine distillate.

7. Test Details:

a. EO 002.03 will be tested in a CEC 002.03.

8. References: C28, A99, and A100.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspect centrifuges, to include:						
1.1 Explaining Purifier and Fuel oil treatment. a. Explain the function of the gravity disk; b. Explains the function of low and high-pressure water; c. Describes the sludge discharge mechanism of an oil purifier; and d. Explain the difference between clarifying and purifying;	100			IL	C28	
1.2 Explaining Purifier Operation. a. Describes the purification process of fuel oil; b. Explain precautions for starting purifier and checking points to ensure a good working order; and c. Describes the correct procedures for the disposal of waste oil, sludge residue, etc.	100			IL	C28	
TP2 Repair centrifuges, to include:						

2.1 Assembling/disassembling Centrifuges to include tasks (AHT0510, AIT0510, AHT0520, AKT2820): a. Disassembling Purifier to include: (1) Required tools for disassembly; (2) Lock-out/tag-out; (3) Disassembly of components to removal of the bowl; (4) Importance of keeping the discs in order; and (5) Approved tools and materials for cleaning the discs and components. b. Reassembling Purifier to include; (1) Bowl alignment; (2) Setting bowl height; and (3) Locking ring and acceptable clearance.	50	50	1000	IL, DP	A99, A100	
TP3 Test centrifuges to include:						
3.1 Operating centrifuges by: a. Ensuring centrifuge operates after maintenance; and b. Ensuring operating parameters: (1) Pressure; (2) Flow rate; (3) Vibration; (4) Cycling; and (5) Timing.		25	100	F2F	A99, A100	

Critical Enabling Check – EO 002.03

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Trainer / workshop.

Arrangements and Procedures

2. CEC 002.03 will be a practical assessment based on content from EO 002.03.

3. Pass/Fail IAW CEC checklist.

4. Time required: 50 minutes.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. CEC 002.03 will be a practical assessment based on content from EO 002.03.

2. During this practical assessment, you will be required to Repair a centrifuge. You must pass this test to pass EO 0002.03. To pass the test, you must complete the centrifuge repair in accordance with the standard.

3. References will be provided.

4. Pass/Fail IAW CEC checklist.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See comments below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
1. Proper PPE is worn. **2.**			PPE is made available, and students have opted to wear it without instruction.
3. Isolated water supply.			
4. Conducted proper lock-out/tag-out. Filled out form.			Forms are made available, instructor to sign commencement of work once satisfied with LO/TO.
5. Open and secure lid.			Lid must be secured to ensure safety.
6. Remove locking ring.			Safety must be adhered to.
7. Remove hood.			
8. Remove disc stack.			
9. Trainees describe the importance of keeping the discs in order.			
10. Heavy components were lifted safely, with chain fall where applicable.			
11. 2 trainees communicated effectively.			
1. *Proper PPE is worn			PPE is made available, and students have opted to wear it without instruction.
2. *Trainees ensure equipment was properly locked out prior to commencing work. Inspect form, and check for lockout devices.			Forms are made available, instructor to hand completed form to students.
3. Installs disc stack			
4. Installs hood			
5. Sets bowl height			
6. *Install locking ring and secure			Trainees should know purpose of markings.
7. Safely secures lid.			
8. Reconnects all lines.			
9. Trainees complete lockout/tag-out forms at completion of work.			Once their work is complete have them remove lockout devices and sign off as complete.
10. Trainees set up centrifuge to run.			
11. Heavy components were lifted safely, with chain fall where applicable.			
12. 2 trainees communicated effectively.			

Instructions to Evaluator:

1. The instructor is to have the trainees, in pairs, disassemble or reassemble a centrifuge. Disassembly shall be up to and including the removal of the disc stack (first part of checklist). Reassembly shall start with the install of the disc stack (second part of checklist);

2. The trainees shall follow the manual during the test;

3. In the event that a trainee misses a step, or are about to cause damage to the centrifuge the instructor is to stop the trainees and mark them for a fault;

4. Trainees begin with 100 points; each fault subtracts 8 from the total;

5. Items with a * are critical items. An UNSAT rating for a critical item constitutes immediate failure of CEC;

6. The trainee must receive more than 75 points in total to pass this CEC; and

7. The only grades awarded for this CEC are PASS (for totals 75 points or greater) or FAIL (for totals less than 75 points).

EO 002.04 Maintain Oil Pollution Abatement Systems

1. Performance: Maintain Oil Pollution Abatement Systems

2. Conditions:

   a. Given:

      (1) References;
      (2) Oil Pollution Abatement System;
      (3) Appropriate tools;
      (4) Supervision; and
      (5) PPE.

   b. Denied: Nil.

   c. Environmental: Classroom; and Workshop.

3. Standard: The Marine Technician Machinery Maintainer shall maintain oil pollution abatement systems under supervision, IAW refs by:

   a. Inspecting oil pollution abatement treatment and collection system components;

   b. Repairing oil pollution abatement treatment and collection system components
      to include:

      (1) Assembling and disassembling oil pollution abatement treatment system components (AYT0040);
      (2) Assembling and disassembling oil pollution abatement collection system components (AYT0030);
      (3) Repairing oil pollution abatement treatment system batch tank (AYT0220); and
      (4) Repairing oil pollution abatement treatment system membranes (AYT0250).

   c. Testing oil pollution abatement system to include:

      (1) Operating oil pollution abatement system locally (UT0250).

4. Time Required:

   a. EO – 500 minutes; and

   b. EC – 50 minutes per trainee.

5. Instructional Methods:

   a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids:

a. Functional oily water separator.

7. Test Details:

a. EO 002.04 will be tested in EC 002.04.

8. References: A2, A96 and C28.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspect oil pollution abatement treatment and collection system components, to include:						
1.1 Discussing Oily water by: a. Defining Oily water; b. Explaining what contaminants that can make separation difficult; c. Discussing sources of bilge water; and d. Explaining the processes that aid in separation: (1) Chemical treatment; (2) Settling Time; and (3) Depth/width of tank. 1.2 Explaining a Coalescing filter type OWS by: a. Describing process of separation; b. Describing advantages; c. Describing disadvantages; d. Describing proper handling and storage; and e. Describing possible incompatible bilge contents. 1.3 Explaining an Evaporator type OWS by: a. Describing the process of separation; b. Describing advantages; c. Describing disadvantages; and d. Describing typical configurations. 1.4 Explaining a Centrifugal type OWS by: a. Describing the process of separation; b. Describing the advantages; c. Describing the disadvantages; and d. Describing the typical configurations. 1.5 Discussing MARPOL and SEMS by: a. Explaining MARPOL exclusion zones; b. Describing Minimum distances from shore; c. Describing Minimum Speed of advance (SOA); d. Explaining the requirements of an Oil Content Monitor (OCM);	150			IL	A2 C28	

	e. Discussing the maximum PPM allowable; and f. Explaining personal liability.					
1.6	Explaining OCM operation by: a. Explaining how it detects the PPM of oil in water; b. Explaining how it must automatically recirculate or prevent discharge overboard if PPM exceeds limit; and c. Explaining how it's illegal to bypass OCM.					
TP2 Repair oil pollution abatement treatment and collection system components, to include:						
2.1	Performing routine maintenance by: a. Conducting filter change; and b. Performing grease routine.		100	DP	A96	Using either the HYDROMEM, or SAREX type OWS conduct up to 6-month maintenance routine.
TP3 Test oil pollution abatement system to include:						
3.1	Operating oil pollution abatement system locally by: a. Flashing up the OWS; b. Monitoring the operation of the OWS; and c. Ensuring proper operating parameters.	50	200	IL / DP	A96	Operate OWS using either the HYDROMEM, or SAREX type.

Enabling Check – EO 002.04

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Trainer.

Arrangements and Procedures

2. EC 002.04 will be a practical assessment based on content from EO 002.04.

3. Pass/Fail.

4. Time required: 50 minutes.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 002.04 will be a practical assessment based on content from EO 002.04.

2. During this practical assessment, you will be required to operate the OWS locally. You must pass this test to pass EO 002.04. To pass the test, you must complete the local operation of the OWS in accordance with the standard.

3. References will not be provided.

4. Pass/Fail IAW EC checklist.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See comments below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
1. Trainee properly sets Collection system. (instructor to set 3 faults for the trainee to find)			
2. Trainee properly sets treatment **3.** System. (instructor to set 3 faults for the trainee to find)			
4. Trainee properly sets Membrane system. (instructor to set 3 faults for the trainee to find)			
5. Trainee properly sets Compressed air system.			
6. Trainee properly sets control panel prior to starting.			
7. Trainee ensures OCM and pump is set.			
8. Trainee starts OWS			
9. Trainee identifies what OCM should be set to.			
10. Trainee identifies 3 conditions which aid in separation			

Instructions to Evaluator:

1. The instructor is to have the trainee set-up and run an OWS;
2. 7-8 improperly set components (valves open instead of closed, air isolated, etc.) in total for the OWS prior to the trainee attempting to set-up;
3. Trainees begin with 100 points; each UNSAT on the checklist subtracts 8 points from the total;
4. The trainee must receive more than 75 points in total to pass this EC; and
5. The only grades awarded for this EC are PASS (for totals 75 points or greater) or FAIL (for totals less than 75 points).

PERFORMANCE CHECK – PO 002

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Trainees will be provided feedback throughout the PO. No formal PC is included in PO 002.

PO 003 MAINTAIN DAMAGE CONTROL SYSTEMS

EO 003.01 Maintain Twin Agent Unit System

1. Performance: Maintain Twin Agent Unit System.

2. Conditions:

   a. Given:

      (1) PPE;
      (2) Functioning Twin Agent Unit;
      (3) Applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom; and Workshop.

3. Standard: The Marine Technician Machinery Maintainer shall independently maintain the Twin Agent Unit (TAU) IAW refs, by:

   a. Inspecting twin agent unit (ANT0590), to include:

      (1) Surveying twin agent unit (ANT1180); and
      (2) Inspecting valves (ATT0570).

   b. Operating twin agent unit (FT0440);

   c. Repairing twin agent unit (ANT1110) to include:

      (1) Troubleshooting twin agent unit (ANT1740); and
      (2) Assembling and disassembling twin agent unit components (ANT0110).

   d. Testing twin agent unit (ANT1460) to include:

      (1) Trialing twin agent unit (ANT1520).

4. Time Required:

   a. EO - 675 minutes; and

   b. EC - 50 minutes per trainee.

5. Instructional Methods:

   a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids:

a. TAU mock up;

b. TAU Test fittings (NSN 4210-01-535-7128);

c. Multimedia projector;

d. Job aid; and

e. Applicable tools.

7. Test Details:

a. The content of EO 003.01 will be assessed in EC 003.01.

8. References: A58 and A43.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspect twin agent unit by:						
1.1 Introducing components to include: a. Propellant cylinders; b. Agent storage tanks; c. Fitted valves, to include: (1) Pressure relief valves; (2) Pressure gauges; and (3) Ball valves. d. Activation points to include: (1) Manual Pull station; and (2) Local/remote. e. Corner pulleys; f. Quick opening levers; g. Safety link pin; h. Hose reel assembly; and i. Hose line nozzle assembly. 1.2 Describing the operation of the TAU system to include: a. Arrangement of piping; b. Direction of flow; c. Operating pressures; d. Nozzle operation.	50	50		IL DP	A58 A43	
TP2 Operating the twin agent unit by:						
2.1 Pre activation set-up to include: a. Proper valve positioning; b. Removal of locks; and c. Removal of safety pins. 2.2 Activating the system by:	50		150			

	a. Remote activation (Pull station); and b. Local Activation (Quick opening lever).						
2.3	Operating the Nozzle, to include: a. AFFF Activation; and b. Dry Chemical Activation.						
TP3 Repair twin agent unit by:							
3.1	Repairing TAU to include: a. Troubleshooting; and b. Assembling and disassembling.	25	50	50	IL DP		
3.2	Accounting for TAU Safety and environmental precautions and considerations.	25			IL		
TP4 Test twin agent unit by:							
4.1	Describing the testing routines, to include: a. PM routines; and b. Trail Routines.	50	25	150	IL DP		C-97-325-000/NY-001; and C-97-325-000/NK-001
4.2	Post activation to include: a. Purging system; and b. Re-setting/Refilling after discharge.						

Enabling Check – EO 003.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Trainer.

Arrangements and Procedures

2. EC 003.01 will be a practical assessment based on content from EO 003.01.

3. Pass/Fail.

4. Time required: 50 minutes per trainee.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.01 will be a practical assessment based on content from EO 003.01.

2. During this practical assessment, you will be required to perform the following tasks in regards to the firefighting system twin agent unit (TAU):

    a. Pre-Activation check;
    b. Activate and operate the system; and
    c. Post-activation routine.

3. You must pass this test to pass EO 003. To pass the test, you must complete the above mentioned required tasks in accordance with the standard.

4. Pass/Fail IAW EC checklist.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. See comments below.

PART IV – PERFORMANCE CHECKLIST

PERFORMANCE CHECKLIST			
Date:	SN/PRI:	Rank:	Name:

TASKS: Successfully and safely complete the following tasks in regards to the firefighting system twin agent unit (TAU):

a. Pre-Activation check;
b. Activate and operate the system; and
c. Post-activation routine.

Administrative Instructions:

a. The Twin Agent Unit (TAU) practical assessment checklist has been created to limit subjectivity during the assessment on each of the performance elements listed. The trainee is to be measured according to the checklist which has been derived from the standard statements from EO 003.01

b. **Standard The Marine Technician Machinery Maintainer shall independently maintain the Twin Agent Unit (TAU)**

In determining the level of assistance an instructor or assessor may provide a trainee during their assessment the following should be considered;

a. Try to ensure the trainee is given an opportunity for the Enabling Check to progress to its conclusion;
b. If the trainee experiences difficulty, provide minimal assistance to maintain progress through the EC;
c. **Physical safety shall be observed throughout for both the trainee and the instructor**. If a safety infraction is committed or about to be committed the testing should be either:
   (1) Paused and corrective action taken before continuing, or
   (2) Halted if the infraction is serious in nature.

CONDITIONS and EQUIPMENT:

a. Twin Agent Unit System;
b. TAU Test fittings (NSN 4210-01-535-7128);
c. Appropriate tools;
d. PPE (glasses, gloves, ear protection);
e. Replacement agent for both AFFF and DC; and
f. Supervision.

TEST PROCEDURES	SAT	UNSAT
Ensure proper valve configuration for activation;		
Ensure **safety pins (or locks)** for the nitrogen cylinder valves are removed;		
Activate the system safely;		
Once activated, verify the valves on the nitrogen cylinders have lifted approximately 100 degrees upwards;		
Operate the hose nozzle assembly properly, utilizing both AFFF and DC;		
Reset the nitrogen cylinder valve and reengage the **safety pins (or locks)**;		
Ensure proper valve configuration for drainage;		

Safely drain the system of excess propellant and agent;		
Open agent cylinders and replace missing contents;		
Ensure agent cylinders are returned to normal after re-filled;		
Remove the **safety pins** and verify the system is returned to normal operation.		
	/11	
SAFETY (CRITICAL FACTOR) UNSAT REPRESENTS AN AUTOMATIC FAILURE OF EC 003.01	**SAT**	**UNSAT**
Safety Pins must be adhered to during the appropriate timings.		
Safely drain the system of excess propellant and agent; and		
Wear protective safety glasses and hearing protection.		

*** Trainees must achieve 8 of the 11 points under the category “Test Procedures” to pass with no safety infractions***

Overall Assessment / 11	**Result (Circle one)** **PASS: P** **FAIL: F**

REMARKS	
Strengths:	Area of improvement:

Date:	**Trainee’s Signature:**	**Instructor’s Signature:**

EO 003.02 Maintain Portable Damage Control Equipment

1. Performance: Maintain Portable Damage Control Equipment.

2. Conditions:

   a. Given:

      (1) Applicable equipment;
      (2) PPE;
      (3) Testing equipment;
      (4) Applicable tools;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom; and Workshop.

3. Standard: The Marine Technician Machinery Maintainer shall independently maintain portable Damage Control Equipment, IAW refs by:

   a. Inspecting portable damage control equipment to include:

      (1) Damage Control Locker contents (IAW C-23-045-000/AG-001);
      (2) Fire Fighting locker contents (IAW C-23-045-000/AG-001);
      (3) Electrical Repair Locker contents (IAW C-23-045-000/AG-001);
      (4) Flood Repair Kit contents (IAW C-23-045-000/AG-001);
      (5) Electrical Submersible Pump;
      (6) Metal Shoring;
      (7) Portable Fire Pump;
      (8) Emergency Burning and Welding Kit;
      (9) Casualty Power Cables and terminals;
      (10) Hilti Gun;
      (11) Splinter boxes;
      (12) Band-it clamp kit box;
      (13) Incident and status boards physical condition;
      (14) Gas Detectors;
      (15) DB Fan;
      (16) Smoke Curtains; and
      (17) Ram Fan.

   b. Repairing portable damage control equipment (ANT1000) to include all items listed in para 3a, 1 to 17; and

   c. Testing portable damage control equipment (ANT1000) to include all items listed in para 3a, 1 to 17.

4. Time Required:

   a. EO – 550 minutes; and

   b. EC – 30 minutes per trainee.

5. Instructional Methods:

   a. Interactive lecture; and

   b. Demonstration – performance.

6. Training Aids:

   a. All items listed in para 3a, 1-17;

   b. Applicable tools; and

   c. PPE.

7. Test Details:

   a. The content of EO 003.02 will be assessed in EC 003.02.

8. References: A43.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Ref	Comments
	T	D	P			
TP1 Inspect portable damage control equipment by:						
1.1 Inspecting portable damage control equipment to include: a. Damage Control Locker contents (IAW C-23-045-000/AG-001); b. Fire Fighting locker contents (IAW C-23-045-000/AG-001); c. Electrical Repair Locker contents (IAW C-23-045-000/AG-001); d. Flood Repair Kit contents (IAW C-23-045-000/AG-001); e. Electrical Submersible Pump; f. Metal Shoring; g. Portable Fire Pump; h. Emergency Burning and Welding Kit; i. Casualty Power Cables and terminals; j. Hilti Gun; k. Splinter boxes;	50	150		IL DP	A43	

l. Band-it clamp kit; m. Incident and status boards physical condition; n. Gas Detectors; o. DB Fan; p. Smoke Curtains; and q. Ram Fan.						
TP2 Repair portable damage control equipment by:						
2.1 Repairing/replacing portable damage control equipment to include all items listed in TP1, a. 1-17.	50		150	IL DP		
TP3 Test portable damage control equipment by:						
3.1 Testing portable damage control equipment to include all items listed in TP1, a. 1- 17.	50		100	IL DP		

Enabling Check – EO 003.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop.

Arrangements and Procedures

2. EC 003.02 will be a practical assessment based on content from EO 003.02.

3. Pass/Fail.

4. Time required: 30 minutes per trainee.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.02 will be a practical assessment based on content from EO 003.02.

2. During this practical assessment, you will be required to maintain two of the following pieces of equipment:

    a. Electrical Submersible Pump;
    b. Portable Fire Pump;
    c. Emergency Burning and Welding Kit;
    d. Hilti Gun;
    e. Gas Detectors;
    f. DB Fan; and
    g. Ram Fan.

3. Your instructor will choose two of the above mentioned portable damage control equipment at random to assess you on.

4. You must pass this test to pass EO 003. To pass the test, you must complete the above mentioned required tasks in accordance with the standard.

5. Pass/Fail IAW EC checklist.

6. You will have 30 minutes to complete this assessment.

7. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. See comments below.

PART IV – PERFORMANCE CHECKLIST

PERFORMANCE CHECKLIST			
Date:	SN/PRI:	Rank:	Name:

TASKS: Successfully and safely maintain two of the portable damage control equipment listed below:

a. Electrical Submersible Pump;
b. Portable Fire Pump;
c. Emergency Burning and Welding Kit;
d. Hilti Gun;
e. Gas Detectors;
f. DB Fan; and
g. Ram Fan.

Administrative Instructions:

a. The portable damage control equipment practical assessment checklist has been created to limit subjectivity during the assessment on each of the performance elements listed. The trainee is to be measured according to the checklist which has been derived from the standard statements from EO 003.02

b. **Standard The Marine Technician Machinery Maintainer shall independently maintain portable damage control equipment.**

In determining the level of assistance an instructor or assessor may provide a trainee during their assessment the following should be considered;

a. Try to ensure the trainee is given an opportunity for the Enabling Check to progress to its conclusion;
b. If the trainee experiences difficulty, provide minimal assistance to maintain progress through the EC;
c. **Physical safety shall be observed throughout for both the trainee and the instructor**. If a safety infraction is committed or about to be committed the testing should be either:
   (1) Paused and corrective action taken before continuing, or
   (2) Halted if the infraction is serious in nature.

CONDITIONS and EQUIPMENT:

a. Electrical Submersible Pump;
b. Portable Fire Pump;
c. Emergency Burning and Welding Kit;
d. Hilti Gun;
e. Gas Detectors;
f. DB Fan;
g. Ram Fan;
h. Appropriate tools;
i. PPE (glasses, gloves, ear protection); and
j. Supervision.

FIRST EQUIPMENT - TEST PROCEDURES	SAT	UNSAT
Ensure equipment is both electrically and mechanically **safe** to start maintenance on, ie. LOTO;		
Perform a cursory inspection while verbally telegraphing each step;		
Remove/Disassemble equipment while verbally telegraphing each step;		
Complete required maintenance while verbally telegraphing each step;		
Replace/Assemble equipment while verbally telegraphing each step; and		
Ensure equipment is returned to functioning order and stored for future emergency use.		
SECOND EQUIPMENT - TEST PROCEDURES	**SAT**	**UNSAT**
Ensure equipment is both electrically and mechanically **safe** to start maintenance on, ie. LOTO;		
Perform a cursory inspection while verbally telegraphing each step;		
Remove/Disassemble equipment while verbally telegraphing each step;		
Complete required maintenance while verbally telegraphing each step;		
Replace/Assemble equipment while verbally telegraphing each step; and		
Ensure equipment is returned to functioning order and stored for future emergency use.		
	/12	
SAFETY (CRITICAL FACTOR) UNSAT REPRESENTS AN AUTOMATIC FAILURE OF EC 003.02	**SAT**	**UNSAT**
Ensure equipment is made "SAFE" before commencing with the maintenance.		
Ensure positive control of equipment and tools at all times.		
Wear protective safety glasses and hearing protection.		
*** Trainees must achieve 10 of the 12 points under the category "Test Procedures" to pass with no safety infractions***		
Overall Assessment / 12	**Result (Circle one)** **PASS: P** **FAIL: F**	

REMARKS	
Strengths:	Area of improvement:

Date:	Trainee's Signature:	Instructor's Signature:

EO 003.03 Maintain Portable Rescue Equipment

1. Performance: Maintain Portable Rescue Equipment.

2. Conditions:

   a. Given:

      (1) Applicable equipment;
      (2) PPE;
      (3) Testing equipment;
      (4) Applicable tools;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom; and Workshop.

3. Standard: The Marine Technician Machinery Maintainer shall independently maintain portable rescue equipment, IAW refs by:

   a. Inspecting portable rescue equipment (AJT0180) to include:

      (1) Power hawk;
      (2) Exothermic Torch kit;
      (3) Self-contained breathing apparatus (SCBA);
      (4) Respirators; and
      (5) Inspecting visually, emergency escape breathing device (EEBD) (ANT0620 and ANT0390).

   b. Performing preventative maintenance routines on portable rescue equipment, to include:

      (1) Troubleshooting self-contained breathing apparatus (SCBA) (ANT1730);
      (2) Troubleshooting respirators (AYT0430); and
      (3) Repairing emergency escape breathing device (EEBD) (ANT0840) (Assume repair means repair by replacement).

   c. Repairing portable rescue equipment (AJT0370) to include:

      (1) Power Hawk;
      (2) Exothermic Torch kit;
      (3) Adjusting portable rescue equipment (AJT0010); and
      (4) Troubleshooting portable rescue equipment (AJT0620).

   d. Testing portable rescue equipment (AJT0520) to include:

      (1) Power Hawk; and

(2) Exothermic Torch kit.

4. Time Required:

a. EO - 650 minutes; and

b. EC – 30 minutes per trainee.

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids:

a. Power hawk;

b. Testing material i.e. scrap car;

c. Exothermic Torch kit;

d. Emergency Escape Breathing Device (EEBD);

e. SCBA;

f. Respirators;

g. Applicable tools; and

h. Job aid.

7. Test Details:

a. The content of EO 003.03 will be assessed in EC 003.03. There is no formal PC for PO 003.

8. References: A43.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Ref	Comments
	T	D	P			
TP1 Inspect portable rescue equipment by:						
1.1 Inspecting portable rescue equipment to include: a. Power hawk; b. Exothermic Torch kit;	50	50	100	IL DP	A43	

c. EEBD; d. SCBA; and e. Respirators.						
TP2 Perform Preventative Maintenance routines on portable rescue equipment by:						
2.1 Perform Preventative Maintenance routines on the following: a. Power hawk; b. Exothermic Torch kit; c. EEBD; d. SCBA; and e. Respirators.	50		150	IL DP		
TP3 Repair portable rescue equipment by:						
3.1 Repairing portable rescue equipment to include: a. Power hawk; and b. Exothermic Torch kit.	50		100	IL DP		
TP4 Test portable rescue equipment by:						
4.1 Testing portable rescue equipment to include: a. Power hawk; and b. Exothermic Torch kit.	50		50	IL DP		

Enabling Check – EO 003.03

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop.

Arrangements and Procedures

2. EC 003.03 will be a practical assessment based on content from EO 003.03.

3. Pass/Fail.

4. Time required: 30 minutes per trainee.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.03 will be a practical assessment based on content from EO 003.03.

2. During this practical assessment, you will be required to complete a preventative maintenance routine on two of the following pieces of equipment:

    a. Power hawk;
    b. Exothermic Torch kit;
    c. EEBD;
    d. SCBA; and
    e. Respirators.

3. Your instructor will choose two of the above mentioned portable rescue equipment at random to assess you on.

4. You must pass this test to pass EO 003.03. To pass the test, you must complete the above mentioned required tasks in accordance with the standard.

5. Pass/Fail IAW EC checklist.

6. You will have 30 minutes to complete this assessment.

7. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. See comments below.

PART IV – PERFORMANCE CHECKLIST

PERFORMANCE CHECKLIST			
Date:	SN/PRI:	Rank:	Name:

TASKS: Successfully and safely complete a preventative maintenance routine on two of the portable rescue equipment listed below:

a. Power hawk;
b. Exothermic Torch kit;
c. EEBD;
d. SCBA; and
e. Respirators.

Administrative Instructions:

a. The portable rescue equipment practical assessment checklist has been created to limit subjectivity during the assessment on each of the performance elements listed. The trainee is to be measured according to the checklist which has been derived from the standard statements from EO 003.03.

b. **Standard: The Marine Technician Machinery Maintainer shall independently maintain portable rescue equipment.**

In determining the level of assistance an instructor or assessor may provide a trainee during their assessment the following should be considered;

a. Try to ensure the trainee is given an opportunity for the Enabling Check to progress to its conclusion;
b. If the trainee experiences difficulty, provide minimal assistance to maintain progress through the EC;
c. **Physical safety shall be observed throughout for both the trainee and the instructor**. If a safety infraction is committed or about to be committed the testing should be either:
   (1) Paused and corrective action taken before continuing, or
   (2) Halted if the infraction is serious in nature.

CONDITIONS and EQUIPMENT:

a. Power hawk;
b. Exothermic Torch kit;
c. EEBD;
d. SCBA;
e. Respirators;
f. Appropriate References (CFTO);
g. Appropriate tools;
h. PPE (glasses, gloves, ear protection); and
i. Supervision.

FIRST EQUIPMENT - TEST PROCEDURES	SAT	UNSAT
Ensure equipment is both electrically and mechanically **safe** to start maintenance on, ie. LOTO;		
Perform a cursory inspection while verbally telegraphing each step;		
Remove/Disassemble equipment while verbally telegraphing each step;		

Complete required maintenance while verbally telegraphing each step;		
Replace/Assemble equipment while verbally telegraphing each step; and		
Ensure equipment is returned to functioning order and stored for future emergency use.		
SECOND EQUIPMENT - TEST PROCEDURES	**SAT**	**UNSAT**
Ensure equipment is both electrically and mechanically **safe** to start maintenance on, ie. LOTO;		
Perform a cursory inspection while verbally telegraphing each step;		
Remove/Disassemble equipment while verbally telegraphing each step;		
Complete required maintenance while verbally telegraphing each step;		
Replace/Assemble equipment while verbally telegraphing each step; and		
Ensure equipment is returned to functioning order and stored for future emergency use.		
	/12	
SAFETY (CRITICAL FACTOR) UNSAT REPRESENTS AN AUTOMATIC FAILURE OF EC 003.03	**SAT**	**UNSAT**
Ensure equipment is made "SAFE" before commencing with the maintenance.		
Ensure positive control of equipment and tools at all times.		
Wear protective safety glasses and hearing protection.		

*** Trainees must achieve 10 of the 12 points under the category "Test Procedures" to pass with no safety infractions***

Overall Assessment / 12	**Result (Circle one)** **PASS: P** **FAIL: F**

REMARKS	
Strengths:	Area of improvement:

Date:	**Trainee's Signature:**	**Instructor's Signature:**

PERFORMANCE CHECK – PO 003

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Trainees will be given feedback throughout the entirety of this PO. There is no formal PC in PO 003.

PO 004 MAINTAIN SHIPBOARD COMPONENTS

EO 004.01 Inspect Ships Structures and Hull

1. Performance: Inspect Ships structures and hull.

2. Conditions:

   a. Given:

      (1) HMC Ship and applicable equipment and/or system
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom/lab and HMC Ship.

3. Standard: The Marine Technician Machinery Maintainer shall independently maintain ship structures and hull IAW refs, by:

   a. Inspecting ships structure and hull;

   b. Inspecting visually, ships structure (BBT0350) to include:

      (1) Inspect visually, ships minor structure (BBT0340); and
      (2) Inspect visually, ships appendages (BBT0320).

   c. Inspecting visually, ships fluid tanks (BBT0330) by;

      (1) Investigating ships fluid tanks (BBT0430).

   d. Explaining confined space entry procedures.

4. Time Required:

   a. EO – 600 minutes; and

   b. EC – 50 minutes.

5. Instructional Methods:

   a. Interactive lecture;

   b. Demonstration – performance; and

   c. Field Trip.

6. Training Aids:

   a. Job aid.

7. Test Details:

   a. EO 004.01 will be assessed in EC 004.01. The PC for PO 004 will not contain content from EO 004.01.

8. References: A43.

9. Remarks: The practical time allotted towards this EO will be utilized for a field trip (FT) down to an HMC Ship to cover the points discussed.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspect visually ships structure by:						
1.1 Inspecting visually ships structure to include: a. Keel; b. Frames; c. Longitudinal; d. Shell Plating; e. Pillars; f. Decks; and g. Bulkheads.	50		100	IL DP FT	A43	Field Trip to HMC Ship during the practical time allotted for this EO.
1.2 Inspecting minor structures to include: a. Ships furniture; b. Deck gratings; c. Ladders; d. Superstructure; e. Hanger; and f. Masts.						
1.3 Inspecting appendages to include: a. Rudder; b. Shaft line; c. Bilge keels; d. “A” brackets; and e. Stern flap.						
TP2 Inspect visually ships fluid tanks to include:						
2.1 Inspecting visually ships fluid tanks to include: a. Fuel or diesel oil; b. Lube oil;	50		100	IL DP	A43	

c. Fresh water; d. Void; and e. SWB.						
TP3 Explain confined space entry, to include:						
3.1 Entering a confined space to include: a. Tank and space type; b. Hazards; c. Safety precautions; and d. Entry.	100		200	IL DP	A43	

Enabling Check – EO 004.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom.

Arrangements and Procedures

2. EC 004.01 will be a written assessment based on content from EO 004.01.

3. Pass mark will be 60%.

4. Time required: 50 minutes.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. EC 004.01 will be a written assessment based on content from EO 004.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. Pass mark is 60%.

10. You will have 50 minutes to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. This EC will consist of no more than 20 questions.

EO 004.02 Manufacture Materials

1. Performance: Manufacture Materials.

2. Conditions:

   a. Given:

      (1) Metal working shop;
      (2) Wood working shop;
      (3) Plastic shop (Can potentially be located within Wood or Metal shops);
      (4) Applicable materials;
      (5) PPE;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil

   c. Environmental: Classroom; and Workshop

3. Standard: The Marine Technician Machinery Maintainer shall independently maintain shipboard components IAW refs, by:

   a. Fabricating wood parts and components (AMT0370) to include:

      (1) Describing shop safety brief general safety rules;
      (2) Inspecting wood;
      (3) Identifying types of construction joints;
      (4) Operating wood working tools; and
      (5) Performing basics of framing construction.

   b. Installing wood parts and components (AMT1010) to include:

      (1) Describing Fasteners/Bonding materials.

   c. Repairing wood parts and components (AMT0720) to include:

      (1) Fabricating construction joints and braces for repair.

   d. Fabricating metal parts and components (AMT0250) to include:

      (1) Describing shop safety brief and general safety rules;
      (2) Describing common shapes and dimensions of metal; and
      (3) Operating metal working tools.

   e. Installing metal parts and components, to include:

      (1) Demonstrating bending allowances.

f. Repairing metal parts and components to include:

(1) Performing mild steel and aluminum flat bar bending;
(2) Performing aluminum and mild steel angle bar bending; and
(3) Cleaning and Finishing of metal parts and components.

g. Fabricating plastic shapes (AMT0300) to include:

(1) Describing shop safety brief and general safety rules; and
(2) Performing plexi-glass bending.

h. Installing plastic parts and components to include:

(1) Identifying properties of plexi-glass.

i. Assembling/disassembling plastic parts and components (AMT0070) to include:

(1) Scribing and breaking;
(2) Cutting with Saber, Band, and Jig Saw;
(3) Cutting with a circular saw;
(4) Drilling; and
(5) Bending.

j. Fabricating plastic regalia/memento (AMT0290);

k. Repairing plastic parts and components (AMT0670), to include:

(1) Fabricating plastic sheet template;
(2) Cutting plexi-glass;
(3) Bending plexi-glass; and
(4) Finishing plexi-glass.

l. Assembling/Disassembling shock mount arrangements (AMT0100), by:

(1) Repairing by replacement.

4. Time Required:

a. EO – 3000 minutes;

b. EC:

(1) Part A (General shop safety) – 50 minutes;
(2) Part B (Project bracket) – 50 minutes;
(3) Part C (Project tray) – 800 minutes;
(4) Part D (Project plastic bracket) – 200 minutes; and
(5) Part E (Project picture frame) – 250 minutes.

c. PC:

(1) Part A (Wood shop phase) – 700 minutes;
(2) Part B (Metal shop phase) – 350 minutes; and
(3) Part C (Plastic shop phase) – 200 minutes.

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids:

a. Applicable tools/materials; and

b. Job aid.

7. Test Details:

a. EO 004.02 will be assessed in EC 004.02 (A), EC 004.02 (B), EC 004.02 (C), EC 004.02 (D) and EC 004.02 (E); and

b. The content of PC 004 will be derived from EO 004.02. It will be broken into 4 parts.

8. References: A67, C7, C8, C9, C16 and C18.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Ref	Comments
	T	D	P			
TP1 Conducting Shop safety rules and general safety brief by:						
1.1 General safety	25			IL		This TP must be covered first, regardless of which shop the trainee's enter first. General safety brief. EC 004.01 (25 minutes)
1.2 Shop machinery and machine tool safety: General						
1.3 Identifying shop safety to include: a. PPE; b. Reporting any injuries; c. Handling shop materials; and d. Proper disposal of waste.						
TP2 Fabricate wood parts and components by:						
2.1 Receiving shop safety brief, safety rules and Environmental concerns with regards to working in a wood workshop.	25			IL	A67 C7 C8 C18	

2.2	Operating wood working tools; a. layout/measurement tools; b. hand tools; c. Circular saw; d. Corded drill / Cordless drill; e. Drill press; f. Band saw; g. Chop saw; h. Miscellaneous tools; and i. Tool box muster.	25			IL		50 minutes Tool box muster and tool demo to be conducted after IL portion.
TP3 Inspecting wood to include:							
3.1	Identifying types of construction materials: a. Common defects found in lumber; b. Plywood; c. Composite boards; and d. Framing joints.	25			IL		
3.2	Describing Fasteners/Bonding materials to include: a. Adhesive; b. Cements and Glues; c. Nails; d. Screws; e. Special fasteners; and f. Others.	50					
3.3	Identifying types of construction joints to include: a. Framing joints; and b. Composite boards.						
TP4 Install wood parts and components by:							
4.1	Fabricating various components to include: a. Joints; b. Walls; c. Ceilings; and d. Floors.	75			IL DP		
TP5 Repair wood parts and components by:							
5.1	Basic Framing construction techniques		50		DP		
5.2	Construction of a Saw horse to include: a. Layout for cutting; b. Cut; c. Test fit; and d. Secure together using screws. This project will be built IAW drawing “Project Saw Horse”			350	P		This project will be built IAW drawing “Project Saw Horse”
5.3	Construction of a Green House to include: a. Layout for cutting; b. Cut; c. Test fit; and d. Secure together using screws. This project will be built IAW drawing “Project Green House”			700	P		Group project This project will be built IAW drawing “Project Green House”
5.4	Construction of a Personal Project to include: a. Blueprint Drafting; b. Layout for cutting; c. Cut; d. Test fit; and e. Secure together using screws.			400	P		This project will be built IAW drawing “Personal Project”

This project will be built IAW drawing “Personal Project”						
TP6 Fabricate metal parts and components to include:						
1.1 Receiving shop safety brief general safety rules to include: a. Tool box muster; b. Sharpening tools with bench/pedestal grinder; c. Mounting a grinding wheel; d. Truing grinding wheels; and; e. Demonstrates dressing and truing grinding wheel.	25		50	IL DP	C9 C16	50 minutes Tool box muster and tool demo to be conducted after IL portion.
1.2 Operating metal working tools to include: a. Rules and tapes; b. Machinist square; c. Combination squares to include: (1) Centre head; (2) Square head; and (3) Protractor head. d. Awls/scribers; e. Centre punch/prick punch; f. Hacksaw; g. Ball peen hammer; h. Machinist hammer; i. Horizontal band saw; and j. Bench grinder.	25					
1.3 Describing common shapes and dimensions of metal to include: a. Properties of metal; b. Classification of metals; and c. Common shapes.	100					
TP7 Install metal parts and components by:						
7.1 Demonstrating bending allowances: a. Explain the importance of identifying the order of bends, in most cases commence at one end and work to the other end. (Angle Bending, Bending Order); and b. Describe and explain the color range of metals when heated.	100	50	100	IL DP		Practice bending 1/8, ¼ inch aluminum and steel “See video”
7.2 Instructor to play videos a. Methods of Bending Flat Bar Mild Steel; b. Methods of Bending Flat Bat Aluminium; and c. Methods of Bending Angle Material.						
TP8 Repair metal parts and components by:						
8.1 Performing mild steel and aluminum flat bar bending using Oxy-Acetylene torch: a. Bend mild steel flat bar using Oxy-Acetylene torch; b. Bend aluminum flat bar using Oxy-Acetylene torch using heat stick/wood to identify required temperature for bending; and c. Trainees to practice.		50	100	DP		
8.2 Performing aluminum and mild steel angle bar bending using the Oxy-Acetylene torch:		50	100			

	a. Layout cutting angle bar for bending; b. Perform bending angle aluminum and mild steel, using Oxy-Acetylene torch; and c. Trainees to practice.						
8.3	Cleaning and Finishing of metal parts and components by removing: a. Burs; b. Sharp edges; c. Tool marks; and d. Grease and oils.		25	50			
TP9 Fabricate plastic parts and components by:							
9.1	Shop safety brief and general safety rules	50			IL		
9.2	Performing Plexiglas bending to include: a. Bending with heat; and b. Strip heat forming.		25		DP		
TP10 Install plastic parts and components by:							
10.1	Identifying properties of Plexiglas to include: a. Weight; b. Electrical insulation; c. Forming qualities; d. Thermoplastic; and e. Thermosetting.	50			IL		
TP11 Assemble/disassemble plastic parts and components by:							
11.1	Cutting Plexiglas to include: a. Scribing and breaking; b. Cutting with Saber, Band, and Jig Saw; c. Cutting with a circular saw; d. Drilling; and e. Bending.		25	125	DP		
TP12 Fabricate plastic regalia/memento by:							
12.1	Repair plastic parts and components a. Fabricating plastic sheet template IAW pattern: b. Cutting plastic sheet as required; c. Bending plastic sheet as required; and d. Finishing Plexiglas.		25	125	DP		
TP13 Assemble/Disassemble shock mount arrangements by:							
13.1	Repairing by replacement.	25			IL		

Enabling Check – EO 004.02

Part A - General Safety

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom.

Arrangements and Procedures

2. EC 004.02 (General Safety) will be a written assessment based on content from EO 004.02.

3. Pass mark is 80%.

4. Time required: 50 minutes.

5. References will not be provided.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. EC 004.02 (General Safety) will be a written assessment based on content from EO 004.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. Pass mark is 80%.

10. You will have 50 minutes to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Nil.

Enabling Check – EO 004.02

Part B - Project Bracket

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Approved Metal Workshop

Arrangements and Procedures

2. EC 004.02 (Project Bracket) will be a practical assessment based on content from EO 004.02.

3. Pass/Fail.

4. Time required: 50 minutes.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 004.02 (Project Bracket) will be a practical assessment based on content from EO 004.02.

2. During this practical assessment, you will be required to construct a project out of steel IAW the provided drawing. You must pass this test to pass EO 004.02. To pass the test, you must complete the project in accordance with the given drawing in the time frame provided.

3. References will be provided.

4. Pass/Fail IAW provided marking guide.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See marking guide for the “Project Bracket” with attached drawing.

Enabling Check – EO 004.02

Part C - Project Tray

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Approved Wood and Metal Workshop

Arrangements and Procedures

2. EC 004.02 (Project Tray) will be a practical assessment based on content from EO 004.02.

3. Pass/Fail.

4. Time required: 800 minutes.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 004.02 (Project Tray) will be a practical assessment based on content from EO 004.02.

2. During this practical assessment, you will be required to construct a project IAW the provided drawing. You must pass this test to pass EO 004.02. To pass the test, you must complete the project in accordance with the given drawing in the time frame provided.

3. References will be provided.

4. Pass/Fail IAW provided marking guide.

5. You will have 800 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See marking guide for the “Tray Project” with attached drawing.

Enabling Check – EO 004.02

Part D - Project Plastic Bracket

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Approved Plastic Workshop

Arrangements and Procedures

2. EC 004.02 (Project Plastic Bracket) will be a practical assessment based on content from EO 004.02.

3. Pass/Fail.

4. Time required: 200 minutes.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 004.02 (Project Plastic Bracket) will be a practical assessment based on content from EO 004.02.

2. During this practical assessment, you will be required to construct a project IAW the provided drawing. You must pass this test to pass EO 004.02. To pass the test, you must complete the project in accordance with the given drawing in the time frame provided.

3. References will be provided.

4. Pass/Fail IAW provided marking guide.

5. You will have 200 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See marking guide for the “Project Plastic Bracket” with attached drawing.

Enabling Check – EO 004.02

Part E - Project Picture Stand

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Approved Plastic Workshop.

Arrangements and Procedures

2. EC 004.02 (Project Picture Stand) will be a practical assessment based on content from EO 004.02.

3. Pass/Fail.

4. Time required: 250 minutes.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 004.02 (Project Picture Stand) will be a practical assessment based on content from EO 004.02.

2. During this practical assessment, you will be required to construct a project IAW the provided drawing. You must pass this test to pass EO 004.01. To pass the test, you must complete the project in accordance with the given drawing in the time frame provided.

3. References will be provided.

4. Pass/Fail IAW provided marking guide.

5. You will have 250 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See marking guide for the “Project Picture Stand” with attached drawing.

PERFORMANCE CHECK – PO 004

Part A – Wood Shop Phase (Chest)

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Approved Wood Workshop.

Arrangements and Procedures

2. PC 004 Part A will be a practical assessment based on content from EO 004.02.

3. Pass/Fail.

4. Time required: 700 minutes.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 004 Part A will be a practical assessment based on content from EO 004.02.

2. During this practical assessment, you will be required to fabricate a project IAW supplied drawings and marking guide.

3. Pass/Fail IAW provided marking guide.

4. You will have 700 minutes to complete this assessment.

5. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Project shall be fabricated IAW provided marking guide and associated drawings.

2. This Part A will make up 30% of the overall mark for PC 004.

PERFORMANCE CHECK – PO 004

Part B – Metal Shop Phase (Chest)

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Approved Metal Workshop.

Arrangements and Procedures

2. PC 004 Part B will be a practical assessment based on content from EO 004.02.

3. Pass/Fail.

4. Time required: 350 minutes.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 004 Part B will be a practical assessment based on content from EO 004.02.

2. During this practical assessment, you will be required to fabricate a project IAW supplied drawings and marking guide.

3. Pass/Fail IAW provided marking guide.

4. You will have 350 minutes to complete this assessment.

5. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Project shall be fabricated IAW provided marking guide and associated drawings.

2. This Part B will make up 30% of the overall mark for PC 004.

PERFORMANCE CHECK – PO 004

Part C – Plastic Shop Phase (Chest)

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop.

Arrangements and Procedures

2. PC 004 Part C will be a practical assessment based on content from EO 004.02.

3. Pass/Fail.

4. Time required: 200 minutes.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 004 Part C will be a practical assessment based on content from EO 004.02.

2. During this practical assessment, you will be required to fabricate a project IAW supplied drawings and marking guide.

3. Pass/Fail IAW provided marking guide.

4. You will have 200 minutes to complete this assessment.

5. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Project shall be fabricated IAW provided marking guide and associated drawings.

2. Part C will make up 30% of the overall mark for PC 004.

3. Once Part A, Part B and Part C are completed, the project shall be fully assembled and marked IAW marking guide Part D.

4. The fully assembled project, they will make up 10% of the overall mark for PC 004. If the project cannot be fully assembled, the Trainee shall lose the above mentioned 10%.

PO 005 LEAD A DAMAGE CONTROL TEAM

EO 005.01 Attack Team Leader (ATL) Course

1. Performance: The Enabling Objective and associated tasks shall be trained and assessed IAW the QSP for ATL, MITE ID 118787.

2. Remarks: Any considerations or special instructions (ie: Instructor requirements, etc) are IAW ATL QSP.

EO 005.02 Attack Live Fires

1. Performance: Attack Live Fires

2. Conditions:

   a. Given:

      (1) Live Fire training area;
      (2) Associated FF equipment;
      (3) PPE;
      (4) Testing equipment and applicable tools;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Live fire training area.

3. Standard: The Marine Technician Basic Maintainer shall independently Attack Live Fires IAW refs by:

   a. Utilizing RCN firefighting techniques to attack live fires to include:

      (1) Attacking class B fires (also known as a fuel pan burn);
      (2) Attacking class B fires utilizing Aqueous Film Forming Foam (AFFF); and
      (3) Advancing on a live fire.

   b. Explaining the effectiveness of a sprinkler system on a fire at different stages; and

   c. Explaining use and effectiveness of hydraulic ventilation.

4. Time Required:

   a. EO - 400 minutes.

5. Instructional Methods:

   a. Demonstration Performance.

6. Training Aids (To be transported to live fire facility):

   a. FFE with SCBA;

   b. Thermal Imaging Camera (TIC) x 2 with 2 spare batteries;

   c. SCBA bottles x 2 per person;

d. VARI nozzles x 2;

e. Forced Rescue Tool x 2;

f. Rescue Strap x 4;

g. Tri-Gate x 2; and

h. Bottled drinking water x 3 flats.

7. Test Details:

a. All assessments will be covered by the staff at the Live Fire facility.

8. References: A43 and A103.

9. Remarks: Course taught at Civilian Live Fire Training Facility under contract. Considerations to be included are: travel times, meals, and transportation. All safety parameters will be covered by civilian agency. Additional driver with Transportation Dangerous Goods (TDG) will be needed to ensure that all personal safety equipment is transported to and from facility.

10. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Utilizing RCN firefighting techniques to attack live fires							
1.1 Utilizing RCN firefighting techniques to attack live fires to include: a. Attacking Class B fires; b. Attacking Class B fires utilizing AFFF; and c. Advancing on live fires.				350	DP		Course taught at Civilian Live Fire Training Facility under contract
1.2 Explaining the effectiveness of a sprinkler system on a fire at different stages.			25				
1.3 Explaining use and effectiveness of hydraulic ventilation.			25				

Enabling Check – EO 005.02

PART I – ADMINISTRATIVE INSTRUCTIONS

1. The contents of EO 005.02 will be assessed during testing provided by the Live Fire facility.

EO 005.03 Lead a Flood Control Team

1. Performance: Lead a flood control team

2. Conditions:

   a. Given:

      (1) Incident Board;
      (2) Plotting Markers;
      (3) D/C legend;
      (4) D/C equipment;
      (5) Appropriate tools;
      (6) PPE;
      (7) References; and
      (8) Supervision.

   b. Denied: Nil.

   c. Environmental: DCTF with a Flood trainer.

3. Standard: The Marine Technician Basic Maintainer shall independently, adhering to all safety regulations, lead a flood control team IAW refs by:

   a. Selecting flood control team requirements/positions, for the appropriate:

      (1) Shoring incident;
      (2) Pipe repair incident; and
      (3) Pumping incident.

   b. Prioritizing on site repairs, to include:

      (1) Identifying hazards;
      (2) Identifying source of flooding; and
      (3) Choosing the appropriate flood control method.

   c. Delivering initial brief to Section Base I/C, to include:

      (1) Requesting power isolation;
      (2) Requesting piping system isolation as required;
      (3) Requesting permission to pump as required; and
      (4) Making recommendations based on priorities and repair requirements.

   d. Leading the flood control team, to include:

      (1) Briefing the flood control team, by:

         i. Ensuring members know their given roles/responsibilities;

ii. Ensuring all members use appropriate PPE;
iii. Ensuring team member's safety throughout; and
iv. Identifying when member's require to be replaced.

(2) Leading a shoring team, by:

i. Selecting the appropriate shoring method; and
ii. Ensuring all shoring benchmarks are met.

(3) Leading a pipe repair team, by:

i. Selecting the appropriate pipe repair method;
ii. Selecting the appropriate jumping method/route; and
iii. Ensuring all pipe repair/jumping benchmarks are met.

(4) Lead a pumping evolution, by:

i. Selecting the appropriate pumping method; and
ii. Ensuring all pumping evolution benchmarks are met.

4. Time Required:

a. EO - 875 minutes; and

b. EC - 50 minutes per trainee.

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aid:

a. DCTF;

b. DC Equipment/materials;

c. Applicable tools;

d. Classroom;

e. Multimedia projector; and

f. Job aid.

7. Test Details:

a. EO 005.03 will be assessed in EC 005.03.

8. References: A43 and A103.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Selecting flood control team requirements/positions, for the appropriate:						
1.1 Shoring incident, to include: a. Shoring items, to include: (1) Softwood plugs and wedges; (2) Flood repair kits; (3) Pre-cut pieces; (4) Metal shore; (5) Wooden shoring; (6) Shoring caps; (7) Splinter boxes; (8) Improvised materials (for large holes); (9) Appropriate tools; (10) Appropriate PPE. b. Shoring team make-up, to include: (1) Shoring team leader; (2) Cutter; (3) Runner; and (4) Builders.		100		D	A43 A103	
1.2 Pipe repair incident, to include: a. Pipe repair kits, to include: (1) Softwood plugs and wedges; (2) Soft patches; (3) Band-it Clamps; (4) Skinner seals; (5) Chain Vice-Grips; (6) Wraps; (7) Additional tools; and (8) Appropriate PPE. b. Jumping/bridging items, to include: (1) Hose sizes; (2) Fittings; and (3) Appropriate tools.						
1.3 Pumping incident, to include: a. Portable pumps, to include: (1) Submersible pumps; (2) WAJAX pumps; (3) Appropriate tools; (4) Appropriate hoses/fittings; and (5) Appropriate PPE. b. Brief on appropriate pump safety, to include: (1) Lifting/carrying requirements; and (2) Up/Down ladder safety.						

c. Internal Pumping equipment, to include: (1) Individual pump suctions; and (2) Eductor system.						
TP2 Prioritizing on site repairs, to include:						
2.1 Identifying hazards, to include: a. Hazards to team; and b. Hazards to ship.	25			IL		
2.2 Identifying source of flooding, to include: a. Internal flooding, ie Pipe break; and b. External flooding.						
2.3 Choosing the appropriate flood control method, to include: a. When/What (type) of repair; and b. When not to repair.						
TP3 Delivering initial brief to Section Base I/C, to include:						
3.1 Requesting power isolation, to include: a. Identifying appropriate space/zone to have isolated.	25			IL		
3.2 Requesting piping system isolation as required, to include: a. When/when not is it required to ask for permission.						
3.3 Requesting permission to pump as required, to include: a. When/when not is it required to ask for permission.						
3.4 Making recommendations based on priorities and repair requirements, to include: a. Utilizing information gathered from TP2.						
TP4 Leading the flood control team, to include:						
4.1 Briefing the flood control team, by: a. Ensuring members know their given roles/responsibilities; b. Ensuring all members use appropriate PPE; c. Ensuring team member's safety throughout; d. Identifying when member's require to be replaced, to include: (1) Identifying signs of hypothermia; (2) Identifying signs of fatigue; (3) Identifying signs of stress; (4) Reacting to stress injuries; and (5) Reacting to physical Injuries.	25	25	100	IL DP		
4.2 Leading a shoring team, by: a. Selecting the appropriate shoring method, to include: (1) Vertical; (2) Square; (3) "V"; (4) Splinter box; (5) EMMS clamp; and (6) Metal shore. b. Ensuring all shoring benchmarks are met, to include: (1) Shoring commenced; (2) 50%;		25	200	DP		

	(3) 100%; and (4) Sentry posted. c. Briefing the sentry, to include: (1) Position of flood / pipe break; (2) Fitted systems activated; (3) Actions if incident reoccurs; (4) Personnel that may enter (CO, XO, DCO, Snr FF, or OOD); and (5) Evidence preservation.						
4.3	Leading a pipe repair team, by: a. Selecting the appropriate pipe repair method, to include: (1) Band-it Clamps; (2) Skinner seals; (3) Chain Vice-Grips; (4) Wraps; and (5) Jumping/bridging. b. Selecting the appropriate jumping method/route, to include: (1) Utilizing hydrants; (2) Utilizing down-comers; (3) Emergency cooling/ flooding considerations; and (4) Affected equipment considerations. c. Ensuring all pipe repair/jumping benchmarks are met, to include: (1) Piping isolated/not-isolated; (2) Piping repaired, or (3) Jump/bridge commenced; (4) Jump/bridge complete, charged; and (5) Sentry posted. d. Brief the sentry, as per TP 4.2 (c).		25	200	DP		
4.4	Lead a pumping evolution, by: a. Selecting the appropriate pumping method, to include: (1) Fixed internal pumping system; and (2) Portable pumping equipment. b. Ensuring all pumping evolution benchmarks are met, to include: (1) Continual reports on water levels; and (2) Pumping commenced; (3) Pumping complete; and (4) Sentry posted. c. Brief the sentry, as per TP 4.2 (c).		25	100	DP		

Enabling check – EO 005.03

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Damage Control Training Facility.

2. Assessments to be conducted during the practical stages of EO 005.03. One trainee will be chosen to act as the Flood Team Leader (FTL) for each instructor designed scenario. To maximize assessment opportunities, multiple scenarios should be conducted simultaneously with multiple assessors. The following are examples of FTL scenario's:

    a. Act as a shoring leader;
    b. Act as a pipe repair team leader; and
    c. Act as a firemain bridging team leader.

Arrangements and Procedures

1. EC 005.03 will be a practical test based on content from EO 005.03.

2. Pass/Fail IAW EC checklist.

3. Time required: 50 minutes per trainee.

4. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 005.03 will be a practical assessment based on content from EO 005.03.

2. During this practical assessment, you will be required to lead a flood control team. You must pass this test to complete EO 005.03. To pass this test, you must complete performance statement in accordance with the standard.

3. References will not be provided.

4. Pass/Fail.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Failure of a critical safety factor will represent an automatic failure of EC 005.03.

PART IV – PERFORMANCE CHECK LIST

ENABLING CHECKLIST			
Date:	SN/PRI:	Rank:	Name:

TASKS: Successfully lead a Flood Team. Trainee to be assessed as a Flood Team Leader (**FTL**) for one of the following scenarios:

a. Shoring;
b. Pipe repair; and
c. Firemain bridging.

Administrative Instructions:

The Flood Team Leader (FTL) practical assessment checklist has been created to limit subjectivity during the assessment of each of the performance elements listed. The trainee is to be measured according to the checklist which has been derived from the standard statements of EO 005.03.

The Marine Technician Basic Maintainer shall independently, adhering to all safety regulations, lead a flood control team.

In determining the level of assistance an instructor or assessor may provide a trainee during their assessment the following should be considered;

a. Try to ensure the trainee and flood team are given an opportunity for the scenario to progress to its conclusion;
b. If the trainee experiences difficulty, provide minimal assistance to maintain progress through the PC;
c. Team safety shall be observed throughout. If a safety infraction is committed or about to be committed the scenario should be either;
   (1) Paused and corrective action taken before continuing, or
   (2) Halted if the infraction is serious in nature.

CONDITIONS and EQUIPMENT:

a. Flood equipment;
b. Dark or lit space;
c. PPE (glasses, gloves, submersion suit, ear protection, BA);
d. Major flood/Major pipe break;
e. Smoke filled compartment (potentially); and
f. Supervision.

OBJECTIVES	SAT	UNSAT
Brief flood team: (Designation of positions)**.**		
Conduct equipment check: a. All personnel have appropriate PPE; b. Submersion suits; c. Appropriate leak stopping equipment based on given scenario; and d. Portable communication (Radio check).	**/4**	

Check teams dress and confirm positions prior to being dispatched: (Team is in the ready position, air turned on IF SMOKE FILLED SPACE)		
LEAKSTOPPING PROCEDURES	**SAT**	**UNSAT**
Consideration for fitted systems or portable pumps. (If applicable)		
Correct procedures adhered to for pumping to commence (permission).		
Selected the appropriate method for Leak stopping.		
Verify plotting board is accurate during brief to I/C.		
Words of Command / Reports: a. Shoring/Bridging/Pipe repair commenced (specific location); b. 50% (if applicable); c. 100%; and d. Sentry briefed and posted.	**/4**	
SENTRY	**SAT**	**UNSAT**
Verbally announce: a. Verify that the sentry has appropriate PPE; and b. Verify that the sentry has appropriate leak stopping equipment. **Brief to sentry:** a. Position of flood / pipe break; b. Fitted systems activated; c. Actions if incident reoccurs; d. Personnel that may enter (CO, XO, DCO, Snr FF, or OOD); and e. Evidence preservation.	**/7**	
BRIEF TO SECTION BASE I/C	**SAT**	**UNSAT**
Brief I/C on relevant info. a. Location of flood; b. Depth of Water; c. Damage repair recommendation; and d. Progress report of water depth.	**/4**	
Verified priority of incidents for repair from I/C		
SAFETY (CRITICAL FACTOR) UNSAT REPRESENTS AN AUTOMATIC FAILURE OF EC 005.03	**SAT**	**UNSAT**
Trainee led the Flood team safely and did NOT place themselves, the team or the casualty (if applicable) at risk through their actions during the assessment.		
Proper use of PPE and SCBA (if applicable)		
Student requires 70 % to pass with no safety infractions		

Overall Assessment / 26	**Result** **PASS: P** **FAIL: F**

REMARKS

Strengths:	Areas for improvement:

Date:	**Trainee's Signature:**	**Instructor's Signature:**

EO 005.04 Lead a Casualty Power Team

1. Performance: Lead a casualty power team

2. Conditions:

   a. Given:

      (1) Incident Board;
      (2) Plotting Markers;
      (3) D/C legend;
      (4) Casualty power equipment;
      (5) PPE;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: DCTF and HMC Ship.

3. Standard: The Marine Technician Basic Maintainer shall independently Lead a casualty power team IAW refs by:

   a. Asses Casualty Power Team Requirements, to include:

      (1) Prioritizing Casualty Power routes.

   b. Tasking of Casualty Power personnel; and

   c. Performing Casualty Power safety practices, to include:

      (1) Configuring casualty power cable route (ANT0200); and
      (2) Inspecting casualty power riser terminal cables (ANT0340).

4. Time Required:

   a. EO - 700 minutes; and

   b. EC – 50 minutes per trainee.

5. Instructional Methods:

   a. Interactive lecture; and

   b. Demonstration Performance.

6. Training Aids:

   a. DCTF;

b. Casualty Power Equipment/tools/materials;

c. 100/400 Amp CP cable;

d. Risers and Bulkhead feed through;

e. Casualty Power route cards;

f. Electrical PPE; and

g. Electrical test Equipment.

7. Test Details:

a. EO 005.04 will be assessed in EC 005.04.

8. References: A43 and A103.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Assessing Casualty Power Team Requirements, to include:						
1.1 Prioritizing CP route by: a. Assessing hazards by receiving electrical manager's brief; b. Brief CP team on route and safety; c. Ensuring appropriately qualified personnel have been tasked to make affected system electrical safe; and d. Liaises with ERT on casualty power route.	20	20	200	IL DP		TP1, TP2 and TP3 should be instructed together within the DCTF.
TP2 Tasking of CP personnel, by:						
2.1 Acting as a CP Team leader by: a. Identifying the key personnel; b. Tasking personnel which section to set up CP route; c. Ensure personnel tasked with CP run in smoke filled compartments are wearing SCBA; and d. Ensure appropriate PPE is available and worn.	15	15	200	IL DP		
TP3 Performing CP Safety practices, by:						
3.1 Ensuring safe electrical practice are followed, by: a. Proper warning signs are posted; b. Cables are safe and tied off correctly; and c. Sentry is brief of duties and responsibilities correctly.	15	15	200	IL DP		

Enabling Check – EO 005.04

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Damage Control Training Facility.

2. Assessments to be conducted during the practical stages of EO 005.04. One trainee will be chosen to act as the casualty power team leader for each instructor designed scenario. To maximize assessment opportunities, multiple scenarios should be conducted simultaneously with multiple assessors.

Arrangements and Procedures

1. EC 005.04 will be a practical test based on content from EO 005.04.

2. Pass/Fail IAW EC checklist.

3. Time required: 50 minutes per trainee.

4. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 005.04 will be a practical assessment based on content from EO 005.04.

2. During this practical assessment, you will be required to lead a casualty power team. You must pass this test to complete EO 005.04. To pass this test, you must complete performance statement in accordance with the standard.

3. References will not be provided.

4. Pass/Fail.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Failure of a critical safety factor will represent an automatic failure of EC 005.04.

PART IV – PERFORMANCE CHECK LIST

ENABLING CHECKLIST			
Date:	SN/PRI:	Rank:	Name:

TASKS: Successfully lead a Casualty Power Team (CPT). Trainee to be assessed as a Casualty Power Leader (**CPL**) for one casualty power layout.

Administrative Instructions:

The CPL practical assessment checklist has been created to limit subjectivity during the assessment of each of the performance elements listed. The trainee is to be measured according to the checklist which has been derived from the standard statements of EO 005.03.

The Marine Technician Basic Maintainer shall independently, adhering to all safety regulations, lead a CPT.

In determining the level of assistance an instructor or assessor may provide a trainee during their assessment the following should be considered:

a. Try to ensure the trainee and CPT are given an opportunity for the scenario to progress to its conclusion;
b. If the trainee experiences difficulty, provide minimal assistance to maintain progress through the EC; and
c. Team safety shall be observed throughout. If a safety infraction is committed or about to be committed the scenario should be either:
   (1) Paused and corrective action taken before continuing, or
   (2) Halted if the infraction is serious in nature.

CONDITIONS and EQUIPMENT:

a. Casualty power equipment;
b. Dark or lit space;
c. PPE (glasses, electrical gloves, rubber mats, ear protection, BA);
d. Multiple casualty power runs;
e. Smoke filled compartment (potentially); and
f. Supervision.

OBJECTIVES	SAT	UNSAT
Receives Brief: a. Ask appropriate question and understands tasking; b. Demonstrate knowledge of tasking; and c. Walk the complete route (if safe to do so) to ensure no unforeseen hazard exist.	**/3**	
Brief CPT: a. Follow brief outline; and b. Task personnel appropriately, (i.e. not tasking individual not familiar with CP system).	**/2**	
Safety: a. Ensure that all electrical safety practices are being followed; (1) Electrical rubber gloves, leather protectors and rubber matting. b. Ensure load is made electrical safe by qualified personnel; and c. PPE is being used.	**/6**	

Conduct equipment check: a. All personnel have appropriate PPE; b. Casualty Power cards; c. Cable, taps, bulk head feed thoughts and risers; and d. Casualty power equipment is dispatched: (2) Twine and other securing devices; (3) Line voltage testers; and (4) Danger tags.	**/7**	
CORRECTLY SUPERVISES	**SAT**	**UNSAT**
Check teams dress and confirm positions prior to being dispatched: (If team is in the ready position, air turned on IF SMOKE FILLED SPACE)	**/1**	
Ensure CPT team hooks up cables correctly.	**/1**	
Check on CPT at intervals to ensure safe practices are being followed.	**/1**	
Allows CPT to work without becoming too hands on the equipment	**/1**	
Correct procedures for layout and securing of cables	**/1**	
Report to HQ1 when run is complete and ready for inspection	**/1**	
SAFETY (CRITICAL FACTOR) UNSAT REPRESENTS AN AUTOMATIC FAILURE OF EC 005.04	**SAT**	**UNSAT**
Trainee led the Casualty Power team safely and did NOT place themselves, the team or the casualty (if applicable) at risk through their actions during the assessment.		
Proper use of PPE and SCBA (if applicable)		
Student requires 70 % to pass with no safety infractions		
Overall Assessment / 25	**Result** **PASS: P** **FAIL: F**	
REMARKS		
Strengths:	Areas for improvement:	
Date:	**Trainee's Signature:**	**Instructor's Signature:**

PERFORMANCE CHECK – PO 005

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Trainees will be given feedback throughout the entirety of this PO. There is no formal PC in PO 005.

PO 006 MAINTAIN COMPRESSED AIR SYSTEMS

EO 006.01 Apply General Mathematics

1. Performance: Apply general mathematics

2. Conditions:

   a. Given:

      (1) References;
      (2) Textbooks;
      (3) Supervision;
      (4) Calculators; and
      (5) Writing materials.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard: The Marine Technician Machinery Maintainer shall apply general mathematics IAW refs, by:

   a. Utilizing fundamental algebra to solve equations and apply trigonometry, to include:

      (1) Describing algebra;
      (2) Solving linear equations;
      (3) Factoring;
      (4) Explaining angles and their positioning;
      (5) Graphing;
      (6) Solving right triangles;
      (7) Solving Trigonometric Equations; and
      (8) Solving Simultaneous Linear Equations.

   b. Performing operations involving complex number, to include:

      (1) Basic definitions;
      (2) Graphical Representation of Complex Numbers/Representing complex numbers as a point on the complex plane/Argand plane; and
      (3) Adding, subtracting, multiplying and dividing complex numbers.

   c. Solving exponential and logarithmic equations, to include:

      (1) Introduction;
      (2) Solving logarithmic functions; and
      (3) Solving exponential and logarithmic equations.

d. Utilizing algebra to solve word and geometry problems, to include:

(1) Solving problems involving word problems, ratio and proportion, similar triangles and variation; and

(2) Solving problems involving perimeter, area, volume, and Radians.

4. Time Required:

a. EO – 2875 minutes; and

b. CEC – 100 minutes. Additional 50 minutes required for CEC review.

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids: Nil.

7. Test Details:

a. The content of EO 006.01 will be assessed in CEC 006.01. An additional 50 minutes will be required prior to the CEC for review.

8. References: C3.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Utilizing fundamental algebra to solve equations and apply trigonometry by:						
1.1 Describing algebra to include: a. Defining algebraic expressions and associated terminology (Stress difference between terms and factors); b. Performing addition and subtraction of algebraic expressions; c. Performing multiplication of algebraic terms to include: (1) Monomial by a monomial, (2) Monomial by a multinomial, and (3) Multinomial by a multinomial. d. Performing division of algebraic expressions to include:	130		170	IL P		

	(1) Monomial by a monomial; and (2) Multinomial by a monomial.						
1.2	Solving linear equations by: a. Defining linear equation and the four types of linear equations: (1) Conditional; (2) Identity; (3) Impossible (contradiction); and (4) Equivalent. b. Rearranging a formula to solve for a given variable; and c. Isolating specific letters or symbols in literal equations.	150		200	IL P		
1.3	Factoring to include: a. Introduction; and b. Factoring Out (Removing) a Common Term/Factor.	40		60	IL P		
1.4	Explaining angles and their positioning to include: a. Explaining the rectangular coordinate system; b. Defining points in the rectangular coordinate system; c. Defining initial side, terminal side, and vertex; d. Explaining labelling angles; e. Defining positive angle and negative angle; f. Defining coterminal angles; g. Defining quadrantal angle; h. Defining right angle, acute angle, and obtuse angle; i. Explaining angles in standard position; and j. Sketching angles in standard position such that the terminal side passes through a given point.	75		25	IL P		
1.5	Graphing, to include: a. Terminology, and b. Graphing Functions.	40		60	IL P		
1.6	Solving right triangles by: a. Defining right triangle; b. Explaining how to draw and label a right triangle; c. Explaining the relationship between corresponding sides and corresponding angles; d. Describing sine, cosine, and tangent; e. Describing Pythagorean theorem; f. Utilizing trigonometric functions and Pythagorean theorem to solve for unknown angles and sides of a right triangle; and g. Solving applications of right triangles.	90		110	IL P		
1.7	Solving trigonometric equations, to include:	70		80	IL P		

	a. Introduction; b. Reference Angles; c. Explaining the signs of the trigonometric functions (CAST rule); and a. Solving for all possible angle θ's given a specific trigonometric function.						
1.8	Solving simultaneous linear equations by: a. Introduction; and b. Substitution.	70		100	IL DP		
TP2 Performing operations involving complex numbers, to include:							
2.1	Solving complex number problems by: a. Basic Definitions; b. Graphical Representation of Complex Numbers/ Representing Complex Numbers as a Point on the Complex Plane/Argand Plane/ Solving basic word problems; and c. Adding, Subtracting, Multiplying and Dividing Complex Numbers.	100		150	IL P		
TP3 Solving exponential and logarithmic equations, to include:							
3.1	Solving exponential and logarithmic equations by: a. Introduction of exponential functions; b. Logarithmic Functions; and c. Exponential and Logarithmic Equations.	130		200	IL P		
TP4 Utilizing fundamental algebra to solve word and geometry problems, to include:							
4.1	Solving word problems, ratio and proportion, similar triangles and variations by: a. Solving word problems; b. Solving ratio and proportion problems; c. Solving similar triangle problems; and d. Solving variation problems.	180		320	IL P		
4.2	Solving problems involving perimeter, area, volume and Radians by: a. Calculating perimeter; b. Calculating Area; c. Calculating Volume; and d. Converting Radians to Degrees.	150		175	IL P		

Critical Enabling Check – EO 006.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom.

Arrangements and Procedures

2. CEC 006.01 will be a written assessment based on content from EO 006.01.

3. Pass mark will be 60%.

4. Time required: 100 minutes.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 006.01 will be a written assessment based on content from EO 006.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will be provided.

9. Pass mark is 60%.

10. You will have 100 minutes to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Nil.

EO 006.02 Perform General Applied Physics

1. Performance: Perform general applied physics.

2. Conditions:

   a. Given:

      (1) References;
      (2) Textbooks;
      (3) Calculators;
      (4) Supervision; and
      (5) Writing materials.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard: The Marine Technician Machinery Maintainer shall independently perform general applied physics, IAW refs, by:

   a. Applying technical mathematics to solve physics problems, to include;

      (1) Defining physics;
      (2) Explaining technical measurement;
      (3) Performing unit conversions;
      (4) Calculating precision, accuracy, and significant digits;
      (5) Explaining scientific and engineering notation; and
      (6) Explaining basic mathematical concepts.

   b. Calculating the resultant of several vectors using the polygon and component methods of vector addition, to include:

      (1) Explaining scalar and vector quantities; and
      (2) Explaining the Component Method of Vector Addition through Calculations.

   c. Explaining Linear Motion through Calculation by:

      (1) Motion, Average Speed, Average Velocity and Uniform Acceleration;
      (2) Solving Uniform Acceleration Problems; and
      (3) Relative Velocity and Velocity-Time Graphs.

   d. Analyze systems in equilibrium and non-equilibrium conditions using free-body diagrams by:

      (1) Explaining Types of Forces;
      (2) Employing Newton’s First and Second Laws of Motion in Free-Body Diagrams; and

(3) Newton's Third Law and Calculating Frictional Forces Acting on a Body.

e. Explaining torque and conditions of rotational equilibrium by:

(1) Torque (Moment), Couples and Moment Arms;
(2) Resultant Torque and Rotational Equilibrium; and
(3) Explaining Center of Gravity and Equilibrium.

f. Explaining the relationship between work, energy and power using calculations by:

(1) Explaining Work and Resultant Work Through Calculations;
(2) Explaining Forms of Energy and the Relationship Between Kinetic Energy and Resultant Work Through Calculations
(3) Explaining Conservation of Energy through Calculations; and
(4) Explaining Power and Efficiency.

4. Time Required:

a. EO – 2845 minutes; and

b. CEC – 100 minutes. Additional 200 mins required for CEC review.

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids: Nil.

7. Test Details:

a. EO 006.02 will be assessed in CEC 006.02. 200 minutes shall be allotted for CEC review prior to the exam.

8. References: C3.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Applying technical mathematics to solve physics problems to include:						
1.1 Defining physics to include: a. Introduction to physics.	20			IL		
1.2 Explaining technical measurement to include:	20			IL		

	a. Seven fundamental physical quantities, such as: (1) Explaining fundamental units of measurement; and (2) Explaining derived units of measurement. b. Listing SI and USCS units.						
1.3	Performing unit conversions to include: a. Powers of 10; and b. Convert units by Multiplication.	125			IL		
1.4	Calculating precision, accuracy, and significant digits to include: a. Precision; b. Accuracy; c. Significant digits; and d. Rules for counting significant digits.	100			IL		
1.5	Explaining scientific and engineering notation by: a. Explaining scientific notation; b. Rules of converting a decimal number; c. Scientific notation to decimal form; d. Distinguish between scientific notation and engineering notation; and e. Using a calculator.	100			IL		
1.6	Explaining basic mathematical concepts to include: a. Basic algebra concepts; b. Explaining basic geometrical concepts, such as: (1) Angle; (2) Parallel lines; (3) Right angle; (4) Perpendicular lines; (5) Opposing angles; (6) Alternate interior angles; and (7) Triangle. c. Explaining right triangle trigonometry.	155			IL		
TP2 Calculating the resultant of several vectors using the polygon and component methods of vector addition by:							
2.1	Explaining scalar and vector quantities by: a. Defining magnitude, to include: (1) Defining scalar quantities; and (2) Defining vector quantities.	50			IL		
2.2	Explaining the Component Method of Vector Addition Through Calculations: a. Resultant Vector; b. Polygon method; c. Single force vector; d. Concurrent forces; e. Component method; and f. Determine magnitude and direction of vector.	390			IL		
TP3 Explaining Linear Motion through Calculation, by:							
3.1	Motion, Average Speed, Average Velocity and Uniform Acceleration, to include: a. Motion; b. Distance and displacement;	100			IL		

	c. Average speed; d. Average velocity; e. Velocity vs Speed; f. Average velocity vs instantaneous velocity; and g. Acceleration.						
3.2	Solving Uniform Acceleration Problems by: a. Uniformly accelerated motion; b. Average velocity (given uniform acceleration); c. Final velocity (given uniform acceleration); d. Displacement (given uniform acceleration); e. Solve linear motion; f. Free-fall; and g. Apply gravitational acceleration concepts.	200			IL		
3.3	Relative Velocity and Velocity-Time Graphs by: a. Concept of velocity relative to inertial frames; and b. Concept of velocity-time graphs.	150			IL		
TP4 Analyze systems in equilibrium and non-equilibrium conditions using free-body diagrams, by:							
4.1	Explaining Types of Forces by: a. Define forces; b. Define relationship between weight and mass; c. Concept of a free body diagram; and d. Total force or net force, to include: (1) One dimension; and (2) Two dimensions.	100			IL		
4.2	Employing Newton's First and Second Laws of Motion in Free-Body Diagrams by: a. Newton's First Law; b. Equilibrant force; and c. Newton's second law.	200			IL		
4.3	Newton's Third Law and Calculating Frictional Forces Acting on a Body by: a. Stating Newton's Third Law b. Define friction c. Define static friction d. Define kinetic friction e. Coefficient of friction f. Calculating Frictional forces	150			IL		
TP5 Explain torque and conditions of rotational equilibrium, by:							
5.1	Torque (Moment), Couples and Moment Arms by: a. Line of action of a force; b. Axis of rotation; c. Moment arm; and d. Torque.	100			IL		
5.2	Resultant Torque and Rotational Equilibrium by: a. Resultant torque; b. Couple; c. Equivalent resultant force and equivalent couple torque; d. Rotational equilibrium; and						

	e. Solve rotational equilibrium problems.						
5.3	Explaining Center of Gravity and Equilibrium by: a. Define Center of gravity; b. Center of gravity in uniformly shaped objects; and c. Center of gravity in non-uniformly shaped objects.	230			IL		
TP6 Explain the relationship between work, energy and power using calculations by:							
6.1	Explaining Work and Resultant Work Through Calculations by: a. Defining work; b. Positive work vs Negative work; c. Resultant work; and d. Area under a force-displacement curve.	160			IL		
6.2	Explaining Forms of Energy and the Relationship Between Kinetic Energy and Resultant Work Through Calculations by: a. Defining energy; b. Defining kinetic energy; c. Kinetic energy of the body; and d. Gravitational potential energy.	130			IL		
6.3	Explaining Conservation of Energy through Calculations by: a. Law of conservation of mechanical energy; and b. Conversion of energy formula.	170			IL		
6.4	Explaining Power and Efficiency by: a. Defining power; b. Unit of power; c. Formulas for power; d. Defining a machine; and e. Defining efficiency, to include: (1) Unit less ratio; and (2) Elapsed time.	195			IL		

Critical Enabling Check – EO 006.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom.

Arrangements and Procedures

2. CEC 006.02 will be a written assessment based on content from EO 006.02.

3. Pass mark will be 60%.

4. Time required: 100 minutes.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 006.02 will be a written assessment based on content from EO 006.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will be provided.

9. Pass mark is 60%.

10. You will have 100 minutes to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Nil.

EO 006.03 Maintain Compressed Air System Components

1. Performance: Maintain compressed air system components.

2. Conditions:

   a. Given:

      (1) Applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment;
      (4) Applicable tools;
      (5) Lock-out/tag-out equipment;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop.

3. Standard: The Marine Technician Machinery Maintainer shall maintain compressed air system components under supervision IAW refs, by:

   a. Inspecting compressed air system components, to include:

      (1) Inspecting HP air system components by:

         i. Inspecting HP air compressor.

      (2) Inspecting LP air system components by:

         i. Inspecting LP air compressor;
         ii. Inspecting compressed air auto lubrication system;
         iii. Inspecting LP air control stations; and
         iv. Inspecting pneumatic valves.

   b. Repairing compressed air system components, to include:

      (1) Installing auxiliary systems components (AIT0590) by:

         i. Assembling and disassembling HP air compressor (AIT0530); and
         ii. Assembling and disassembling LP air compressor (AIT0550).
         iii. Repairing compressed air automatic lubrication system (AHT0690);
         iv. Assembling and disassembling HP air system components (AIT0540);

v. Assembling and disassembling LP air system components (AIT0560);
vi. Repairing low pressure compressed air control station (AIT0810); and
vii. Assembling and disassembling pneumatic valve (ATT0140).

b. Testing compressed air system components, to include:

(1) Testing HP air system components by:

i. Testing HP air compressor.

(2) Testing LP air system components by:

i. Testing compressed air auto lubrication system;
ii. Testing LP air compressor;
iii. Testing LP air control stations; and
iv. Testing pneumatic valves.

4. Time Required:

a. EO – 2200 minutes; and

b. EC – 75 minutes per trainee.

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids:

a. Reciprocating piston type compressor;

b. Rotary vane type compressor;

c. HP air;

d. LP air;

e. Compressed air lubricators;

f. LP compressed air control stations; and

g. Pneumatic valves.

7. Test Details:

a. EO 006.03 will be assessed in EC 006.03.

8. References: A94 and A95.

9. Remarks: Requires at least 3 compressors of each type and 8 lubricators, control air stations, reducing stations, and pneumatic valves.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspect Compressed air systems to include:						
1.1 Inspecting High Pressure (HP) air system components by: a. Inspecting HP air compressor by: (1) Identifying the main components of a multi-stage reciprocating compressor; (2) Interpreting OEM manual or CFTO; (3) Identifying each compression stage; (4) Identifying path of coolant; (5) Identifying inter-stage cooler; (6) Identifying suction/discharge valves; (7) Identifying safety valves; and (8) Explaining operation from stationary to loaded, unloaded and shut down. b. Inspecting an HP air reducing station by: (1) Identifying components of an HP air reducing station. c. Identifying indications of a malfunctioning reducing station: (1) Icing; (2) Relief valve lifting; (3) Improperly set pressure; and (4) Clogged strainer.	75			IL	A95	Instructor to use OEM manual or CFTO to familiarize trainees with equipment to be assembled/disassembled.
1.2 Inspecting Low Pressure (LP) air system components by: a. Inspecting LP air compressor by: (1) Identifying the main components of a rotary vane compressor; (2) Interpreting the OEM manual or CFTO; (3) Identifying the path of oil, coolant and air through the compressor; (4) Identifying safety valves; (5) Identifying separators; (6) Identifying heat exchangers; and (7) Explaining operation from stationary, to operating	75			IL	A94	Instructor to use OEM manual or CFTO to familiarize trainees with equipment to be assembled/disassembled.

loaded/unloaded, to shut down. b. Inspecting compressed air auto-lubrication systems by: (1) Identifying level of lubricant in reservoir; (2) Checking for leaks with liquid leak detect solution; and (3) Checking visual indicators of lubrication. c. Inspecting LP air compressed air control stations by: (1) Checking if the solenoid blocks are set for remote control; (2) Investigating if proper pressure is supplied; (3) Checking physical integrity of the components; and (4) Checking for moisture in the station by operating drain valve. d. Inspecting pneumatic valves by: (1) Investigating proper operation.						
TP2 Repair Compressed air systems to include:						
2.1 Assembling / disassembling HP air compressor by: a. Conducting a proper lock-out/tag-out by: (1) Isolating 2 valves upstream; (2) Isolating 1 valve downstream; and (3) Isolating electrically. b. Utilizing equipment specific tools; c. Disconnecting lines; d. Repairing suction & discharge valves by: (1) Removing suction and discharge valves; (2) Lapping discs; (3) Measuring spring size and replacing if required; and (4) Replacing gaskets. e. Repairing intercoolers by: (1) Removing intercooler covers; (2) Inspecting intercooler tubes; (3) Inspecting corrosion pieces; and (4) Reassemble all components. f. Repairing an HP air reducing station by: (1) Isolating and bleeding off air pressure; (2) Removing and cleaning strainer; and (3) Reinstalling strainer. g. Assembling/disassembling a control air station by: (1) Removing solenoids; (2) Disassembling block; (3) Replacing o-rings;		1600	DP	A94, A95	Instructor to walk students through overhaul, and assist as required.

(4) Cleaning block assembly; and (5) Reassembling block with solenoids. h. Assembling disassembling a pneumatic operated valve by: (1) Isolating air supply; (2) Bleeding off pressure on diaphragm; (3) Disassembling valve; (4) Inspecting valve internal components for defects (5) Replacing diaphragm; and (6) Reassembling valve.					
2.2 Assembling / disassembling LP air compressor by: a. Conducting a proper lock-out/tag-out by: (1) 1 valve upstream isolation; (2) 1 valve downstream isolation; and (3) De-isolating electrically. b. Disassembling all minor components on the casing; c. Disassembling compressor up to removing rotor and changing vanes; d. Reassembling rotor with vanes and installing into casing; and e. Reassembling all minor components.					
TP3 Test compressed air systems to include:					
3.1 Testing HP air compressed systems by: a. Testing HP air compressor by: (1) Removing lock-out/tag-out devices; (2) De-isolating valves; (3) Setting up compressor to flash-up; and (4) Starting compressor. b. Recognizing normal operating parameters by: (1) Identifying proper pressure; (2) Identifying proper temperature; (3) Inspecting for leaks; and (4) Identifying normal vibration. c. Testing HP air reducing station by: (1) Setting HP air reducing station dome pressure; and (2) Checking proper pressure is regulated.		225	DP	A95	
3.2 Testing LP air system components by: a. Testing LP air compressor by: (1) Removing lock-out/tag-out devices; (2) De-isolating valves; (3) Setting up compressor to flash-up; and (4) Starting compressor. b. Recognizing normal operating parameters by: (1) Identifying proper pressure;		225		A94	

(2) Identifying proper temperature; (3) Inspecting for leaks; and (4) Identifying normal vibration. c. Testing operation of pneumatic valve with control air station; and d. Testing compressed air automatic lubrication system.					

Enabling Check – EO 006.03

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop with one invigilator per compressor

Arrangements and Procedures

2. EC 006.03 will be a practical assessment based on content from EO 006.03.

3. Pass/Fail.

4. Time required: 75 minutes per trainee X 6 compressors, 3 of each style (HP and LP), with two trainees per comp.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 006.03 will be a practical assessment based on content from EO 006.03.

2. During this practical assessment, you will be required to maintain compressed air system components. You must pass this test to pass EO 006.03. To pass the test, you must maintain compressed air system components in accordance with the standard.

3. References will be provided.

4. Pass/Fail as per EC checklist.

5. You will have 75 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See comments in the table below

PART IV – ENABLING CHECK LIST

Check List (time in min) EC 006.03	SAT	UN-SAT	Comments (for invigilator, before printing, these comments to be deleted and the space is used for comments on trainees)
1. Inspect HP Air Compressor (10)			The invigilator is to have the trainee inspect the HP Air Comp and then after successfully telling the invigilator what they are looking for the invigilator to tell the trainee the fault.
2. Inspect LP Air Compressor (10)			The invigilator is to have the trainee inspect the LP Air Comp and then after successfully telling the invigilator what they are looking for the invigilator to tell the trainee the fault.
3. Inspect pneumatic valves (5)			The invigilator is to have the trainee inspect the pneumatic valve and then after successfully telling the invigilator what they are looking for the invigilator to tell the trainee the fault.
4. Inspect LP air control stations (5)			The invigilator is to have the trainee inspect the LP air control stations and then after successfully telling the invigilator what they are looking for the invigilator to tell the trainee the fault.
5. Assemble and disassemble LP air compressor (25)			The trainee is to take apart the LP air comp and fix the fault that the invigilator gave them during the inspect task.
6. Repair compressed air automatic lubrication system (10)			The repair of the auto lube system can be repair by replacement or refilling it with new oil.
7. Assemble and disassemble HP air system components (25)			The components that need assembling and disassembling will be told to the trainee by the invigilator.
8. Assemble and disassemble LP air system components (15)			The components that need assembling and disassembling will be told to the trainee by the invigilator.
9. Repair low pressure compressed air control station (15)			The repair will be to the fault that was found during the inspect task.
10. Assemble and disassemble pneumatic valve (15)			This is a basic assemble/disassemble taken from the inspect.
11. Test HP air compressor (10)			This is a run up of the comp to prove function
12. Test LP air system components (10)			The LP air components can be tested when the LP air compressor is run up for testing
13. Test compressed air auto lubrication system (10)			The test for this is to have an air gun on the exit side of the lubricator. Spray a piece of wood or a rag to see if oil comes out.

14.	Test LP air compressor (10)			This is a run up of the comp to prove function
15.	Test LP air control stations (10)			A signal will have to be sent to the control air station to prove function
16.	Test pneumatic valves (10)			The signal sent to the control air station in the previous test can be to operator the valve in this task.

Instructions to Evaluator:

1. For this EC, if there are enough invigilators to have one per compressor, then the total time for the EC will be 75 mins. If there are not enough compressors and/or invigilators, then the time for this EC will increase.
2. Trainees must attempt 2 of the 4 inspect tasks and successful pass both to move to the repair part of the EC.
3. Trainees must attempt 4 of the 6 repair/assemble and disassemble tasks and successful pass 3 to move to the test part of the EC.
4. Trainees must attempt 3 of the 5 test tasks and successful pass 3 to complete this EC. The test tasks should be testing of the repairs done during the repair part of the EC.
5. The only grades awarded for this EC are PASS or FAIL.

PERFORMANCE CHECK - PO 006

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Trainees will be provided feedback throughout the PO. No formal PC is included in PO 006.

PO 007 MAINTAIN AUXILIARY STEAM SYSTEMS

EO 007.01 Maintain Auxiliary Steam Generator

1. Performance: Maintain Auxiliary Steam Generator

2. Conditions:

   a. Given:

      (1) References;
      (2) Steam generator;
      (3) Lock out/ Tag out equipment;
      (4) Appropriate tools;
      (5) PPE; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Workshop and Classroom.

3. Standard: The Marine Technician Machinery Maintainer shall maintain auxiliary steam generator under supervision IAW refs, by:

   a. Inspecting steam generator sub-system components, to include:

      (1) The Feedwater System:
      (2) The Fuel System; and
      (3) The Control System.

   b. Repairing steam generator sub-system components, to include:

      (1) Repairing steam generator fuel igniter (AHT0780);
      (2) Repairing auxiliary steam generator fuel nozzle (AIT0670);
      (3) Repairing auxiliary steam generator forced air supply fan (AIT0660);
      (4) Repairing auxiliary steam generator air damper (AIT0650);
      (5) Repairing feed water chemical injection system (AHT0700); and
      (6) Cleaning auxiliary steam generator tubes (AHT0550).

   c. Testing steam generator sub-system components, to include:

      (1) Operating auxiliary steam generator remotely (ET0120);
      (2) Operating auxiliary steam generator locally (ET0100);
      (3) Testing auxiliary steam generator high fire electrode (AHT0810);
      (4) Testing auxiliary steam generator electronic safety control (AHT0800); and
      (5) Conducting salinity test (AT0090).

4. Time Required:

   a. EO – 1025 minutes;

   b. EC – 30 minutes per 2 trainees; and

   c. CEC – 50 minutes per trainee.

5. Instructional Methods:

   a. Interactive lecture; and

   b. Demonstration – performance.

6. Training Aids:

   a. Functional steam generator;

   b. Burner assembly with nozzles, igniters and fire-eye;

   c. Multi meter; and

   d. Feed water test kits.

7. Test Details:

   a. Content from EO 007.01 will be tested in EC 007.01 and CEC 007.01.

8. References: C28 and A97.

9. Remarks: A functional steam generator will be required to ensure each student's repairs are properly done. 1 burner assemblies per pair of student (6 total) will drastically decrease testing time.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Ref	Comments
	T	D	P			
TP1 Inspecting steam generator sub-system components, by:						
1.1 Inspecting the feed water system: a. Flow; b. Hot well; c. Feedwater pump; d. Heating coils; and e. Steam Separator.	100			IL		
1.2 Inspecting the fuel system: a. Fuel system; b. Air Supply; and c. Burner.	100			IL		
1.3 Inspecting the control system: a. Main Temperature limit controller	150			IL		

	b. Primary Main Temperature Limit Controller (MTLC2); c. Secondary Main Temperature Limit Controller (MTLC1); d. Automatic Burner modulator; e. Air Pressure Switch; f. Aux. Thermostat Switch; g. Electric Interlock Switch; h. Limit Pressure Switch; i. Modulating Pressure Switch; j. Electronic Safety Controls; k. Fuel Pressure Switch; l. Main Oil Valve; m. High Fire Valve; n. High Fire Timer; o. Holding Relay; p. Ignition Transformer; q. Limit Pressure Switch; r. Operating Pressure Switch; and s. Annunciation System.						
TP2 Repairing steam generator sub-system components, by:							
2.1	Perform the following Preventative Maintenance routines: a. Daily; b. Weekly; c. Monthly; d. Annual or Semi-Annual; e. Manual Blow; and f. Soot Blow.	50	25	25	IL DP		
2.2	Performing troubleshooting on the following: a. Steam Trap; and b. Annunciator System.	25			IL		
TP3 Testing steam generator sub-systems, by:							
3.1	Performing the Boiler Feed water Testing	25	50	250	IL DP		
3.2	Performing water treatment practices.	25	25	175	IL DP		

Enabling Check – EO 007.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop

Arrangements and Procedures

2. EC 007.01 will be a practical assessment based on content from EO 007.01.

3. Pass/Fail IAW EC checklist.

4. Time required: 30 minutes per 2 trainees

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 007.01 will be a practical assessment based on content from EO 007.01.

2. During this practical assessment, you will be required to service a steam generator fuel burner assembly. You must pass this test to pass EO 007.01. To pass the test, you must complete the servicing of a steam generator fuel burner assembly in accordance with the standard.

3. References will be provided.

4. Pass/Fail IAW EC checklist

5. You will have 30 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See checklist below

PART IV – ENABLING CHECK LIST

Check List		SAT	UNSAT	Comments
1.	PPE is worn			Minimum nitrile gloves and safety glasses when working with equipment that has had POLs.
2.	Trainee fills out a lockout / tag-out form.			Trainee indicating what needs to be locked to ensure safety while conducting maintenance.
3.	Trainee properly disconnects all lines and removes burner from steam generator.			
4.	Disassembled burner assembly fully.			Trainee is given a workbench and appropriate tools.
5.	Inspects fuel nozzles.			
6.	Inspects electrodes.			
7.	Inspects mirror and sight glass.			
8.	Inspects fire eye.			
9.	Identifies defect.			Points out to the instructor the defect
10.	Remedies defect.			To the satisfaction of the instructor the trainee knows how to rectify the defect, ideally without repair by replacement.
11.	Reassembles the burner assembly.			
12.	Sets the electrode gap			Gap is set within tolerance IAW OEM specs.
13.	Reinstalls burner assembly correctly connecting each line and electrical connectors.			
14.	Trainee completes LOTO form			

Instructions to Evaluator:

1. The instructor is to have the trainees, in pairs, disassemble and then reassemble a steam generator fuel burner. There should be defective components in the burner;
2. The defect can be anything; as long as it was taught on the EO;
3. Only one defect needs to be given to the trainee, if the instructor has enough time and needs to validate the trainee further, a second defect can be added;
4. Trainees begin with 100 points; each fault subtracts 8 from the total;
5. The trainee must receive more than 75 points in total to pass this EC; and
6. The only grades awarded for this EC are PASS (for totals 75 points or greater) or FAIL (for totals less than 75 points).

Critical Enabling Check – EO 007.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop with functional steam generator

Arrangements and Procedures

2. CEC 007.01 will be a practical assessment based on content from EO 007.01.

3. Pass/Fail IAW CEC checklist.

4. Time required: 50 minutes per trainee

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. CEC 007.01 will be a practical assessment based on content from EO 007.01.

2. During this practical assessment, you will be required to operate a steam generator. You must pass this test to pass EO 007.01. To pass the test, you must operate the steam generator in accordance with the standard.

3. References will not be provided.

4. Pass/Fail IAW CEC checklist.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. See comments in the table below

PART IV – CRITICAL ENABLING CHECKLIST

Checklist	SAT	UNSAT	Comments
1. Wears proper PPE (15)			Trainee dons ear plugs, safety glasses, and steam gloves when told to commence with the test.
2. Inspect the steam generator (15)			Trainee is checking the steam generator's physical integrity and ensuring the steam generator is ready to start. Trainee is to notify instructor of all deficiencies.
3. Starting the steam generator (30)			Trainee follows proper flash-up procedure. Trainee maintains minimum steam pressure during warm-up, Blow-downs as required, proper cycling between hi-low fire, opening/closing drains as required, etc.
4. Monitoring steam generator (10)			Trainee properly fills out logbook and records all required operating parameters.
5. Indicates daily maintenance requirements (20)			Trainee describes soot blowing procedure, where and how to collect feedwater samples. Demonstrates knowledge of each feedwater test and acceptable limits.

Instructions to Evaluator:

1. For this CEC, only 1 invigilator is required due to the limited availability of equipment and systems.
2. Trainees begin with 100 points; per non-answer or missed step/procedure subject to the severity of the fault subtracts 8 from the total.
3. The trainee must receive more than 75 points in total to pass this CEC.
4. The only grades awarded for this CEC are PASS (for totals 75 points or greater) or FAIL (for totals less than 75 points).

EO 007.02 Perform General Chemistry

1. Performance: Perform General Chemistry

2. Conditions:

   a. Given:

      (1) References; and
      (2) Calculator.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard: The Marine Technician Machinery Maintainer shall perform general chemistry IAW refs, by:

   a. Defining nature of matter, to include:

      (1) Definition of chemistry;
      (2) Defining matter and the Kinetic Molecular theory;
      (3) The three states of matter;
      (4) Defining Heterogeneous or homogenous;
      (5) Describing physical and chemical properties; and
      (6) Describing the Laws of conservation of mass/energy.

   b. Explaining models of atomic structure, to include:

      (1) Models of atomic structure; and
      (2) Models of atomic structure – the Bohr model.

   c. Explaining the periodic table;

   d. Explaining bonding-classifying chemicals as ionic or covalent to include:

      (1) Bonding;
      (2) Bonding-ionic compounds; and
      (3) Bonding – covalent compounds.

   e. Explaining chemical reactions, to include:

      (1) Chemical equations;
      (2) The unit factor and the Mole; and
      (3) Quantitative relationships.

   f. Explaining solution chemistry, to include:

      (1) Qualitative;
      (2) Quantitative; and

(3) Reactions in solution.

g. Explaining acids and bases, to include:

(1) Definitions; and
(2) Ionization of water and pH.

h. Explaining redox reactions, to include:

(1) Definitions; and
(2) Reactions.

i. Explaining organic chemistry.

4. Time Required:

a. EO - 1685 minutes; and

b. CEC - 50 minutes. An additional 50 minutes will be required for CEC review.

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids:

a. Textbook.

7. Test Details:

a. EO 007.02 will be assessed in CEC 007.02. An additional 50 minutes shall be allotted prior to the exam for review.

8. References: Nil.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Defining nature of matter, to include:						
1.1 Define chemistry by: a. Branches of Chemistry.	15			IL		
1.2 Defining matter and the Kinetic Molecular theory;	15					
1.3 The three states of matter, to include:	25					

	a. Three common states of matter; b. Physical states of matter; and c. 6 changes in states of matter.						
1.4	Defining Heterogeneous or homogenous by: a. Elements; b. Compounds; and c. Solution.	75					
1.5	Describing physical and chemical properties: a. Quantitative; and b. Qualitative.	45					
1.6	Describing the Laws of conservation with: a. Mass; and b. Energy	15					
TP2 Explaining models of atomic structure, to include:							
2.1	Models of atomic structure by: a. Atomic number; b. Mass number; c. Neutrons; d. Electrons; e. Isotopes; and f. Average atomic Mass.	110			IL		
2.2	Models of atomic structure – the Bohr model by: a. Valence electrons; and b. Electron dot.	65					
TP3 Explaining the periodic table to include:							
3.1	Explaining the periodic table to include: a. Alkali metals; b. Alkali earth metals; c. Halogens, inert gases; d. Transition metals; e. Main group or representative elements; f. Lanthanides; and g. Actinides.	55			IL		
3.2	Explaining the trends in the metallic character across a period and down a group;	15					
3.3	Defining Ionization Energy by: a. Cation Formation (positive ions) and Loss of Electrons; and b. Loss of electrons.	55					
3.4	Explaining the trends in the Ionization by: a. Anion Formation (negative ions) and the Gain of Electrons.	15					
TP4 Explaining bonding, to include:							
4.1	Explaining Ionic Bonding by: a. Explaining the Octet Rule, by: (1) Using the position of the elements on the periodic table; and (2) Using difference in electronegativity. b. Defining ionic bond; c. Describing the formation of cations anions; d. Writing the formula from the name; and e. Naming Ionic compounds	30			IL		

4.2	Explaining Covalent Bonding by; a. Defining formation of covalent bonds; b. Writing the formula from the name; c. Writing the name from the formula; d. Describing polar covalent bonds; and e. Summary of bonding types.	165					
TP5 Explaining chemical reactions, to include:							
5.1	Describing chemical equations by: a. Defining chemical equations; b. Balancing chemical equations; and c. Defining types of chemical reaction.	140			IL		
5.2	Explaining the unit factor and the Mole by: a. Defining the unit factor of problem solving; b. Explaining the Mole; c. Explaining Molar mass; and d. Explaining composition as Mass %.	135					
5.3	Explaining quantitative relationships, by: a. Explaining stoichiometry; b. Using Mole relationships to solve problems involving moles; and c. Using Mole relationships to solve problems involving mass.	150					
TP6 Describing Solution chemistry-qualitative by:							
6.1	Describing solutions by: a. Defining solution; b. Defining solute; c. Defining solvent; d. Defining solubility, e. Unsaturated Solution; f. Saturated solution; and g. Supersaturated.	25			IL		
6.2	Explaining Factors Affecting Solubility by: a. Temperature Dependence of Solubility and Saturation; b. Polarity or "Like dissolves Like"; c. Vocabulary; and d. Pressure.	100					
6.3	Rate of Dissolution by: a. Stirred; and b. Crushed.	5					
6.4	Describing Solutions Quantitatively by: a. Mass/Mass percent Concentration; b. From Solubility Graph; and c. Molarity or Molar.	90					
6.5	Describe dilution;	10					
6.6	Describing reactions in solutions by: a. Ionic Solutes Dissociate in Water; and b. Hydration of Ions.	35		15	IL P		
6.7	Explaining ionic equations by: a. Defining ionic equations; and b. Writing net ionic equations using solubility rules.	85			IL		
TP7 Defining Acids and Bases by:							
7.1	Explaining Acids and Bases: a. Operational Definition: (1) An acid; and (2) A base.	40			IL		
7.2	Arrhenius Definition (1884);	20					

7.3 Neutralization Reactions;	25					
7.4 Auto ionization of water and pH by: a. Defining pH; and b. Measurement of Ph.	15					
7.5 Calculations involving Ph;	20					
7.6 Buffer Solutions;	60					
TP8 Redox Reactions by:						
8.1 Defining redox reactions: a. Oxidation; and b. Reduction.	15			IL		

Critical Enabling Check – EO 007.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom.

Arrangements and Procedures

2. CEC 007.02 will be a written assessment based on content from EO 007.02.

3. Pass mark will be 60%.

4. Time required: 50 minutes.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 007.02 will be a written assessment based on content from EO 007.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. Pass mark is 60%.

10. You will have 50 minutes to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Nil.

PERFORMANCE CHECK- PO 007

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Trainees will be provided feedback throughout the PO. No formal PC is included in PO 007.

PO 008 MAINTAIN DOMESTIC SERVICE SYSTEMS

EO 008.01 Maintain Reverse Osmosis Desalination

1. Performance: Maintain Reverse Osmosis Desalination (ROD)

2. Conditions:

   a. Given:

      (1) Applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard: The Marine Technician Machinery Maintainer shall maintain reverse osmosis desalination under supervision IAW refs, by:

   a. Preserving reverse osmosis desalination membranes (AOT1010);

   b. Inspecting reverse osmosis desalination unit;

   c. Repairing reverse osmosis desalination unit, to include:

      (1) Assembling and disassembling reverse osmosis desalination unit (AOT0190);
      (2) Cleaning reverse osmosis desalination membranes (AOT0310); and
      (3) Troubleshooting fluid piping system (ATT1410).

   d. Testing reverse osmosis desalination unit, to include:

      (1) Testing fluid piping system for leaks; and
      (2) Testing potable water quality (JT0480).

4. Time Required:

   a. EO - 565 minutes.

5. Instructional Methods:

   a. Interactive Lecture;

b. Guided discussion; and

c. Demonstration/Performance.

6. Training Aids:

a. Require ROD unit and portable salinity meter; and

b. Technical publication.

7. Test Details:

8. References: A27 and A28.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Preserving reverse osmosis desalination membranes, by:						
1.1 Mark 5 ROD Unit a. Self-Cleaning strainer; b. UF Modules; c. Backwash and Booster pumps; d. Control panel; and e. Human Machine Interface. 1.2 Cleaning ROD membranes to include: a. When/why to clean; b. CIP tank; c. Products used; and d. Cleaning method. 1.3 Flushing ROD membranes to include: a. When/why to flush; and b. Product used. 1.4 Preservation of ROD membranes to include: a. PPE required; b. When/why preserve; c. Chemicals used; d. Method to preserve; and e. SEMS.	15	50	200	IL DP		Trainees will work in pairs to clean ROD. Instructor to demonstrate procedure for preserving RODs.
TP2 Inspect Reverse Osmosis Desalination, by:						
2.1 Investigating ROD to include: a. Visual inspection; b. Reading data; and c. Analyzing data.		25	25	GD	A27, A28	Trainees to visually inspect ROD unit and analyze data.
TP3 Repair Reverse Osmosis Desalination, by:						
3.1 Assembling and disassembling ROD unit to include: a. Consulting technical publication b. Identifying tools required; and c. Isolating component or sub-system.		25	50	DP		Instructor to point out common faults and repairs with the use of

						the technical pub.
3.2 Troubleshooting fluid piping systems to include: a. Detecting low running pressure; and b. High salinity.			50	P		Trainees to troubleshoot a fluid piping system.
TP4 Test Reverse Osmosis Desalination, by:						
4.1 Testing fluid piping system for leaks to include: a. De-isolating system; and b. Verifying system operation.			50	P		Trainees to flash up and verify operating parameters.
4.2 Testing potable water quality		25	50	DP		Trainees will perform water quality test.

Enabling Check – EO 008.01

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Trainees will be provided feedback throughout the EO. No formal EC or PC is included in EO 008.01.

EO 008.02 Apply General Thermodynamics

1. Performance: Apply general thermodynamics.

2. Conditions:

   a. Given:

      (1) References;
      (2) Textbooks;
      (3) Calculators;
      (4) Supervision; and
      (5) Writing materials.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard: The Marine Technician Machinery Maintainer shall independently perform general thermodynamics IAW refs to include:

   a. Introduction to Thermodynamics;

   b. Temperature;

   c. Heat;

   d. Phase relationships;

   e. Heat transfer;

   f. Gas laws;

   g. First Law of Thermodynamics;

   h. Second Law of Thermodynamics;

   i. Introduction to Thermodynamic Cycles, to include:

      (1) Carnot Cycle;
      (2) Otto Cycle
      (3) Diesel Cycle;
      (4) Brayton Cycle; and
      (5) Rankine Cycle.

   j. Air Compressor Cycle;

   k. Liquid Vapour Tables; and

   l. Refrigeration Cycle.

4. Time Required:

   a. EO – 2955 minutes; and

   b. CEC – 100 minutes. An additional 100 minutes shall be allotted for CEC review.

5. Instructional Methods:

   a. Interactive lecture.

6. Training Aids: Nil.

7. Test Details:

   a. EO 008.02 will be assessed in CEC 008.02. 100 minutes will be allotted prior to the CEC for review.

8. References: Nil.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Delivery Approach	Refs	Comments Supporting the Instructional Strategy
		T	D	P			
TP1 Introduction to thermodynamics, to include:							
1.1	Energy types to include: a. Introduction to mechanical, thermal, chemical, and electrical energy types; b. Mechanical energy; and c. Thermal energy.	50			IL		
TP2 Introduction to Temperature, to include:							
2.1	Temperature to include: a. Temperature measurement; b. Thermometers; c. Temperature scales and conversions; and d. Degrees Kelvin.	150			IL		
TP3 Introduction to heat, to include:							
3.1	The measurement of heat to include: a. BTU; and b. Joules.	10			IL		
3.2	Specific heat;	20					
3.3	Thermal mixtures to include: a. Sensible heat; and b. Latent heat.	120					
TP4 Explain effect of pressure and temperature on phase relationships by:							
4.1	Effect of pressure on solid/liquid phase;	20			IL		
4.2	Effect of pressure on liquid/gas phase;	25					

4.3	Interpret phase diagrams, to include; a. Triple point; b. Critical point, pressure, and temperature; c. Recognize substances which contract or expand upon freezing; d. Solid, liquid, and gaseous regions; and e. Phase change lines.	50					
TP5 Explain principles of heat transfer by:							
5.1	Introduction to types of heat flow to include: a. Conduction; b. Convection; and c. Radiation.	30			IL		
5.2	Explain thermal conductance of materials;	20					
5.3	Explain conduction;	30					
5.4	Convection to include: a. Natural convection; and b. Forced convection.	30					
5.5	Explain radiation;	20					
5.6	Heat exchanger a. Heat flow through a dividing wall; b. Flow characteristics; c. Fluid velocity and relative direction of flow; and d. Types of heat exchangers.	130					
TP6 Gas laws, to include:							
6.1	Pressure measurement and scales;	50			IL		
6.2	Boyle's law;	25					
6.3	Charles's law;	25					
6.4	Gay Lussac's Law;	25					
6.5	Combined Gas Laws;	25					
6.6	Ideal Gas Laws; and	25					
6.7	Dalton's Law of Partial Pressures.	25					
TP7 First Law of Thermodynamics, to include:							
7.1	Explain properties, states and processes;	50			IL		
7.2	Explain First Law of Thermodynamics;	50					
7.3	Apply 1st Law of thermodynamics to non-flow processes;	200					
7.4	Apply 1st Law of thermodynamics to steady flow processes;	250					
TP8 Second Law of thermodynamics, to include:							
8.1	Explain limitations of 1st law of thermodynamics a. Explain definitions of the 2nd law; b. Explain thermal efficiency; c. Explain Reversibility; d. Explain what contributes to irreversibility; e. Explain Entropy; and f. Explain use of Temperature-Entropy (T-s) diagram.	150			IL		
TP9 Introduction to Thermodynamic Cycles, to include:							
9.1	Explain thermodynamic cycles;	300			IL		
9.2	Reasons for studying thermodynamic cycles;						
9.3	Describe internal vs external combustion; and						
9.4	Analyse the following thermodynamic cycles: a. Carnot Cycle;						

	b. Otto cycle; c. Diesel cycle; and d. Brayton cycle.						
TP10 Introduction to Air compressor Cycles, to include:							
10.1	Explain air to compressor cycle to include: a. Cycle P-v diagram; b. Clearance volume; c. Effect of clearance on volumetric efficiency; d. Determine volumetric efficiency; e. Explain compression characteristics; and f. Explain multi-stage compression.	200			IL		
TP11 Introduction to Refrigeration Cycles, to include:							
11.1	Explain refrigeration cycle to include: a. Ideal refrigeration cycle; b. Refrigeration cycle process; and c. Refrigeration cycle P-h and T-s diagrams.	250			IL		
11.2	Determine the refrigeration rating (in tons) of an ideal refrigeration cycle; and						
11.3	Determine the coefficient of performance (COP) of an ideal refrigeration cycle.						
TP12 Analyze the Otto Cycle by:							
12.1	Explain air standard assumptions;	150			IL		
12.2	Explain cycle processes;						
12.3	Explain cycle P-v and T-s diagrams;						
12.4	Explain thermal efficiency of Otto cycle;						
12.5	Explain Indicated Mean Effective Pressure; and						
12.6	Analyze Otto cycle given initial conditions.						
TP13 Analyze the Diesel Cycle by:							
13.1	Explain cycle processes;	150			IL		
13.2	Explain cycle P-v and T-s diagrams;						
13.3	Explain thermal efficiency of Diesel cycle;						
13.4	Explain cut-off ratio; and						
13.5	Analyze Diesel cycle given initial conditions.						
TP14 Analyze the Brayton Cycle by:							
14.1	Describe gas turbine block diagram, to include: a. Compressor; b. Combustion chamber; c. Turbine; and d. Heat exchanger (to return to initial ideal conditions).	150			IL		
14.2	Explain cycle processes;						
14.3	Explain cycle P-v and T-s diagrams;						
14.4	Explain thermal efficiency of Brayton cycle; and						
14.5	Analyze Brayton cycle given initial conditions.						
TP15 Explain Liquid-Vapour tables by:							
15.1	Interpret Liquid-Vapour tables, to include: a. Explain liquid-vapour T-s diagram; b. Explain lines on diagram; c. Saturated liquid line; d. Saturated vapour line; and e. Lines of constant pressure.	150			IL		
15.2	Explain T-s diagram regions: a. Compressed liquid region;						

b. Saturate liquid-vapour region; c. Superheated steam region; and d. Correlate T-s diagram to liquid-vapour tables.						

Critical Enabling Check – EO 008.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom.

Arrangements and Procedures

2. CEC 008.02 will be a written assessment based on content from EO 008.02

3. Pass mark will be 60%.

4. Time required: 100 minutes.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 008.02 will be a written assessment based on content from EO 008.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. Pass mark is 60%.

10. You will have 100 minutes to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Nil.

EO 008.03 Maintain Refrigeration Components

1. Performance: Maintain refrigeration components

2. Conditions:

   a. Given:

      (1) Applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment;
      (4) Applicable tools;
      (5) Lock-out/tag-out equipment;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop.

3. Standard: The Marine Technician Machinery Maintainer shall maintain refrigeration components under supervision, IAW refs by:

   a. Inspecting refrigeration components, to include:

      (1) Inspecting chiller units;
      (2) Inspecting double seated refrigeration service valve; and
      (3) Inspecting pilot operated thermostatic control valve.

   b. Repairing refrigeration components, to include:

      (1) Adjusting chiller unit (AOT0040); and
      (2) Repairing double seated refrigeration service valve (AWT0120).

   c. Testing refrigeration components, to include:

      (1) Conducting pressure test (AT0060);
      (2) Conducting vacuum test (AT0160);
      (3) Testing pilot operated thermostatic control valve (AT0280); and
      (4) Testing mechanical thermostatic control valve (AT0270).

4. Time Required:

   a. EO - 545 minutes.

5. Instructional Methods:

   a. Interactive lecture; and

b. Demonstration/Performance.

6. Training Aids:

a. Vacuum pump;

b. Recovery unit; and

c. ODS documentation.

7. Test Details:

a. There is no formal EC for EO 008.03. Trainees will be given feedback throughout the EO.

8. References: A2, A80 and A81.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspect Refrigeration components, by:						
1.1 Inspecting chiller units to include: a. Visual inspection; b. Verifying system configuration; c. Checking logs and operating data.	25	50	50	IL DP	A2 A80 A81	After brief refresher on chiller units, instructor to demo inspecting refrigeration components.
1.2 Inspecting double seated refrigeration service valve to include: a. Visually inspecting for correct operation.		15	25	DP		
1.3 Inspecting pilot operated thermostatic control valve to include: a. Visually inspecting for correct operation.			25	P		
TP2 Repair Refrigeration components, by:						
2.1 Recovering refrigerant to include: a. Introduction to recovery unit; b. Vapor/Liquid Recovery; and c. Liquid Push-Pull Recovery		50		D		Inspector to demo use of recovery unit
2.2 Adjusting chiller unit to include: a. Troubleshooting; b. Confirming operating parameters; and c. Checking for faulty operation.			50	P		Trainees to troubleshoot chiller unit with assistance from the instructor.
2.3 Repairing double seated refrigeration service valve to include: a. Troubleshooting; b. Installing new component IAW technical publications; and c. Adjusting/calibrating new component.			50	P		

TP3 Test Refrigeration components, by:							
3.1	Charging system to include: a. Safety and environmental regulations; and b. Completing equipment/ODS documentation.	15	25				Instructor to discuss regulations and demonstrate how to document maintenance
3.2	Testing pilot operated thermostatic control valve to include: a. Ensuring it is set to maintain 5°C superheat.		15	50			
3.3	Testing mechanical thermostatic control valve						
3.4	Conducting vacuum test to include: b. Using vacuum pump; c. Vacuum pump connections; and d. Analyzing vacuum		50	50			Instructor to demo use of vacuum pump.
3.5	Conducting pressure test to include: a. Disconnecting elements rated at a lower pressure than test; b. Using recommended 150psi for testing; and c. Test procedures.						Instructor to demo pressure test.

Enabling Check – EO 008.03

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Trainees will be provided feedback throughout the EO. No formal EC or PC is included in EO 008.03.

EO 008.04 Maintain Heating, Ventilation and Air Conditioning

1. Performance: Maintain heating, ventilation and air conditioning (HVAC)

2. Conditions:

   a. Given:

      (1) Access to applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment;
      (4) Applicable tools;
      (5) Lock-out/tag-out equipment;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom, HMC Ship and Workshop.

3. Standard: The Marine Technician Machinery Maintainer shall maintain heating, ventilation, and air conditioning systems under supervision IAW refs, by:

   a. Inspecting hydronic and HVAC/CBRN system components, to include:

      (1) Inspecting chilled water and hydronic system components (AOT0770) to include:

         i. Pneumatic Valve assemblies;
         ii. Balancing valves;
         iii. Expansion tanks;
         iv. Bridging (jumper) locations;
         v. Main isolation valves; and
         vi. Pressure relief valves (if fitted) (maid-o-mist).

      (2) Inspecting HVAC and CBRN air distribution system (AOT0780) to include:

         i. Different types of air filters;
         ii. HVAC control cabinets (thermostats);
         iii. Location of Magnehilic gauges;
         iv. Pneumatic valve assemblies;
         v. A/C Plants;
         vi. CBRN Compartments;
         vii. Smoke EVAC/ Smoke control system; and
         viii. Natural ventilation.

      (3) Adjusting chilled water and hydronic systems components (AOT0030) to include:

i. Obtaining chilled water specific gravity readings;
ii. Replenish glycol and corrosion inhibitor in the hydronic system;
iii. Top up the hydronic system via the fresh water filling; and
iv. Inspect and replace chilled water filters (socks).

b. Repair or Replace HVAC system components to include:

(1) pressure relief valves;
(2) pneumatic valve actuators;
(3) valve assembly controllers;
(4) HVAC terminal units and components; and
(5) Repair HVAC air flow switch (Magnehilic gauges) (AKT1820).

c. Testing heating, ventilation, and air conditioning systems, to include:

(1) Performing calibration of HVAC air flow switch (Magnehilic gauge)

4. Time Required:

a. EO – 625 minutes; and

b. EC – 50 minutes.

5. Instructional Methods:

a. Interactive lecture;

b. Demonstration – Performance; and

c. Field trip.

6. Training Aids:

a. HMC Ship with a functional HVAC and Chilled water system; and

b. Multimedia projector.

7. Test Details:

a. EO 008.04 will be assessed in EC 008.04.

8. References: A62, A63 and A102.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspect hydronic and HVAC/CBRN system components, by:						
1.1 System configuration: a. Zones; b. Visually inspect Balancing Values; c. Filter units; d. Expansion tanks; e. Air bleed off locations; f. Bridging (jumper) locations; and g. Main isolation valves.	50		200	IL FT	A62 A63	
1.2 Investigating HVAC and CBRN air distribution system to include: a. Different types of air filters; b. HVAC control cabinets (thermostats); c. A/C Plants; d. CBRN Compartments; e. Smoke EVAC/ Smoke control system; and f. Natural ventilation.	50			IL FT	A62 A63 A102	
1.3 Adjust chilled water and hydronic systems components to include: a. Obtaining chilled water specific gravity readings; b. Replenish glycol and corrosion inhibitor in the hydronic system; c. Replenish the hydronic system via the fresh water filling; and d. Inspect and replace chilled water filters (socks).		25	50	DP FT	A62 A63	
TP2 Repair or Replace HVAC system components, by:						
2.1 Repair or replace HVAC systems and components to include: a. pressure relief valves; b. pneumatic valve actuators; c. valve assembly controllers; d. HVAC terminal units and components; and e. Repair HVAC air flow switch (Magnehilic gauges)	50	50	100	IL DP	A62 A63 A102	
TP3 Testing heating, ventilation, and air conditioning systems						
3.1 Conduct testing on the magnehilic valve.			50		A102	

Enabling Check – EO 008.04

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Trainer / Classroom / Ship.

Arrangements and Procedures

2. EC 008.04 will be a written assessment based on content from EO 008.04.

3. Pass Mark 60%.

4. Time required: 50 minutes.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 008.04 will be a written assessment based on content from EO 008.04.

2. During this written assessment, you will be required to demonstrate your comprehension of the material taught in EO 008.04. You must pass this test to pass EO 008.04. To pass the test, achieve a 60% mark or higher.

3. References will not be provided.

4. Pass Mark 60%.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Nil.

PERFORMANCE CHECK- PO 008

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Trainees will be provided feedback throughout the PO. No formal PC is included in PO 008.

PO 009 PERFORM SMALL BOAT MAINTENANCE

EO 009.01 Maintain Small Boat Gasoline Engines

1. Performance: Maintain small boat gasoline engines

2. Conditions:

   a. Given:

      (1) Applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop.

3. Standard: The Marine Technician Machinery Maintainer shall independently maintain small boat gasoline engines, IAW refs, by:

   a. Inspecting gasoline engines, to include:

      (1) Investigating operation of small gasoline engine (AZT0120);
      (2) Inspect spark plug gaps (AZT0100);
      (3) Investigating small boat outboard engine; and
      (4) Investigating small boat fuel pump.

   b. Repairing gasoline engines, to include:

      (1) Troubleshooting small gasoline engine (AZT0390); and
      (2) Repairing small gasoline engine (AZT0290).

   c. Testing gasoline engines, to include:

      (1) Operating small gasoline engine (VT0020);
      (2) Operating small boat fuel pump; and
      (3) Operating small boat outboard motor.

4. Time Required:

   a. EO - 340 minutes; and

   b. EC – 50 minutes (2 trainees per engine X 6 engines).

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids:

a. Small gasoline engine (lawn mower/wajax) 2 trainees per engine; and

b. Outboard engine 2 trainees per engine.

7. Test Details:

a. EO 009.01 will be assessed in EC 009.01.

8. References: A83 and A82.

9. Remarks: This EO can be taught with any small gas engine, it is the basics that need to be taught, not a specific engine. The outboard engine part of this EO can be taught using old outboards taken off the ships. The timings for the demo are based the demo and practical being done and that same time.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspect gasoline engines, by:						
1.1 Investigating operation of small gasoline engine, by: a. Describing 2 stroke engine principles, to include: (1) Direct scavenging; (2) Intake and compression stroke; and (3) Power and exhaust stroke. b. Describing 4 strokes engine principle; c. Describing ignition systems, to include: (1) Magneto ignition system; (2) Spark plug; and (3) Spark plug gap setting. d. Describing carburetors, to include: (1) Purpose; (2) Operation; (3) Components; and (4) Circuits. e. Describing the cooling systems, to include: (1) Liquid cooled; and	20			IL	A82 A83	

(2) Air cooled. f. Describing engine lubrication, to include: (1) 4 Stroke engine lubrication; and (2) 2 Stroke engine lubrication.						
1.2 Investigating small boat outboard engine.	20			IL		Instructor to teach what the faults are over and above the small gas engines, eg: the leg part of outboard.
1.3 Investigating small boat fuel pump.		25		DP		Instructor to show generic fuel pump that is found on a small engine.
TP2 Repair gasoline engines, by:						
2.1 Troubleshooting small gasoline engine.		25		DP		Instructor to demo what to listen and look for on the small engines.
2.2 Repairing small gasoline engine.		40		DP		Instructor to demo how to disassemble and assemble a small engine off of its frame for repair by replacement.
2.3 Repairing small boat outboard engine.		100		DP		Instructor to demo small repairs on outboard engines.
2.4 Adjusting small boat fuel pump.		30		DP		Instructor to demo how to adjust a fuel pump for a small boat engine.
2.5 Repairing small boat engine fuel pump.		30		DP		If the small boat fuel pump cannot be adjusted, the instructor shall show how to repair by replacement.
TP3 Test gasoline engines, by:						
3.1 Operating small gasoline engine.		15		DP		After repairs are done, instructor to demo running up of engine to check on repairs complete.
3.2 Operating small boat fuel pump.		10		DP		This TP can be done at the same time as the operating the small gasoline engine.
3.3 Operating small boat outboard motor.		25		DP		Instructor to demo flashing up of small boat outboard motor to validate repairs done.

Enabling Check – EO 009.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Small Engine Workshop, six small gas engines, six outboard engines, and six invigilators.

Arrangements and Procedures

2. EC 009.01 will be a practical assessment based on content from EO 009.01.

3. Pass/Fail.

4. Time required: 50 minutes (2 trainees per engine X 6 engines X 6 invigilators).

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 009.01 will be a practical assessment based on content from EO 009.01.

2. During this practical assessment, you will be required to maintain gasoline engines. You must pass this test to pass EO 009.01. To pass the test, you must maintain gasoline engines in accordance with the standard.

3. References will be provided.

4. Pass/Fail.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See instructions to the evaluator located in the table below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
1. Investigating operation of small gasoline engine or outboard.			
2. Repair small gasoline engine or outboard to include;			
3. Faulty fuel pump			
4. Fouled spark plug			
5. Fouled impeller			
6. Confirm repairs by operating small gas engine or outboard			

Instructions to Evaluator:

1. The instructor to have the trainee flash up the small gas engine or outboard and look and listen for any faults. The instructor is to tell the trainee what the fault is and have them fix it.
2. The fault can be a bad fuel pump, impeller, spark plug; as long as it was taught on the EO.
3. Each fault is to be repaired by the trainee.
4. The only grade awarded for this EC is a PASS or FAIL.

EO 009.02 Perform First Line Maintenance on Small Boat Diesel Engines

1. Performance: Perform first line maintenance on small boat diesel engines

2. Conditions:

   a. Given:

      (1) Applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop.

3. Standard: The Marine Technician Machinery Maintainer shall independently perform first line maintenance on small boat diesel engines, IAW refs, by:

   a. Describing small boat diesel engine theory, to include:

      (1) Operating principles of a four stroke diesel engine;
      (2) Technical data derived from the engine model numbers;
      (3) Deriving technical engine data from the maintenance manual;
      (4) Function of major engine components; and
      (5) Engine alarm lights and indicators.

   b. Performing maintenance routines on small boat diesel engines, to include:

      (1) Changing engine oil and filter;
      (2) Changing engine air filter;
      (3) Changing fuel/water separator elements;
      (4) Changing fuel filter;
      (5) Repairing small boat engine fuel pump (AZT0200);
      (6) Repairing small boat outboard engine (AZT0240);
      (7) Venting the fuel system following maintenance routines;
      (8) Adjusting valve clearances;
      (9) Inspecting and replace belts; and
      (10) Inspecting/replacing cooling systems.

4. Time Required:

   a. EO – 200 minutes; and

   b. EC:

(1) Part A – 50 minutes; and
(2) Part B – 100 minutes per trainee.

5. Instructional Methods:

a. Demonstration – performance; and

b. Interactive Lecture.

6. Training Aids:

a. Diesel engine static display with mounts.

7. Test Details:

a. EO 009.02 will be assessed in EC 009.02.

8. References: A35 C-23-343-000/MS-001 Rigid Hull Inflatable Boat Workshop Maintenance Manual.

9. Remarks: This EO can be taught with any small boat platform, it is based on the RHIB currently used by the RCN and it is the basics that need to be taught, not a specific small boat.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Describing small boat diesel engine theory, to include:						
1.1 Operating principles of a four stroke diesel engine;	75			IL	A35	
1.2 Technical data derived from the engine model numbers;						
1.3 Deriving technical engine data from the maintenance manual;						
1.4 Function of major engine components, to include: a. Cylinder Block; b. Cylinder Head; c. Pistons; and d. Cylinder Liners.						
1.5 Engine alarm lights and indicators, to include: a. Rev counter; b. Operating Hours; c. Temperature Gauge; d. Oil Pressure Gauge; e. Voltmeter; f. General Audible Alarm; g. Visual Warning Display; h. Test/Acknowledge; and i. Troubleshooting.						

TP2 Perform maintenance routines on small boat diesel engines, to include:						
2.1	Complete the following maintenance routines: a. Change engine oil and filter; b. Change engine air filter; c. Change fuel/water separator elements; d. Change fuel filter; e. Vent the fuel system following maintenance routines; f. Adjust valve clearances; and g. Inspect and replace belts.		50	75	DP		
2.2	Inspect/replace cooling systems, to include: a. Raw water pump impeller; b. Thermostats; and c. Coolant reservoir						

Enabling Check – EO 009.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Part A - Classroom.

2. Part B -

   a. Workshop;
   b. Diesel test cell;
   c. Six operational small boat diesel engines;
   d. Lubricants and spare parts;
   e. PPE;
   f. Testing equipment and applicable tools;
   g. Lock-out/tag-out equipment; and
   h. References.

Arrangements and Procedures

Part A – Written Assessment

1. EC 009.02 Part A will be a written assessment based on content from EO 009.02.

2. Pass mark will be 60%.

3. Time required: 50 minutes.

4. References will not be required.

Part B – Practical Assessment

1. EC 009.02 Part B will be a practical assessment based on content from EO 009.02.

2. Pass/Fail.

3. Time required: 100 minutes.

4. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT – PART A

1. EC 009.02 Part A will be a written assessment based on content from EO 009.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment

necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. Pass mark is 60%.

10. You will have 50 minutes to complete this assessment.

11. Do you have any questions before you start?

PRACTICAL ASSESSMENT – PART B

1. EC 009.02 Part B will be a practical assessment based on content from EO 009.02.

2. During this practical assessment, you will be required to maintain small boat diesel engines. You must pass this test to pass EO 010.02. To pass the test, you must maintain small boat diesel engines in accordance with the standard.

3. References will be provided.

4. Pass/Fail.

5. You will have 100 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. The instructor will give each trainee a list of line maintenance actions to complete.

2. During the oil change, after satisfactory demonstration of the task at hand, the instructor will tell the trainee when sufficient oil has been drained. The oil removed will be re-introduced.

3. All filters will be re-installed unless due for replacement.

4. Items #1, 2, 3 and 4 from the checklist will be done in the workshop, items #5 to 8 in the test cell.

5. All reference material will be readily available.

6. The instructor may be asked to clarify requirements of maintenance task but will not lend assistance.

7. Trainees will be assessed individually with the instructor checklist detailed below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
1. **Change engine oil and filter**: a. Find appropriate references; b. Suck oil out through appropriate tube; c. Unscrew and remove oil filter using filter wrench; d. Coat a new filter rubber seal with oil before installing; e. Check engine contact surface and thread on the filter by hand until contact is made with the surface; f. Turn filter one half turn after contact made; g. Replace engine oil; h. Start engine, run at idle, and immediately check for correct oil pressure; and i. Check oil level and for leaks around filter			
2. Change engine air filter: a. Find appropriate references; b. Remove crankcase ventilation filter, if c. fitted; d. Remove air filter housing cover; e. Remove oil air filer; f. Install new air filter; g. Reinstall housing cover; and h. Reinstall crankcase ventilation filter, if fitted.			
3. Change fuel filter: a. Find appropriate references; b. Unscrew fuel filter (taking care not to spill fuel); c. Dispose of fuel filter in designated container; d. Inspect new filter for faults; e. Coat filter seal with small amount of oil; f. Check engine contact surface and thread on the filter by hand until contact made with the surface; and g. Turn filter one half turn after contact is made.			
4. Adjust valves: a. Find appropriate references; b. Turn battery switch off; c. Remove kill switch lanyard; d. Remove rocker gear covers;			

	e. Turn engine over by turning crankshaft pulley clockwise looking aft; f. Close intake valve for #1 cylinder; g. Stop turning crank pulley when intake valve for next cylinder in firing order sequence is fully closed; h. Set cylinder #1 valves; i. Check out-of-contact clearances using feeler gauges; j. Adjust clearances by loosening locking nut and turning tappet screw; k. Tighten locking nut; l. Re-check clearance; and m. Replace rocker covers at 72 in. lbs torque			
CARE TAKEN TO ENSURE NO FOREIGN MATERAIL ENTERS ENGINE				
5.	**Inspect belts:** a. Find appropriate references; b. Check belt play between F/W pump and alternator (3/8”); c. Check for belt wear; d. Loosen three bolts; e. Tighten belts; and f. Tighten three bolts.			
6.	**Check raw water pump impeller:** a. Find appropriate references; b. Remove pump cover plate; c. Remove impeller; d. Inspect impeller and replace if necessary; e. Reinstall impeller; f. Replace cover plate; and g. Replace gasket (if damaged)			
7.	**Change fuel/water separator element:** a. Find appropriate references; b. Isolate fuel valves; c. Check bowl for contamination; d. Place container below filter; e. Have absorbent pads ready to clean up spillage; f. Drain fuel from separator bowl using drain plug; g. Undo T-handle on the top of filter and remove cover; h. Remove bowl from casting and gaskets; i. Remove old filter element (avoiding spillage of fuel);			

	j. Wipe out case and bowl using lint free rags; k. Insert new filter elements; l. Install new gaskets and o - rings; m. Reinstall bowl to casting; n. Install drain plug (do not overtighten); o. Replace top cover and hand tighten Handle; and p. De-isolate fuel valves.			
8.	**Purge fuel system after maintenance:** a. Find appropriate references; b. Open venting screw on top of fuel filter (4 turns); c. Use absorbent pad around vent screw to catch fuel spillage; d. Pump fuel using hand primer until fuel runs free or air bubbles appear; e. Close vent screw; f. Pump hand primer for 30 seconds to vent the injector pump; g. Loosen all injector delivery pipe nuts; h. Crank engine, with control lever in neutral, full RPM setting unit fuel flows out of delivery pipes (use absorbent pads to catch fuel leakage); i. Tighten delivery pipe nuts; j. Discard absorbent pads in an approved container; and k. Start engine and check for fuel leaks at all vent locations.			

Instructions to Evaluator:

1. The instructor will give each trainee a list of line maintenance actions to complete. (The provided checklist titles above, i.e. Change engine oil and filter.)
2. During the oil change, after satisfactory demonstration of the task at hand, the instructor will tell the trainee when sufficient oil has been drained. The oil removed will be re-introduced.
3. All filters will be re-installed unless due for replacement.
4. Items #1, 2, 3 and 4 from the checklist will be done in the workshop, items #5 to 8 in the test cell.
5. All reference material will be readily available.
6. The instructor may be asked to clarify requirements of maintenance task but will not lend assistance.
7. The only grade awarded for this EC is a PASS or FAIL.

EO 009.03 Troubleshoot Small Boat Diesel Engines

1. Performance: Troubleshoot small boat diesel engines

2. Conditions:

   a. Given:

      (1) Applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environment: Classroom and Workshop/ lab.

3. Standard: The Marine Technician Machinery Maintainer shall independently troubleshoot small boat diesel engines, IAW refs, by:

   a. Performing small boat diesel engine troubleshooting techniques;

   b. Adjusting small boat fuel pump (AZT0020); and

   c. Troubleshooting common small boat diesel engine faults.

4. Time Required:

   a. EO – 250 minutes; and

   b. EC – 20 minutes per trainee.

5. Instructional Methods:

   a. Interactive Lecture; and

   b. Demonstration – performance.

6. Training Aids:

   a. Small boat diesel engine; and

   b. Small boat.

7. Test Details:

   a. EO 009.03 will be assessed in EC 009.03.

8. References: C25.

9. Remarks: The demo timings include demo/practical. All the small diesel engine demo/practical and EC/PC can be done in the diesel workshop and classroom. The small boat demo/practical and EC/PC can be done with a small boat on a trailer or at the boat shed.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Perform small boat diesel engine troubleshooting techniques, by:						
1.1 Troubleshooting the following sub-systems: a. Cooling system; b. Fuel system; c. Starting system; d. Air Induction; e. Emissions system; f. Electrical system; and g. Lube Oil system.	25	50	50	DP		
TP2 Troubleshooting common small boat diesel engine faults, to include:						
2.1 Troubleshooting the following common faults: a. High crank case pressure; b. Fuel blockages; c. Restricted inlet; d. Low lube oil pressure; e. High lube oil pressure; f. High water temperature alarms; g. Faulty start relay; and h. No power.		50	75	DP		

Enabling Check – EO 009.03

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop;
2. Diesel test cell;
3. Six operational small boat diesel engines;
4. Lubricants and spare parts;
5. PPE;
6. Testing equipment and applicable tools;
7. Lock-out/tag-out equipment; and
8. References.

Arrangements and Procedures

1. EC 009.03 will be a practical assessment based on content from EO 009.03.
2. Pass/Fail.
3. Time required: 20 minutes per trainee.
4. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 009.03 will be a practical assessment based on content from EO 009.03.
2. During this practical assessment, you will be required to troubleshoot small boat diesel engines components. You must pass this test to pass EO 009.03. To pass the test, you must troubleshoot small boat diesel engines components in accordance with the standard.
3. References will be provided.

4. Pass/Fail.

5. You will have 20 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See comments in the table below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
1. High Crankcase pressure: a. Find appropriate references; b. Identify fault; c. Check manometer operation; d. Check vent line filter in valve cover; e. Check vent line for blockage; and			
2. Fuel Blockage: a. Find appropriate references; b. Identify fault; c. Loosen fuel filter vent screw to verify fuel flow; d. Pump primary pump to determine blockage location; and e. Locate blockage.			
3. Restricted inlet: a. Find appropriate references; b. Identify fault; and c. Check inlet filter for obstruction.			
4. Low Lube Oil pressure alarm: a. Find appropriate references; b. Identify fault; c. Verify oil pressure, level and system integrity; and d. Test alarm switch.			
5. High Lube Oil pressure alarm: a. Find appropriate references; b. Identify fault; c. Verify oil pressure; and d. Test alarm switch.			
6. High water temperature alarm: a. find appropriate references; b. identify fault; c. verify water temperature; d. verify water level; and e. test alarm switch			
7. Fault start relay: a. find appropriate references; b. identify fault; c. verify power available; d. trip reset; and e. Jumper start relay.			
8. No Power: a. find appropriate references; and b. Identify fault.			
Instructions to Evaluator: **1.** The instructor will give each trainee a list of issues for them to rectify. (The provided checklist titles above, i.e. High crankcase pressure.) **2.** All reference material will be readily available.			

3.	The instructor may be asked to clarify requirements of the troubleshooting tasks but will not lend assistance.
4.	The only grade awarded for this EC is a PASS or FAIL.

EO 009.04 Maintain Small Boat Stern Drive

1. Performance: Maintain small boat stern drive

2. Conditions:

   a. Given:

      (1) References;
      (2) Appropriate equipment;
      (3) Lock out/ Tag out equipment;
      (4) Appropriate tools;
      (5) PPE; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop.

3. Standard: The Marine Technician Machinery Maintainer shall maintain small boat stern drive under supervision IAW refs, by:

   a. Describing the theory of stern drive operation;

   b. Inspecting the small boat stern drive, to include:

      (1) Inspect small boat propeller (AZT0400).

   c. Performing first line maintenance on a small boat stern drive, to include:

      (1) Repairing small boat saltwater impeller (AZT0250);
      (2) Repairing small boat shaftline shear coupling (AZT0260);
      (3) Repairing small boat throttle assembly (AZT0270); and
      (4) Repairing steering system of a small boat (AZT0300).

4. Time Required:

   a. EO – 300 minutes; and

   b. EC:

      (1) Part A – 50 minutes; and
      (2) Part B – 30 minutes per trainee.

5. Instructional Methods:

   a. Interactive lecture; and

   b. Demonstration – performance.

6. Training Aids:

   a. Volvo training engines (3 trainees per engine)

7. Test Details:

   a. EO 009.04 will be assessed in EC 009.04.

8. References: Volvo engine maintenance manual when added to references.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Describing the theory of stern drive operation, by:						
1.1 Identifying general characteristics; and	50			IL DP		
1.2 Identifying impact damage analysis.						
TP2 Inspecting the small boat stern drive, by:						
2.1 Inspecting the following stern drive sub-systems and components: a. Cooling system; b. Lubrications; c. Zinc anodes; d. Transmission, to include: (1) Inspecting the u-joint; (2) Inspecting the upper gear housing/clutch mechanism; and (3) Inspecting the vertical shaft with splined sleeve. e. Propeller, to include: (1) Propeller shaft. f. Power control and linkage.		50	50	DP		
TP3 Performing first line maintenance on a small boat stern drive, to include:						
3.1 Changing stern drive oil and oil filter		50	100	DP		
3.2 Performing a propeller replacement;						
3.3 Performing a trueness test of a propeller shaft; and						
3.4 Performing a splined sleeve replacement.						

Enabling Check – EO 009.04

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Part A: Classroom

2. Part B:

   a. Workshop;
   b. Diesel test cell;
   c. Six operational small boat diesel engines;
   d. Lubricants and spare parts;
   e. PPE;
   f. Testing equipment and applicable tools;
   g. Lock-out/tag-out equipment; and
   h. References.

Arrangements and Procedures

Part A

1. EC 009.04 Part A will be a written assessment based on content from EO 009.04.

2. Pass mark will be 60%.

3. Time required: 50 minutes.

4. References will not be required.

Part B

1. EC 009.04 will be a practical assessment based on content from EO 009.04.

2. Pass/Fail.

3. Time required: 30 minutes per trainee.

4. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT – PART A

1. EC 009.04 Part A will be a written assessment based on content from EO 009.04.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. Pass mark is 60%.

10. You will have 50 minutes to complete this assessment.

11. Do you have any questions before you start?

PRACTICAL ASSESSMENT

1. EC 009.04 will be a practical assessment based on content from EO 009.04.

2. During this practical assessment, you will be required to maintain small boat stern drive. You must pass this test to pass EO 009.04. To pass the test, you must complete the maintain engine sub-system components in accordance with the standard.

3. References will be provided.

4. Pass/Fail.

5. You will have 30 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See the comments in the table below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments (for invigilator, before printing, these comments to be deleted and the space is used for comments on trainees)
1. Change stern drive oil: a. Find appropriate references; b. Remove dipstick; c. Remove oil drain plug at base of drive leg; d. Drain oil into a waste oil container; e. Reinstall drain plug (make sure O-ring is properly seated); f. Remove clutch assembly cover; g. Remove oil fill plug and tilt drive; h. Refill with oil to upper mark on dipstick; i. Check oil level; j. Drain excess oil if level above upper mark on dipstick;			
2. Inspect/remove propeller: a. Find appropriate references b. Lock drive to prevent turning c. Turn propeller manually to check for “slip”. d. Remove propeller cone bolt; e. Insert Allen key through hole in propeller cone and remove cone; f. Remove Teflon washer; g. Remove propeller; h. Remove line cutter; and i. Inspect sacrificial anode and replace if reduced in mass by 50%.			
3. Check trueness of propeller shaft: a. Find appropriate references; b. Set up dial indicator; c. Measure run out at two places (allow-able run out 0.002 inches close to bearing box and 0.010 inches end of shaft); and d. Remove dial indicator.			
4. Re-install propeller a. Grease the propeller shaft with a thin layer of water-resistant grease; b. Re-install line cutter;			

c. Re-install propeller; d. Re-install Teflon washer; e. Re-install and tighten propeller cone; f. Re-install and tighten propeller cone bolt.			

Instructions to Evaluator:

1. The instructor will give each trainee a list of stern drive maintenance actions to complete. (The provided checklist titles above, i.e. Change stern drive oil.)

2. During the oil change, after satisfactory demonstration of the task at hand, the instructor will tell the trainee when sufficient oil has been drained. The oil removed will be re-introduced.

3. All reference material will be readily available.

4. The instructor may be asked to clarify requirements of the stern drive maintenance task but will not lend assistance.

5. The only grade awarded for this EC is a PASS or FAIL.

EO 009.05 Maintain Small Boat Components

1. Performance: Maintain small boat components

2. Conditions:

   a. Given:

      (1) References;
      (2) Appropriate equipment;
      (3) Lock out/ Tag out equipment;
      (4) Appropriate tools;
      (5) PPE; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop.

3. Standard. The Marine Technician Machinery Maintainer shall independently Maintain small boat components IAW refs, by:

   a. Describing the function of small boat sub-systems;

   b. Performing administrative maintenance on small boat components; and

   c. Performing maintenance on small boat components, to include:

      (1) Repairing rigid inflatable boat electrical systems (AZT0190);
      (2) Repairing small boat hull (AZT0210); and
      (3) Testing rigid inflatable boat electrical systems (AZT0370).

4. Time Required:

   a. EO – 415 minutes; and

   b. EC:

      (1) Part A – 50 minutes; and
      (2) Part B – 100 minutes per trainee.

   c. PC – 50 minutes per trainee.

5. Instructional Methods:

   a. Interactive lecture; and

   b. Demonstration – performance.

6. Training Aids:

a. Volvo training engines (3 trainees per engine).

7. Test Details:
   a. EO 009.05 will be assessed in EC 009.05.

8. References: Volvo engine maintenance manual when added to references

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Describing the function of small boat sub-systems, to include:						
1.1 Identifying the following sub-system functions: a. Steering system; b. Electrical system; c. Fuel system; d. Bilge pump system; and e. Stern drive lifting system.	50	25		IL D		
TP2 Performing administrative maintenance on small boat components, to include:						
2.1 Maintaining operation, maintenance routines and logs.	20			IL		
2.2 Performing cold weather precautions.	5	15		D		
TP3 Performing maintenance on small boat components, to include:						
3.1 Conducting the following maintenance routines: a. pre-start checks; b. grease routines and husbandry; c. Inspecting/cleaning electrical contacts; d. Maintaining steering system; and e. Inspecting drive shaft.		100	200	DP		

Enabling Check – EO 009.05

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Part A: Classroom

2. Part B:

   a. Workshop;
   b. Diesel test cell;
   c. Six operational small boat diesel engines;
   d. Lubricants and spare parts;
   e. PPE;
   f. Testing equipment and applicable tools;
   g. Lock-out/tag-out equipment; and
   h. References.

Arrangements and Procedures

Part A

1. EC 009.05 Part A will be a written assessment based on content from EO 009.05.

2. Pass mark will be 60%.

3. Time required: 50 minutes.

4. References will not be required.

Part B

1. EC 009.05 will be a practical assessment based on content from EO 009.05.

2. Pass/Fail.

3. Time required: 100 minutes per trainee.

4. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT – PART A

1. EC 009.05 Part A will be a written assessment based on content from EO 009.05.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. Pass mark is 60%.

10. You will have 50 minutes to complete this assessment.

11. Do you have any questions before you start?

PRACTICAL ASSESSMENT

1. EC 009.05 will be a practical assessment based on content from EO 009.05.

2. During this practical assessment, you will be required to maintain small boat components. You must pass this test to pass EO 009.05. To pass the test, you must maintain small boat components in accordance with the standard.

3. References will be provided.

4. Pass/Fail.

5. You will have 100 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. See the comments in the table below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments (for invigilator, before printing, these comments to be deleted and the space is used for comments on trainees)
1. Perform general grease routines: a. Find appropriate references; b. Grease throttle, stop, air flap and fire extinguisher cables; c. Grease steering ram; d. Grease upper and lower drive bushings (if fitted); e. Grease stowage box hinges; f. Grease console hinges; and g. Grease forward hatch plug.			
2. Inspect and clean electrical connections: a. Find appropriate references; b. Inspect, clean and tighten alternator connections; c. Inspect, clean and tighten starter connections; d. Inspect battery boxes, covers and tie downs; e. Check battery electrolyte level (if applicable); f. Inspect, clean, tighten and lightly grease battery connections; and g. Inspect and clean trim wiring connections and relays.			
3. Inspect shaft components: a. Find appropriate references; b. Inspect transom shield studs, nuts and washers for irregularities; c. Tighten studs to 45 ft. lbs; d. Remove shaft shroud; e. Check condition of flexible coupling bolts; and f. Check for play in drive shaft.			
4. Flush raw water system: a. Find appropriate references; b. Hook up fresh water hose; c. Open appropriate valves; d. Close appropriate valves; e. Start engine; and f. Return engine to normal condition.			
5. Winterize RHIB: a. Find appropriate references; b. Plug in to 110 volts; c. Remove flushing connection cap; d. Open flushing connection valve;			

e. Open drain manifold valve; f. Remove strainer plug; g. Drain bilge; and h. Properly dispose of contaminated water.			

Instructions to Evaluator:

1. The instructor will give each trainee a list of line maintenance actions to complete. (The provided checklist titles above, i.e. Perform general grease routines.)
2. All reference material will be readily available.
3. The instructor may be asked to clarify requirements of maintenance task but will not lend assistance.
4. The only grade awarded for this EC is a PASS or FAIL.

PERFORMANCE CHECK – PO 009

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop;
2. Diesel test cell;
3. Six operational small boat diesel engines;
4. Lubricants and spare parts;
5. PPE;
6. Testing equipment and applicable tools;
7. Lock-out/tag-out equipment; and
8. References.

Arrangements and Procedures

1. PC 009 will be a practical assessment based on content from PO 009.
2. Pass/Fail.
3. Time required: 50 minutes per trainee.
4. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. PC 009 will be a practical assessment based on content from PO 009.
2. During this practical assessment, you will be required to perform small boat maintenance. You must pass this test to pass PO 009. To pass the test, you must perform a pre-launch checklist and maintain a small boat, in accordance with the standard.
3. References will be provided.
4. Pass/Fail.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. There are two enabling checklists the trainee must complete. They are the Pre-Launch checklist and the small boat maintenance enabling check list.

PART IV – PRE-LAUNCH ENABLING CHECK LIST

PRE-LAUNCH CHECKLIST			
Check List	**SAT**	**UNSAT**	**Comments**
1. Bow Area: a. Inflatable collar at operating pressure; b. Contents of forward locker; c. Locker dry; d. Secure forward hatch; and e. Caps on relief fuel.			
2. Fuel level: a. Fuel tank full; and b. Fuel cap secure.			
3. Engine compartment: a. Visual inspection; b. Stripping valve open; c. Stripping pump operation; d. Stripping valve closed; e. Fuel/water separator bowl clear; f. Fuel supply valve open; g. Fuel return valve open; h. Belt tension; i. Coolant level; j. Oil level; k. Sea water flushing and drain valves closed; l. Console/engine cover secured; m. Lifeline on console replaced; n. Air flaps open; and o. Fire extinguisher pin out.			
4. Console/engine cover: a. Shore power disconnected; b. Battery switch to position #1; c. Voltmeter reading greater than 12 volts; d. Battery switch re-set to position #2 e. Voltmeter reading greater than 12 volts; f. Battery switch re-set to position #1; g. Outdrive trim pump (5 to 6 degrees); h. Electric bilge pump: i. Manual; and j. Switch set to automatic. k. Lights; l. Horn; and m. Steering wheel operational.			
5. After stowage locker: a. Sea water strainer clean;			

b. Water intake valve open (outdrive); c. Manual bilge pump; d. Float switch; e. Reserve valve pressure 20 to 30 psi and fluid level; f. Outdrive trim pump fluid level; and g. Secure after stowage locker.			
6. Ensure plug installed in hull			

Instructions to Evaluator:

1. The instructor will give each trainee a list of pre-launch actions to complete. (The provided checklist titles above, i.e. Bow area.)
2. All reference material will be readily available.
3. The instructor may be asked to clarify requirements of a pre-launch task but will not lend assistance.
4. The only grade awarded for this PC is a PASS or FAIL.

PART IV – PERFORM SMALL BOAT MAINTENANCE ENABLING CHECK LIST

PERFORM SMALL BOAT MAINTENANCE CHECKLIST			
Check List	**SAT**	**UNSAT**	**Comments**
1. Change engine oil and filter: a. Find appropriate references; b. Suck oil out through appropriate tube; c. Unscrew and remove oil filter using filter wrench; d. Coat new filter rubber seal with oil before installing; e. Check engine contact surface and thread on filter by hand until contact made with surface; f. Turn filter one half turn after contact made; g. Replace engine oil; h. Start engine, run at idle, and check oil pressure correct; i. Check oil level; and			
2. Change fuel filter: a. Find appropriate references; b. Unscrew fuel filter (no spillage); c. Dispose of filter in designated container; d. Replace fuel filter; e. Inspect new filter seal for faults; f. Coat filter seal with small amount of oil; g. Check engine contact surface and thread on filter by hand until contact made with surface; and h. Turn filter one half turn after contact made.			
3. Purge fuel system after maintenance: a. Find appropriate references; b. Open venting screw on top of fuel filter(4 turns); c. Use absorbent pad around vent screw to catch spillage; d. Pump fuel using hand primer until fuel runs free or air bubbles; e. Close vent screw; f. Pump hand primer for 30 seconds to vent injector pump; g. Loosen all injector delivery pipe nuts;			

	h. Crank engine, with control lever in neutral and full RPM setting, until fuel flows from delivery pipes; i. Use absorbent pads to soak up spillage; j. Tighten delivery pipe nuts; k. Discard absorbent pads in approved container; l. Start engine; and m. Check for fuel leaks at all vent locations.			
4.	**Change stern drive oil:** a. Find appropriate references; b. Remove dipstick and oil drain plug at base of drive leg; c. Drain oil into waste oil container; d. Reinstall drain plug; e. Ensure O-ring properly sealed; f. Remove clutch assembly cover; g. Remove oil fill plug and tilt drive; h. Refill with appropriate oil to upper mark on dipstick; i. Check oil level; j. Drain excess oil if level above upper mark on dipstick; k. Ensure gasket and O-rings are serviceable; l. Re-install fill plug and dipstick; and m. Check for oil leaks at drain plug.			
5.	**Test and remove propeller:** a. Find appropriate references; b. Lock drive to prevent turning; c. Turn propeller manually to check for slip; d. Remove propeller cone belt; e. Insert Allen key through hole in propeller cone and remove; f. Remove Teflon washer; g. Remove propeller; h. Remove line cutter; and i. Inspect sacrificial anode and replace if reduced in mass by 50%			
6.	**Re-install propeller:** a. Grease propeller shaft with thin layer of water- resistant grease;			

	b. Re-install line cutter; c. Re-install propeller; d. Re-install Teflon washer; e. Re-install and tighten propeller cone; and f. Re-install and tighten propeller cone bolt.			
7.	**Inspect transom shield:** a. Find appropriate references; b. Inspect transom shield studs, nuts and washers; and c. Tighten studs to 45ft. lbs			
8.	**Fuel Blockage:** a. Find appropriate references; b. Identify fault; c. Loosen fuel filter vent screw to verify fuel flow; d. Pump primary pump to determine blockage location; and e. Locate blockage.			
9.	**Faulty start relay:** a. Find appropriate references; b. Identify fault; c. Verify power available; d. Trip reset; and e. Jumper start relay.			

Instructions to Evaluator:

1. The instructor will give each trainee a list of small boat maintenance actions to complete. (The provided checklist titles above, i.e. Change engine oil and filter.)
2. During the oil change, after satisfactory demonstration of the task at hand, the instructor will tell the trainee when sufficient oil has been drained. The oil removed will be re-introduced.
3. All filters will be re-installed unless due for replacement.
4. All reference material will be readily available.
5. The instructor may be asked to clarify requirements of the small boat maintenance task but will not lend assistance.
6. The only grade awarded for this PC is a PASS or FAIL.

PO 010 MAINTAIN ANCILLARY SYSTEMS

EO 010.01 Maintain Gas Turbine Water Wash System

1. Performance: Maintain gas turbine water wash systems.

2. Conditions:

   a. Given:

      (1) Applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Workshop and Classroom.

3. Standard: The Marine Technician Machinery Maintainer shall maintain gas turbine water wash systems under supervision, IAW refs, by:

   a. Inspecting gas turbine water wash systems, to include:

      (1) Inspecting gas turbine water wash tank.

   b. Repairing gas turbine water wash systems, to include:

      (1) Troubleshooting gas turbine water wash system (AVT1430);
      (2) Repairing gas turbine water wash tank (AVT1120);
      (3) Installing ancillary systems components (AHT0580); and
      (4) Assembling and disassembling gas turbine water wash system (AVT0390).

   c. Testing gas turbine water wash systems.

4. Time Required:

   a. EO – 300 minutes.

5. Instructional Methods:

   a. Interactive lecture;

   b. Demonstration – performance;

   c. Field trip (only required if there is no functional GT Water Wash system at the TE); and

d. Tutorial.

6. Training Aids:

a. Gas turbine water wash system mock-up.

7. Test Details:

a. EO 010.01 will be assessed during the demo/performance.

8. References: A86.

9. Remarks: NFS may need to purchase a working gas turbine water wash system IOT teach this EO. If there is no working system, a pressure tank with threaded fittings can be used for instruction. The operation part will need to be done onboard the ship.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspect gas turbine water wash systems by:						
1.1 Inspecting gas turbine water wash tank.		15		D	A86 pg 2-106 - 110	This is a thorough visual inspection that is more in depth than just doing rounds. It does not include internal inspection.
TP2 Repair gas turbine water wash systems by:						
2.1 Troubleshooting gas turbine water wash system.		15	30	DP	A6, pg 2-106 - 110	The trainee to troubleshoot the faults that are given to them from the instructor.
2.2 Repairing gas turbine water wash tank.		25	50	DP	A86, pg 2-106 - 110	The common error will be leaking thread on the pipes leading from the tank. The instructor to show how to seal up the threads on the tank
2.3 Installing ancillary systems components.		25	50	DP	A86, pg 2-106 - 110	The instructor to show how to install new valves on the water wash system this can be included in the sealing of threads.
2.4 Assembling and disassembling gas turbine water wash system.		25	50	DP	A86, pg 2-106 - 110	This shall be part of the previous TPs for fixing leaks and installing valves.
TP3 Test gas turbine water wash systems by:						
3.1 Testing the integrity and functionality of the gas turbine water wash system.		15		D	A86, pg 2-106 - 110	After all repairs are done the GT water wash mock-up shall be run up to test for leaks. The instructor will demo running up of the system

Enabling Check – EO 010.01

PART I – ADMINISTRATIVE INSTRUCTIONS

1. As the evaluation is done during the demo/performance of EC 010.02, there will be no EC performed.

EO 010.02 Maintain Main Shaftline

1. Performance: Maintain main shaftline.

2. Conditions:

   a. Given:

      (1) Applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Workshop and Classroom.

3. Standard: The Marine Technician Machinery Maintainer shall maintain main shaftline under supervision, IAW refs, by:

   a. Inspecting main shaftline, to include:

      (1) Inspecting flexible coupling (AUT1150);
      (2) Investigating operation of inflatable seal (XT0020);
      (3) Inspecting stern tube gland; and
      (4) Inspecting steering system components.

   b. Repairing main shaftline, to include:

      (1) Repairing main shaftline bulkhead seal (BT0190);
      (2) Adjusting stern tube gland (BT0010);
      (3) Repairing main shaft line pitch drive belts;
      (4) Adjusting rudder angle indicator (BAT0010); and
      (5) Repairing mechanical rudder locking arrangement (BAT0250).

   c. Testing main shaftline, to include;

      (1) Operating steering system;
      (2) Operating pitch belts; and
      (3) Testing auto steering unit (BAT0300).

4. Time Required:

   a. EO – 480 minutes; and

b. EC – 75 minutes per trainee (this EC can be invigilated at five different stations (bulkhead seal, stern tube gland, steering, flexible coupling, and pitch drive belts). If there are five invigilators, the time will be reduced).

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids:

a. Bulkhead seal static display;

b. Stern seal static display;

c. Steering system static display; and

d. Pitch drive belts static display.

7. Test Details:

a. EO 010.02 will be assessed in EC 010.02.

8. References: A87, A88, A89 and A90.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspecting the Main Shaftline, by:						
1.1 Describing the following Main Shaftline components: a. main shaftline; b. Flexible coupling; c. Inflatable seal; d. Stern tube gland; and e. Steering system components.	50			IL	A87, A88, A89, A90	The lesson shall start off with a power point of flex couplings, inflatable seals, stern tubes, and steering systems.
1.2 Inspecting flexible coupling.		25		DP	A87	The instructor to take the trainees to the flex coupling mock up and show how to inspect the rubber ‘pucks’ and have them use the hardness tester.
1.3 Investigating operation of inflatable seal.		25		DP	A88	The instructor to take the trainees to the inflatable seal mock to have a look at how it

						works and what to look for.
1.4	Inspecting stern tube gland.		25		A88	The instructor to take the trainees to the stern gland tube static display and show what to look for when inspecting it.
1.5	Inspecting steering system components.		35	DP	A90	The instructor to take the trainee to the steering static display and show them what to look for when inspecting it.
TP2 Repair main shaftline by:						
2.1	Repairing main shaftline bulkhead seal.		45 50	DP	A88	The instructor to show what the common repairs are for the bulkhead seal.
2.2	Adjusting stern tube gland.		35		A88	The instructor to walk the trainees through the process of adjusting the stern tube gland.
2.3	Repairing main shaft line pitch drive belts.		50 55		A89	The instructor to use the standby pitch pump belt static display to show the trainees how to adjust belt tension and replace belts. This lesson does not include breaking the shaftline apart to add new belts.
2.4	Adjusting rudder angle indicator.		15		A90	Instructor to walk the trainee through how to adjust the rudder angle indicator
2.5	Repairing mechanical rudder locking arrangement.		25		A90	Instructor to show the trainee how the micro switch works for the rudder locking arrangement and how to adjust it.
TP3 Testing main shaftline to include:						
3.1	Operating steering system.		15	DP	A90	The instructor to walk the trainees through the flash up of the static steering unit
3.2	Operating pitch belts.		15		A89	This can be done with a motor driving a four-belt system to show how multiple belts react differently and the importance of aligning the shafts.
3.3	Testing auto steering unit.		15		A90	Once the steering unit is running the instructor is to walk the trainee through the process of putting an offset on the ships heading to see if the auto unit corrects

Enabling Check – EO 010.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop. 5 invigilators will be required for each station to assess the trainees.

Arrangements and Procedures

2. EC 010.02 will be a practical assessment based on content from EO 010.02.

3. Pass/Fail.

4. Time required: 75 minutes per trainee.

5. References will be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 010.02 will be a practical assessment based on content from EO 010.02.

2. During this practical assessment, you will be required to maintain main shaftline. You must pass this test to pass EO 010.02. To pass the test, you must maintain main shaftline in accordance with the standard.

3. References will be provided.

4. Pass/Fail.

5. You will have 75 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Nil.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
1. Inspecting flexible coupling. (10)			The trainee is to inspect the flexible coupling by using the hardness tool on the rubber pucks.
2. Investigating operation of inflatable seal. (5)			The trainee to inflate the seal and wait two minutes to see if the pressure holds.
3. Inspecting stern tube gland. (10)			The invigilator to ask the trainee what they are looking for when inspecting the stern tube gland. (excess amount of water leaking)
4. Inspecting steering system components. (10)			The trainee is to inspect the steering mock-up and explain to the invigilator on what they are looking for. (leaking hydraulics, loose parts)
5. Repairing main shaftline bulkhead seal. (5)			The trainee is to show the invigilator where the bolts are to remove the bulkhead seal and what procedure they would use to replace the seal.
6. Adjusting stern tube gland. (5)			The trainee is to show the invigilator where the adjustments are on the stern tube gland.
7. Repairing main shaft line pitch drive belts. (10)			The trainee is to show the invigilator how to adjust the pitch belts and how to check for proper tension.
8. Adjusting rudder angle indicator. (10)			The trainee is to show how to adjust the mechanical adjustment at the steering unit to show proper angle indication of the rudder.
9. Testing auto steering unit. (10)			The trainee is to flash the steering unit, then move the rudder manually. They trainee is to then put the unit in auto and set a course that is offset of the ship's heading to see if the auto function will try to move the rudder to correct the heading.

Instructions to Evaluator:

1. Trainees begin with 100 points. Each fault shall subtract 8 points from the total.
2. Failure of any safety related items constitutes a failure of this EC.
3. The trainee must receive more than 75 points in total to pass this EC.
4. The only grades awarded for this EC are PASS (for totals 75 points or greater) or FAIL (for totals less than 75 points).

PERFORMANCE CHECK – PO 010

PART I – ADMINISTRATIVE INSTRUCTIONS

1. There will be no PC for PO 010 as all assessments will be done during the ECs. The trainees shall pass all ECs to pass this PC.

PO 011 MAINTAIN SHIPBOARD ELECTRICAL SYSTEMS

EO 011.01 Perform Electrical Circuit analysis

1. Performance: Perform electrical circuit analysis

2. Conditions:

   a. Given:

      (1) Classroom/Workshop/Lab;
      (2) Textbooks;
      (3) References; and
      (4) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop/ lab.

3. Standard: The Marine Technician Machinery Maintainer shall independently perform electrical circuit analysis, IAW refs.

   a. Explaining electrical principles, to include:

      (1) Electric charge;
      (2) Voltage, current, resistance;
      (3) Symbols and schematics;
      (4) Si units;
      (5) The battery;
      (6) Internal resistance; and
      (7) Chemical action while discharging.

   b. Explaining series circuits, to include:

      (1) The electric circuit;
      (2) Conductors, switches and fuses;
      (3) Ohm's law;
      (4) Meter measurements;
      (5) Energy and power (watt's law);
      (6) Concept of a load;
      (7) Series circuit; and
      (8) Open and short circuits.

   c. Explaining Parallel circuits;

   d. Explaining Series-parallel circuits;

   e. Explaining Basic Magnetism; and

   f. Explaining AC current and voltage.

4. Time Required:

   a. EO - 2450 minutes; and

   b. CEC - 100 minutes. An additional 50 minutes shall be allotted for CEC review.

5. Instructional Methods:

   a. Interactive lecture;

   b. demonstration performance;

   c. Study assignment; and

   d. Guided Discussion.

6. Training Aids:

7. Test Details:

   a. CEC 011.01 will be a written (critical) assessment based on content from EO 011.01.

8. References: Nil.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Explaining Electrical principles, to include:						
1.1 Describe Electric principles by: a. Electric charge; b. Voltage, current, resistance: (1) Electron vs. conventional flow. c. Symbols and schematics; d. Si units; e. The battery: (1) Series connected cells; and (2) Parallel connected cells. f. Internal resistance; and g. Chemical action while discharging.	200			IL		

1.2	Laboratory Exercise: EE-01 - Voltage Sources, Resistance and Measurements.			150	P		
TP2 Explaining Series circuits, to include:							
2.1	Describing Series circuits by: a. The electric circuit; b. Conductors, switches and fuses; c. Ohm's law; d. Meter measurements; e. Energy and power (watt's law); f. Concept of a load; g. Series circuit; and h. Open and short circuits.	400			IL		Confirm understanding by using the example of a cabling run
2.2	Laboratory Exercise: EE-02 - Ohm's law and Series Circuits.			150	P		
TP3 Explaining Parallel circuits, to include:							
3.1	Describing Parallel circuits by: a. Parallel voltages; b. Parallel currents; c. Kirchhoff's current law (kcl); d. Parallel resistance; e. Product over the sum; f. Conductance and rt; g. Current divider rule; and h. Open and short circuits.	200			IL		
TP4 Explaining Series-parallel circuits, to include:							
4.1	Describing Series-parallel circuits by: a. Equivalent series-parallel circuits; b. Current in series-parallel circuits; c. Voltage drops in series-parallel circuits; d. Voltage dividers; e. Unloaded voltage divider; f. Loaded voltage divider; g. Split supply and voltages wrt ground; and h. Open and short circuits.	750					Discussion - circuit analysis when quantities (V, I, and R) are unknown values
4.2	Laboratory Exercise: EE-03 - Parallel and Series-Parallel Circuits.			150	P		
TP5 Explaining Basic Magnetism, by:							
5.1	Describing Basic magnetism, to include : a. Magnets; b. Residual magnetism; c. Right hand rule; d. Conductors; e. Coils; f. Electromagnetism; and g. B-H curve.	150			IL		
TP6 Explaining AC current and voltage, to include:							

6.1	Introduction to AC current and voltage, by: a. The sinusoidal wave; b. Current flow in ac; c. Comparison of ac and dc power; d. Notational methods used in ac circuits; and e. Frequency, period and wavelength.	150			IL		
6.2	Laboratory Exercise: EE-06 - Oscilloscope Familiarization			150	P		

Critical Enabling Check – EO 011.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom.

Arrangements and Procedures

2. CEC 011.01 will be a written assessment based on content from EO 011.01.

3. Pass mark will be 60%.

4. Time required: 100 minutes.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 011.01 will be a written assessment based on content from EO 011.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. Pass mark is 60%.

10. You will have 100 minutes to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Nil.

EO 011.02 Maintain Electrical Components

1. Performance: Maintain Electrical Components

2. Conditions:

   a. Given:

      (1) Pin connectors;
      (2) Cabling (various);
      (3) Cannon plugs;
      (4) Tools and test equipment;
      (5) PPE;
      (6) Textbooks;
      (7) References; and
      (8) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop/ lab.

3. Standard: The Marine Technician Machinery Maintainer Shall Maintain Electrical Components under supervision, IAW refs, by:

   a. Inspecting electrical connectors, to include:

      (1) Inspecting PIN connectors (ALT 0680); and
      (2) Inspecting cannon plugs.

   b. Repairing electrical connectors (AUT2210), to include:

      (1) Repairing PIN connectors (ALT1230);
      (2) Fabricating PIN connectors (AMT0270);
      (3) Repairing cannon plugs (AUT2070); and
      (4) Repairing UPS wiring harness (AUT2870).

   c. Testing electrical connectors to include:

      (1) Testing PIN connectors; and
      (2) Testing cannon plugs (AUT3080).

4. Time Required:

   a. EO - 650 minutes; and

   b. EC - 50 minutes per trainee.

5. Instructional Methods:

   a. Interactive lecture; and

b. Demonstration performance.

6. Training Aids:

a. PIN connector;

b. Canon plug;

c. Crimper; and

d. PIN insertion/extraction tool.

7. Test Details:

a. EC 011.02 will be a Practical EC based on content from EO 011.02.

8. References: A7, A34, A84, C26 and C27.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspecting electrical connectors, by:						
1.1 Inspecting PIN connector to include: a. Mechanical condition; and b. Security.	25		25	IL P		
1.2 Inspecting cannon plugs to include a. Mechanical condition; and b. Security.	25		25	IL P		
TP2 Repairing electrical connectors, by:						
2.1 Fabricating PIN connectors to include: a. Common components of connectors b. Connector part numbers c. Contact identification d. Selection and preparation of wire e. Crimping contacts/tools f. Installing the contact; g. Removing the contact h. Connector types, such as: (1) “P” (2) “J” (3) “C” i. Repair ‘C’ connectors j. Repair ‘P’ connectors k. Repair ‘J’ plug	50		350	IL P	C26	
2.2 Repairing the following connectors: a. PIN connectors; b. Cannon plugs; and c. UPS wiring harness.	50		50	IL P	C26	
TP3 Testing electrical connectors, by:						

3.1	Testing PIN connectors to include: a. Test ‘C’ connectors; and b. Test ‘P’ connectors.			50	DP	C27	
3.2	Testing cannon plugs to include: a. Test ‘J’ plug.					C27	

Enabling Check – EO 011.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Workshop.

Arrangements and Procedures

2. EC 011.02 will be a practical assessment based on content from EO 011.02.

3. Pass/Fail.

4. Time required: 50 minutes per trainee.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 011.02 will be a practical assessment based on content from EO 011.02.

2. During this practical assessment, you will be required to Maintain Electrical Components. You must pass this test to pass EO 011.02. To pass the test, you must complete the Maintain Electrical Components EC in accordance with the standard.

3. References will not be provided.

4. Pass/Fail.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Nil.

PART IV – ENABLING CHECK LIST

Check List		SAT	UNSAT	Comments
PERSONAL SAFETY				
1.	Explain proper PPE			Location of PPE in the room
2.	Work area clean and free of debris			No extra parts or tools on bench
TOOL ASSEMBLY				
3.	Proper assembly/use of AF8 crimping tool.			Failure to follow safe procedures constitutes an automatic fail
WIRE PREPARTION				
4.	Within tolerance. (1/32-1/64)			
5.	No wire strands cut			
6.	Insulation not damaged.			Failure to follow safe procedures constitutes an automatic fail
REMOVE PIN				
7.	Correct Use of Puller			
8.	Pin removed with no wire damage			
REPLACE PIN				
9.	Safety followed for de-soldering/soldering PIN			
10.	Good solder joint			
11.	PIN correctly re-inserted into body			
12.	Correct test upon completion			

Instructor Notes

1. Trainee shall obtain 9 of 12 SAT on check list without receiving an automatic failure for safety/assembly infraction.
2. Total allotted time 50 minutes.
3. This EC is a Pass/Fail.

EO 011.03 Maintain Field Devices

1. Performance: Maintain field devices

2. Conditions:

   a. Given:

      (1) Field devices (various);
      (2) Tools and test equipment;
      (3) PPE;
      (4) Textbooks;
      (5) References; and
      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop/ lab.

3. Standard: The Marine Technician Machinery Maintainer shall maintain field devices under supervision, IAW refs, by:

   a. Inspecting Field Devices, to include:

      (1) Inspecting rheostat (AUT1420);
      (2) Inspecting differential switches (AUT0960);
      (3) Inspecting solenoid (AUT1450); and
      (4) Inspecting Temperature Switches (AUT1540).

   b. Repairing Field Devices (AUT2330) to include:

      (1) Repairing rheostat (AUT2640);
      (2) Repairing micro switch (AUT2470);
      (3) Repairing selector switches (AUT2670);
      (4) Repairing push buttons (AUT2620);
      (5) Repairing limit switches (AUT2440);
      (6) Assembling and disassembling temperature switches (ATT0260);
      (7) Repairing temperature switches (AUT2770);
      (8) Repairing differential switches (ALT1070); and
      (9) Repairing solenoid (AKT2420).

c. Testing Field Devices, to include:

(1) Testing Rheostat;
(2) Testing Solenoid;
(3) Testing differential switches (AUT3200); and
(4) Testing limit switches (AUT3470).

4. Time Required:

a. EO - 600 minutes; and

b. EC - 50 minutes per trainee.

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration performance.

6. Training Aids:

a. Field devices(various); and

b. Lab-volt trainer.

7. Test Details:

a. EC 011.03 will be a Practical EC based on content from EO 011.03.

8. References: A7 and C29.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Delivery Approach	Refs	Comments
	T	D	P			
TP1 Inspecting Field Devices, by:						
1.1 Inspect mechanical condition and security of field devices to include: a. Rheostats; b. Differential switches; c. Solenoids; and d. Temperature Switches.	50		100	IL P	A7 C29	
TP2 Repairing Field Devices, by:						
2.1 Repairing the following field devices: a. Rheostat; b. Micro switch; c. Selector switches; d. Push buttons; and e. Limit switches.	50		250	IL P	A7 C29	

2.2 Assembling and disassembling the following switches: a. Temperature switches; b. Differential switches; and c. Solenoids.						
TP3 Testing Field Devices, by:						
3.1 Testing the following: a. Rheostat; b. Solenoid; c. Differential switches; and d. Limit switches.	50		100	IL P	A7 C29	

Enabling Check – EO 011.03

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Lab Volt.

Arrangements and Procedures

2. EC 011.03 will be a practical assessment based on content from EO 011.03.
3. Pass/Fail.
4. Time required: 50 minutes per trainee.
5. References will be provided.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 011.03 will be a practical assessment based on content from EO 011.03.
2. During this practical assessment, you will be required to maintain field devices. You must pass this test to pass EO 011.03. To pass the test, you must complete the Maintain field devices in accordance with the standard.
3. References will be provided.
4. Pass/Fail.
5. You will have 50 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Nil.

PART IV – ENABLING CHECK LIST

Check List		SAT	UNSAT	Comments
PERSONAL SAFETY				
1.	Explain proper PPE			
2.	Work area clean and free of debris			
TOOL ASSEMBLY				
3.	Proper use of Multi- meter			
TESTING				
4.	Able to test function of rheostat			
5.	Be able to identify NO/NC and test with Multi-meter			
6.	Be able to identify broken wires at connectors			
7.	Inspect wire and look for nicks or exposed copper			
8.	Be able to identify overall condition of field device. ie: dented, cracked broken seal.			
9.	Able to test solenoid coil by either bench testing or resistance.			

Instructor Notes

1. Trainee shall obtain 6 of 9 SAT on check list without receiving an automatic failure for safety/assembly infraction.
2. Total allotted time 50 minutes.
3. This EC is a Pass/Fail.

EO 011.04 Maintain Sensors

1. Performance: Maintain sensors

2. Conditions:

   a. Given:

      (1) Sensors (various);
      (2) Test and calibration equipment;
      (3) PPE;
      (4) IPMS DCC trainer;
      (5) Textbooks;
      (6) References; and
      (7) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop/ lab.

3. Standard: The Marine Technician Machinery Maintainer shall maintain sensors under supervision, IAW refs, by:

   a. Inspecting Sensors, to include:

      (1) Inspecting pressure sensors (AUT1390);
      (2) Inspecting temperature sensors (AUT1530);
      (3) Inspecting thermocouples; and
      (4) Inspecting thermostats (AUT1590).

   b. Repairing Sensors, to include:

      (1) Repairing pressure sensors;
      (2) Repairing vacuum sensors (AKT2530);
      (3) Repairing temperature sensors;
      (4) Repairing thermostats; and
      (5) Repairing thermocouples (AUT2830).

   c. Testing Sensors, to include:

      (1) Testing pressure sensors;
      (2) Testing temperature sensors;
      (3) Testing thermocouples (AUT2960);
      (4) Testing thermostats;
      (5) Testing heat detector (ANT1350);
      (6) Testing smoke detector; and
      (7) Testing explosive vapour detector (hanger).

4. Time Required:

a. EO - 900 minutes; and

b. EC - 50 minutes.

5. Instructional Methods:

a. Interactive lecture; and

b. Demonstration performance.

6. Training Aids:

a. Various sensors; and

b. IPMS trainer.

7. Test Details:

a. EC 011.04 will be a written EC based on content from EO 011.04.

8. References: Nil.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspecting sensors, by:						
1.1 Defining the following: a. Sensors; and b. Transducers.	50			IL		
1.2 Explain temperature sensors to include: a. Resistive Thermal Device (RTD); b. Bimetallic strip; c. Thermometer; d. Thermocouple; and e. Thermistors.	25	25		IL D		
1.3 Explain flow sensors to include: a. Various types of flow devices.	25	25		IL D		
1.4 Explain pressure sensors to include: a. Bellows; b. Bourbon tube; c. Capacitive; d. Capsule; e. Diaphragm; f. Manometer; g. Piezoelectric; and h. Strain tube.	25	25		IL D		
1.5 Explain level sensors to include: a. Float;	25	25		IL D		

	b. Pressure; c. Conductivity; and d. Ultrasonic.						
1.6	Explain position sensors to include: a. Potentiometer; b. Linear Variable Differential Transformer (LVDT); c. Optical encoders; and d. Variable capacitance.	25	25		IL D		
1.7	Explain velocity to include: a. Magnetic pick-ups Accelerometers; b. DC tachometers; and c. AC tachometers.	25	25		IL D		
1.8	Inspect sensors to include: a. Visually; b. Mechanical condition; and c. Security and mounting.			100	P		
TP2 Repairing sensors, by:							
2.1	Performing the following repair techniques: a. Determine the defective sensor; b. Correct use of LOTO; c. Correct PPE; and d. Remove and install new or repaired sensor.			100	P		
TP3 Testing sensors, by:							
3.1 3.2	Use correct test equipment to test sensors. Ensure correct procedures are followed prior to testing sensors (DCS alarm, etc.).	50		150	IL P		
3.3	Laboratory Exercise: EE-05 – Cable Run and Input Device Troubleshooting			150	P		

Enabling Check – EO 011.04

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom

Arrangements and Procedures

2. EC 011.04 will be a written assessment based on content from EO 011.04.

3. Pass Mark 60%.

4. Time required: 50 minutes.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. EC 011.04 will be a written assessment based on content from EO 011.04.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. You will have 50 minutes to complete this assessment.

10. Do you have any questions before you start?

PERFORMANCE CHECK – PO 011

PART I – ADMINISTRATIVE INSTRUCTIONS

4. Trainees will be provided feedback throughout the EO, no formal PC will be provided.

PO 012 MAINTAIN ELECTRICAL ROTATING MACHINERY AND CONTROL

EO 012.01 Perform Rotating Machinery Analysis

1. Performance: Perform rotating machinery analysis

2. Conditions:

   a. Given:

      (1) Textbooks;
      (2) References; and
      (3) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop/ lab.

3. Standard: The Marine Technician Machinery Maintainer shall perform rotating machinery analysis, IAW refs.

   a. Performing rotating machinery analysis, by:

      (1) Discussing Electromagnetic induction analysis; and
      (2) Explaining Relay and contactor construction and operation to include contactor troubleshooting.

   b. Explaining DC generators;

   c. Explaining DC motors;

   d. Explaining transformers;

   e. Explaining synchronous alternators;

   f. Explaining 3-phase induction motors;

   g. Explaining 3-phase synchronous motors;

   h. Explaining Single-phase motors; and

   i. Introducing automatic control systems.

4. Time Required:

   a. EO – 3700 minutes; and

   b. CEC 100 minutes. An additional 50 minutes required for CEC review.

5. Instructional Methods:

   a. Interactive lecture;

   b. Demonstration performance;

   c. Study assignment; and

   d. Guided Discussion.

6. Training Aids:

   a. Lab-volt trainer;

   b. Various Contactors/transformers etc. for training/demonstration;

   c. Various control devices; and

   d. PLC system.

7. Test Details:

   a. CEC 012.01 will be a written (critical) assessment based on content from EO 012.01. An additional 50 minutes shall be allotted for CEC review prior to the exam.

8. References: C5.

9. Remarks: Nil.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Perform Rotating Machinery Analysis, by:						
1.1 Discussing Electromagnetic induction analysis, to include: a. Similarities between magnetism and electricity; b. Flux; c. Magneto motive force; d. Reluctance; e. Non-linear effects of ferromagnetic materials; f. Hysteresis; g. Magnetic circuits; h. Voltage induced in a conductor; i. Fleming’s right hand rule; j. Voltage induced in a coil; k. Lenz's law;	100			IL	C5	

	l. Force produced by a conductor; m. Bio-Savart law; n. Torque developed by a conductor; o. Torque developed by a coil; and p. Back electromotive force.						
1.2	Explaining Relay and contactor construction and operation to include contactor troubleshooting.	50		50	IL P	C5	
TP2 Explain DC Generators, by:							
2.1	Discussing DC Generators, to include: a. Basic theory; b. Armature windings; c. Generator construction; d. Armature reaction; e. Field poles; f. Interpol's; g. Compensating windings; h. Armature windings; i. DC generator characteristics: (1) Basic generator equation; (2) Separately excited; (3) Shunt; (4) Series; and (5) Compound generator. j. Parallel operation; and k. Generator calculations.	450			IL SA	C5	
2.2	Laboratory Exercise: EE-141-11 - DC Machine Construction; and			100	P SA		
2.3	Laboratory Exercise: EE-141-12 - DC Generator Characteristics.			100	P SA		
TP3 Explain DC motors, by:							
3.1	Discussing DC motor characteristics, to include: a. Basic motor equation; b. Back EMF; c. Equivalent circuit of a DC motor; d. Speed regulation; e. Motor efficiency; f. Shunt motor; (1) Operation; (2) Calculations; (3) Torque-speed characteristics; and (4) Speed control. g. Series motor: (1) Operation; (2) Calculations; (3) Torque-speed; Characteristics; and (4) Speed control. h. Compound motor: (1) Operation; and	450			IL SA	C5	

	(2) Torque-speed Characteristics. i. Starting DC motors; and j. Stopping DC motors.						
3.2	Laboratory Exercise: EE-141-13 - DC Motor Characteristics			100	P SA		
TP4 Explain transformers, by:							
4.1	Discussing Transformers, to include: a. Review of single phase AC circuits; b. Measurement of power; c. Basic transformer theory; d. Practical single-phase transformer; e. Efficiency; f. Multiple winding transformers; g. Auto transformer; h. Basic three phase AC theory; i. Power measurement in three phase systems; j. Three phase transformers; k. Basic transformer connections: (1) WYE-WYE; (2) DELTA-DELTA; (3) WYE-DELTA; (4) DELTA-WYE; (5) Load calculations for three phase systems; and (6) Three phase AC power.	300		100	IL P SA	C5	
4.2	Laboratory Exercise: EE-141-14 - Three-Phase AC Circuits.			100	P SA		
TP5 Explain synchronous alternators by:							
5.1	Discussing Synchronous alternators to include: a. Basic construction; b. Frequency relationship; c. Generated voltage; d. 3Φ alternator; e. Alternator equivalent circuit; f. Voltage regulation; g. Efficiency; h. Typical alternator characteristics; and i. Alternators in parallel synchro-scope.	350			IL	C5	
5.2	Laboratory Exercise: EE-141-15 - The Synchronous Alternator.			100	P SA		
TP6 Explain 3-phase induction motors by:							
6.1	Discussing 3-phase induction motors, to include: a. Squirrel cage motor: (1) Construction; (2) Operation; (3) Characteristics; and (4) Applications.	300			IL	C5	

	b. Wound rotor motor: (1) Construction; (2) Operation; and (3) Characteristics. c. Double squirrel cage; and d. Starting techniques.						
6.2	Laboratory Exercise: EE-141-16 - Three Phase Induction Motors.			100	P SA		
TP7 Explain 3-phase synchronous motors by:							
7.1	Discussing 3-phase synchronous motors, to include: a. Construction; b. Theory of operation; c. Synchronous motor starting techniques; d. Single phase synchronous motor; e. Self-synchronous units (Selsyn); and f. Power, efficiency and torque.	200		100	IL P	C5	
7.2	Laboratory Exercise: EE-141-17 - Three Phase Synchronous Motors.			100	P SA		
TP8 Explaining Single-phase motors by:							
8.1	Discussing Single-phase motors, to include: a. Single phase induction motor; b. Split phase motor; c. Shaded pole motor; d. Capacitor start motor; and e. Series motor.	150			IL	C5	
TP9 Introducing automatic control systems by:							
9.1	Defining Fundamentals of Automatic control to include: a. Define and state purpose; b. Describe variety of electronic controllers (PID, PLC, Computer etc) c. Describe how automatic controls are utilized in the ships propulsion machinery (ex. temperature, level control and include control parameters such as time lag, time constant, dead time, 1st and 2nd order lag, disturbance and offset. a. On-Off control: (1) What it means; (2) Characteristics; (3) How it is used; and (4) Use applications as examples. b. Sequential Control: (1) What it means; (2) Characteristics; (3) How it is used; and (4) Use applications as examples.	400			IL		

c. Proportional-Integral-Derivative (PID) Control Introduction: (1) Explain principal/ theory; (2) Characteristics of PID Control (explain the P, I, D, PD, and PID actions respectively using step or ramp input; and (3) Describe components comprising a PID control systems including (in block format) sensing unit, Transducer, manipulator and controller. d. Introduce PLC Controllers.						

Critical Enabling Check – EO 012.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom.

Arrangements and Procedures

2. CEC 012.01 will be a written (critical) assessment based on content from EO 012.01.

3. Pass mark will be 60%.

4. Time required: 100 minutes.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 012.01 will be a written (Critical) assessment based on content from EO 012.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. Pass mark is 60%.

10. You will have 100 minutes to complete this assessment.

11. Do you have any questions before you start?

EO 012.02 Maintain Electric Motors

1. Performance: Maintain electric motors

2. Conditions:

   a. Given:

      (1) Various AC/DC motors;
      (2) Motor control trainer;
      (3) Induction heater;
      (4) Bearing installation kit;
      (5) Bearing Puller Kit
      (6) Bearings;
      (7) Tools and test equipment;
      (8) Textbooks;
      (9) References; and
      (10) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop/ lab.

3. Standard: The Marine Technician Machinery Maintainer shall maintain electric motors under supervision, IAW refs:

   a. Inspecting electrical motors, to include:

      (1) Inspecting motor seals (AUT1290);
      (2) Inspecting magnetic Brake (AKT0690);
      (3) Inspecting vanaxial fan (AUT1660);
      (4) Inspecting enclosure fan (AUT1100);
      (5) Inspecting motor impellers (AUT1280); and
      (6) Inspecting bearings (AKT0530).

   b. Repairing electrical motors (AUT2260), to include:

      (1) Repairing motor impeller;
      (2) Repairing low voltage fans (24vDC) (AUT2450);
      (3) Cleaning electrical motor components (AUT0670);
      (4) Repairing magnetic brake;
      (5) Adjusting Brake pads (AKT2910);
      (6) Repairing enclosure Fan (AUT2320);
      (7) Adjusting magnetic brake (AKT0030); and
      (8) Repairing bearings (AKT1410).

   c. Testing electrical motors (AUT3290), to include:

      (1) Testing DC Motor (AUT3170);
      (2) Testing magnetic brake;

(3) Testing motor impeller; and
(4) Testing bearings.

4. Time Required:

a. EC - 825 Minutes; and

b. EO - 100 Minutes per trainee.

5. Instructional Methods

a. Interactive lecture;

b. Demonstration performance;

c. Study assignment; and

d. Guided discussion.

6. Training Aids:

a. Bearing Trainer (Machined shafts);

b. Motors; and

c. Bearings for replacement.

7. Test Details:

a. EO 012.02 will be a practical EC based on content from EO 012.02. Test should be managed where all trainees carryout the practical EC concurrently.

8. References: A1, A2, A4, A14, A16, A17 and A113.

9. Remarks: For testing, there should be one student per motor, so it can be all done concurrently.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Inspecting electrical motors, by:						
1.1 Discussing bearings, to include: a. Bearing safety; b. Application of bearings; and c. Types: (1) Open; and (2) Sealed seals.	50			IL	A4 A16 A17	

1.2	Discussing bearing Construction types, such as: a. Ball; b. Roller; c. Needle; and d. Thrust.	50			IL	A4 A16 A17	
1.3	Discussing Magnetic Brakes assemblies to include: a. Operation; and b. Adjustment.	50		25	IL P	A113	
1.4	Identifying Bearing failures, such as: a. Normal fatigue; b. Contamination; c. Misalignment; d. Noise; e. Temperature; f. Excessive vibration; g. Lubricant failure; h. Corrosion; and i. Safety.	100		25	IL P	A1 A2 A4 A14 A16 A17	
TP2 Repairing electrical motors, by:							
2.1	Disassembling electrical motors, to include: a. Power OFF / disconnected; b. End bells marked; c. Produce drawing of disassembly; d. Disassemble motor; and e. Secure rotor (protect from vice jaws).		50	100	DP	A1 A2 A4 A14 A16 A17	Trainees to mark end bells with a permanent marker vice a punch in order to preserve training resources. Trainees to produce a sound drawing of the motor disassembly.
2.2	Mounting and Dismounting procedures for Bearings to include: a. Safety procedures; b. Care in Handling; c. Mounting tools; d. Mechanical tools; e. Induction heater: (1) No more than 110 degrees Celsius (230F); and (2) Prevents bearing overheating. f. Removal tools; g. Cold mounting/dismounting method; h. Bearing pullers method; i. Sleeve/slug and hammer method; and j. Heated mounting/dismounting method (no more than 110 degrees Celsius (230F).		50	150	DP	A1 A2 A4 A14 A16 A17	
2.3	Assembling motor a. Bells aligned; and b. Shims/brakes installed correctly.		25	50	DP		
TP3 Testing electrical motors (AUT3290), to include							

3.1 Conducting the following tests: a. Bearing run-in test (C-02-010-002/AB-000); b. Current test; and c. Magnetic brake. 3.2 Explain Vibration analysis.		50	50	DP		Instructor to demonstrate motor test procedure to trainees, going over various testing details as per ref. VA discussion is info only and discussed as part of the test procedure

Enabling Check – EO 012.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom.

Arrangements and Procedures

2. EC 012.02 will be a practical assessment based on content from EO 012.02.

3. Pass/Fail.

4. Time required: 100 minutes.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 012.02 will be a practical assessment based on content from EO 012.02.

2. During this practical assessment, you will be required to maintain electric motors. You must pass this test to pass EO 012.02. To pass the test, you must complete the maintain electric motors in accordance with the standard.

3. References will/ be provided.

4. Pass/Fail.

5. You will have 100 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Trainee shall have 100 minutes to disassemble motor, remove bearing, replace bearings, re-assemble motor, and carry out a free spin test.

2. Trainee shall obtain 12 of 18 sat on check list without receiving an automatic failure for safety/assembly infraction. Total allotted time 100 minutes. This test is a Pass/Fail.

PART IV – ENABLING CHECK LIST

Check List		SAT	UNSAT	Comments
PERSONAL SAFETY				
1.	Explain proper PPE			Location of PPE in the room
2.	Work area clean and free of debris			No extra parts or tools on bench
DISASSEMBLE MOTOR				
3.	Power OFF / disconnected			Failure to follow safe procedures constitutes an automatic fail
4.	End Bells Marked			
5.	Disassemble motor			
6.	Produce drawing of disassembly			
7.	Secure Rotor (protect from vice jaws).			Failure to follow safe procedures constitutes an automatic fail
REMOVE BEARING				
8.	Protect Threaded Shaft (if required)			
9.	Bearing Pullers fit Snugly			
10.	Bearing Pullers Aligned with Shaft			
11.	Bearing Removed Without Struggle			
INSTALL BEARING				
12.	Take Care in Handling the Bearing			
13.	Heat the Bearing			
14.	Mount the Bearing Using Gloves			
REASSEMBLE THE MOTOR				
15.	End Bells Securing nuts / bolts evenly tighten			
16.	All parts correctly assembled CRITICAL			Failure to correctly assemble motor using all parts constitutes an automatic fail
17.	Conduct "MANUAL TEST" (spin by hand)			
18.	Explain "BEARING RUN IN TEST"			
Instructor Notes:				

EO 012.03 Maintain Motor Control Devices

1. Performance: Maintain motor control devices

2. Conditions:

   a. Given:

      (1) Tools and test equipment;
      (2) Applicable motor control drawings;
      (3) Lab volt;
      (4) As fitted ships motor controller (various);
      (5) Lockout/tag out;
      (6) PPE;
      (7) Textbooks;
      (8) References; and
      (9) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom and Workshop/ lab.

3. Standard: The Marine Technician Machinery Maintainer shall maintain motor control devices under supervision, IAW refs, by:

   a. Inspecting motor control devices, to include:

      (1) Interpreting motor control drawings;
      (2) Inspecting electrical contacts (AUT1010);
      (3) Inspecting overload relays (AUT1350);
      (4) Inspecting overload heating elements (AUT1340);
      (5) Inspecting visually, internal wiring (AUT1710);
      (6) Inspecting electrical coil (AUT0980); and
      (7) Inspecting timing relay (AUT1600).

   b. Repairing motor control devices, to include:

      (1) Repairing electrical contacts (AUT2230);
      (2) Repairing overload relay (AUT2550);
      (3) Repairing overload heating elements (AUT2540);
      (4) Repairing internal wiring (AUT2390);
      (5) Assembling/disassembling controller (AUT0350)
      (6) Cleaning electrical controller components (AUT0660);
      (7) Repairing electrical coil (AUT2190); and
      (8) Adjusting timing relay (AUT0270).

   c. Testing motor control devices, to include:

      (1) Testing timing relay;
      (2) Testing electrical coil (AUT3230);

(3) Conducting current test (AKT0470);
(4) Testing electrical contactor (AUT3240); and
(5) Testing electrical motor starter (AZT0350).

4. Time Required:

a. EO - 1250 Minutes;

b. EC - 50 Minutes; and

c. PC - 100 Minutes per trainee.

5. Instructional Methods:

a. Interactive lecture;

b. demonstration performance; and

c. Study assignment.

6. Training Aids:

a. Lab volt;

b. Motor control schematics;

c. Motor control devices;

d. Ships fitted motor controller (as many as possible for varying examples); and

e. Test equipment.

7. Test Details:

a. EO 012.03 will be a written EC based on content from EO 012.03.

8. References: A1, A3, A14, A15, A26 and D2.

9. Remarks: Trainees will use their schematics from TP1 and TP2 to construct and diagnose faults within a motor control circuit. PC 0012 should be tested with multiple lab volt bays to have most if not all students tested concurrently. Safety numbers and invigilators should be planned for in advance to support the number of trainees.

10. Teaching Points:

Teaching Point	Instructional Technique (Time in min)			Methods	Refs	Comments
	T	D	P			

TP1 Inspecting motor control devices, by:							
1.1	Interpreting HMC Ships motor control schematics, to include: a. Symbols and controls, to include: (1) Identification; (2) Relays; (3) Contacts; (4) Automatic controls; and (5) Push buttons, etc. b. Explain schematics, to include: (1) Types (i.e. block/line and ladder diagrams); (2) Parts of title block, legend, etc. c. Examples: Fan controller, whistle controller, fridge compressor, etc.	50		150	IL DP	D2	Instructor to ensure multiple copies of ships fitted drawings of a motor control circuit is available. Instructor to go over analyzing the operation of the CCT
1.2	Producing workable schematics for motor control circuit, to include: a. Start; b. Stop; c. Overload (O/L) protection; d. Control circuit protection; e. One AC/DC motor, such as: (1) Squirrel cage; (2) Wound motor; (3) Synchronous; (4) DC series; (5) DC compound; and (6) DC shunts. f. Two types of motor controls, such as: (1) Directional control; (2) Speed control; (3) Jog control; (4) Brake control; and (5) Timing sequence control.		100	100	DP	A14 A15	Trainees to construct a functioning control circuit using lab-volt components
TP2 Repairing motor control devices, to include:							
2.1	Understanding the Motor/Controller fault finding process, by: a. Maintaining motor/controller: (1) Definitions; (2) Types (pm cm); and (3) Examples. b. Inspecting motor/controller visually for damage of: (1) Sensors; (2) Insulators; (3) Connectors; (4) Cleanliness; (5) Sheathing; (6) Oil ingress; (7) Heat damage; (8) Cabling; (9) Wiring; (10) Physical damage; (11) Vibration; (12) Mounts;	50	100		IL DP	A1 A3 A14 A15 A26 D2	Develop a mockup of a simulated ships motor controller in order to demo to students possible faults that can occur.

	(13) Ingress of liquids; (14) Temperature; (15) Proper locking; and (16) Seal. c. Smell for unusual odors; d. Listen for unusual noise; e. Feel for the following: (1) Temperature; and (2) Vibration. f. Visual examples of damaged components i.e. Pitted contacts, open coil, mechanically stressed components, over current, low voltage, etc. 2.2 Troubleshooting Motor/Controller for the following: a. Check for power, fuses, and overloads; b. Review circuits diagrams; c. Review publications; d. Identify cause of damage of components; e. Seek assistance from supervisor if required; and f. Communicate findings. 2.3 Repairing motor/controller wiring components, such as: a. Wires; b. Terminal board; c. Crimp connectors; and d. Physical security of wires, glands, bracketing, ty-wraps, etc. 2.4 Replacing motor/controller components, such as: a. Fuses; b. Bulbs; c. Screws; d. Bolts; e. Terminal Board segments; f. Gasket; g. Insulators; and h. Formica board.						
2.5	Lab-Volt Control Lab 4 Electromagnetic Contacts and Relays;			100	SA		
2.6	Lab-Volt Control Lab 15 Cam and Drum Switches			100	SA		
2.7	Constructing Motor control circuit, by: a. Assembling circuit based on schematic produced in TP1; and b. Verifying and confirm operation of the circuit.			50	SA		This will be the control circuit trainees will use to troubleshoot
2.8	Fault finding motor control circuit, by: a. Instructor to introduce faults; b. Trainees to diagnose faults; and c. Trainees to repair the faults.			400	DP		Instructor(s) to inject faults in circuit for trainees to troubleshoot.
TP3 Testing motor control devices, to include:							

3.1 Testing Motor control circuit, by: a. Testing motor/controller devices, to include: (1) Fuses; (2) Bulbs; (3) Push buttons; (4) Contacts; (5) Continuity; (6) Overload; and (7) System/device. b. Testing Motor Controller for correct operation.		25	25	SA GD		Trainees to ensure circuit operates as designed i.e. Timing is correct, permissive work etc.

Enabling Check – EO 012.03

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Classroom

Arrangements and Procedures

2. EC 012.03 will be a written assessment based on content from EO 012.03.

3. Pass mark will be 60%.

4. Time required: 50 minutes.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 012.03 will be a written assessment based on content from EO 012.03.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer questions with which you have difficulty. Continue the assessment then come back to the problem questions.

6. Read each question carefully before answering and all responses fully before submitting your test.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. Pass mark is 60%.

10. You will have 50 minutes to complete this assessment.

11. Do you have any questions before you start?

PERFORMANCE CHECK – PO 012

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom one bay per trainee.

Arrangements and Procedures

2. PC 012 will be a practical test based on content from PO 012.

3. Pass/Fail.

4. Time required: 100 minutes per trainee.

5. References will not be required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 012 will be a practical assessment based on content from PO 012.

2. During this practical assessment, you will be required to maintain electrical rotating machinery and control. You must pass this test to complete PO 012. To pass this test, you must complete maintain electrical rotating machinery and control in accordance with the standard.

3. References will not be provided.

4. Pass/Fail.

5. You will have 100 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. 30 mins to construct and become familiar with an operational circuit, supplied by the instructor at the time of the test; trainee must be able to explain the circuit operation from the drawing and then confirm by operating the circuit.

2. Trainee will draw a number from an envelope which contains the faults and leave the room for the instructor to insert the fault without changes to the wiring.

3. Trainee will troubleshoot and correct two single faults with a maximum time allocation of 20 mins each, followed by a single double fault with a maximum time allocation of 30 mins. Instructor is not permitted to give guidance during troubleshooting.

4. Trainee shall obtain 20 of 25 sat on check list without receiving an automatic failure for safety/time infraction. Total allotted time 100 mins. This test is a Pass/Fail.

PART IV – PERFORMANCE CHECK LIST

Check List	SAT	UNSAT	Comments
CONSTRUCTION AND EXPLAINATION OF CIRCUIT (30 MINS) Time:			
1. Construct circuit using supplied drawing			*Failure to construct circuit within time limit constitutes an unsat, but student continues
2. Explain circuit function from drawing			
3. Work station neat and organized			No extra parts on console, leads are orderly
4. Explain proper PPE			Location of PPE in the room
5. Confirm circuit operation			Circuit must be functional before proceeding more than one mistake constitutes an unsat
6. Observe safety at all times			*Failure to follow safe procedures constitutes an automatic fail
Instructor comments:			
FIRST SINGLE FAULT (20 MINS) Time:			
*Trainee picks fault from envelope, instructor inserts the fault, no guidance from instructor.			
Fault chosen:	Fault details (exact location of fault in circuit):		
1. Test circuit to observe fault			
2. Test meter			
3. Use meter correctly			
4. Explain logic behind chosen troubleshooting technique			
5. Correctly identify fault			*Failure to identify fault within time limit constitutes an automatic fail
6. Follow proper course of action to repair Fault			Bench test replacement part
7. Prove fault corrected			
Instructor comments:			
SECOND SINGLE FAULT (20 MINS) Time:			
*Trainee picks fault from envelope, instructor inserts the fault, no guidance from instructor.			
Fault chosen:	Fault details (exact location of fault in circuit):		
1. Test circuit to observe fault			
2. Test meter			
3. Use meter correctly			
4. Explain logic behind chosen troubleshooting technique			
5. Correctly identify fault			*Failure to identify fault within time limit constitutes an automatic fail

Check List	SAT	UNSAT	Comments
6. Follow proper course of action to repair Fault			Bench test replacement part
7. Prove fault corrected			
Instructor comments:			
DOUBLE FAULT (30 MINS) Time 1st Fault: **Time 2nd Fault:**			
*Trainee picks faults from envelope; instructor inserts the faults, no guidance from instructor.			
Faults chosen:	Fault details (exact location of faults in the circuit):		
1. Test circuit to observe fault			
2. Test meter			
3. Use meter correctly			
4. Explain logic behind chosen troubleshooting technique			
5. Correctly identify first fault			*Failure to identify faults within time limit constitutes an automatic fail
6. Follow proper course of action to repair Fault			Bench test replacement part
7. Correctly identify second fault			*Failure to identify faults within time limit constitutes an automatic fail
8. Follow proper course of action to repair Fault			Bench test replacement part
9. Prove complete circuit functional			
Instructor comments:			

ANNEX A – MAIN REFERENCES

Refs Code	Publication Number	Title
A1	C-93-010-007/NY-001	BATTERIES, LEAD ACID, VENTED - APP TO TRL, PTR, AND AUX CLASSES AND CFFS(H)
A2	---	Safety and Environmental Management System (SEMS)
A3	---	MARCORDs
A4	C-27-877-000/MP001	PUMP, DIESEL DRIVEN (FIRE PUMP)
A5	CF-670	Log for Secondary Battery
A6	C-38-138-000/MS-001	HANDBOOK - CATHODIC PROTECTION
A7	D-03-003-005/SF-000	GENERAL ELECTRICAL SPECIFICATION FOR CANADIAN FORCES SHIPS
A8	C-23-HFX-000/MS-001	HALIFAX CLASS SHIP INFORMAITON BOOK - VOLUME 1 - HULL AND MARINE SYSTEMS
A9	C-26-320-000/MS-001	PRIMARY ELECTRICAL POWER GENERATION AND DISTRIBUTION SYSTEM MANUAL APPLICABLE TO HALIFAX CLASS
A10	C-26-321-000/MS-001	TECHNICAL MANUAL VOL 1 OF 2 GENERATOR SET DIESEL 850 KW APPLICABLE TO HFX CLASS
A11	C-26-324-000/MS-001	INSTRUCTION BOOK FOR SHORE CONNECTION AND SHIP ALONGSIDE BOXES AND GENERATOR TEST TERMINALS
A12	C-26-325-000/MS-001	INSTRUCTION BOOK FOR CASUALTY POWER SYSTEM COMPONENTS - APPLICABLE TO HALIFAX CLASS
A13	C-26-383-A00/MS-001	TECHNICAL MANUAL AUTOMATIC CHANGE-OVER SWITCH TYPE ACOS, 450 VOLTS, 660HZ, APPL. 2, 100 AMPERES
A14	C-26-383-B00/MS-001	TECHNICAL MANUAL - AUTOMATIC CHANGE-OVER SWITCH, ASCO 942, CATALOG NO. 435A3D, TYPE ACOS, 450 VOLTS, 60HZ, APPL.2, 250 AMPERES, DRIPPROOF ENCLOSURE
A15	C-26-383-C00/MS-001	MANUAL CHANGE-OVER SWITCH (250 AMPS)
A16	C-27-890-000/MS-002	INSTALLATION, OPERATION, AND MAINTENANCE INSTRUCTIONS FOR SEWAGE TREATMENT PLANT OMNIPURE MODEL 15MX3-1
A17	C-28-398-000/MS-001	CAPSTAN WINDLASS/PEDESTAL/MOTER CONTROLLER P/N 8611172/ 8712581/6968ED500 (HFX)
A18	C-28-399-000/MS-001	CAPSTAN AFTER (MOORING TOWING ASSEMBLY) P/N 8611170/8712580/ 6968ED501 (HFX)
A19	C-38-224-000/MS-001	SECONDARY ELECTRICAL POWER DISTRIBUTION AND LIGHTING SYSTEM APPLICABLE TO HALIFAX CLASS
A20	C-38-231-000/MS-001	TECHNICAL MANUAL CATHODIC PROTECTION (CAPAC) SYSTEM

A21	C-38-232-000/MS-002	OPERATION AND MAINTENANCE INSTRUCTION WITH PARTS IDENTIFICATION LIST HALIFAX CLASS DEGAUSSING POWER SUPPLIES SUPPLIES APPLICABLE TO HALIFAX CLASS
A22	C-51-500-000/MS-001	INSTRUCTION MANUAL SHIPS WHISTLE SYSTEM MODEL EP-130
A23	C-57-574-000/MS-001	NAVIGATION AND WEATHERDECK LIGHTING CONTROL EQUIPMENT
A24	---	NAVORDS
A25	A-LM-007-100/AG-001	http://materiel.mil.ca/en/business-functions-materiel-management/supply-administration-manual-sam.page SUPPLY ADMINISTRATION MANUAL
A26	C-02-005-009/AM-000	MATERIEL MANAGEMENT POLICY INSPECTION AND CONDITIONING OF MATERIEL RETURNED TO AND HELD IN SUPPLY SYSTEM
A27	C-27-948-000/MS-001	Installation, Operation and Maintenance Manual for the DDH 280 Tribal Class/Halifax Class FFH 336-341 Reverse Osmosis Desalination System
A28	C-27-948-000/MS-Z01	Shipboard Reverse Osmosis Desalinator (SROD) NSN 4610-21-904-5434 Membrane Housing Assembly Disassembly, Inspection, Maintenance, Assembly, and Parts
A29	C-23-343-000/NY-001	NAVAL PREVENTIVE MAINTENANCE SCHEDULE (HULL) BOAT, RIGID, INFLATABLE, 24 FT APPLICABLE TO HFX AND IRO CLASSES, CFNOS, CFFS(E) AND NAVAL RESERVE UNITS
A30	C-23-343-000/NY-003	NAVAL PREVENTIVE MAINTENANCE SCHEDULE (ENGINEERING) BOAT, RIGID, INFLATABLE 24 FT (DIESEL ENGINES, RIB-P(C) (VOLVO AQAD 41)) APPLICABLE TO HFX, IRO AND PTR CLASSES,FDU`S, CFNOS, CFFS(E) AND NAVAL RESERVE UNITS
A31	C-23-343-000/NY-004	NAVAL PREAVENTIVE MAINTENANCE SCHEDULE (HULL) - BOAT, RIGID, INFLATABLE 24 FT (WALTER KIDDE FM-200 FIXED FIRE SUPPRESSION SYSTEM) - APPLICABLE TO HFX, IRO AND PTR CLASSES, FDU`S, CFNOS, CFFS(E) AND NAVAL RESERVE UNITS
A32	C-23-343-000/NY-Z03	NAVAL PREVENTIVE MAINTENANCE SCHEDULE (ENGINEERING) BOAT, RIGID, INFLATABLE, 24 FT RIB P(C) (DIESEL ENGINES, RIB P(C) (VOLVO AQAD 41)) APPLICABLE TO HFX, IRO, AND PTR CLASSES, FDU'S, CFNOS, CFFS (E) AND NAVAL RESERVE UNITS
A33	C-23-343-000/NY-Z04	NAVAL PREVENTIVE MAINTENANCE SCHEDULE (HULL) BOAT, RIGID, INFLATABLE, 24 FT RIB P(C) APPLICABLE TO HFX, IRO, PTR CLASSES, FDUs, CFNOS, CFFS(E) AND NAVAL RESERVE UNITS
A34	A-GG-040-003-AG-001	General Safety Training Program
A35	C-23-343-000/MS-001	MAINTENANCE MANUAL 24 FOOT RIGID INFLATABLE BOAT
A36	C-23-343-000/NY-002	NAVAL PREVENTIVE MAINTENANCE SCHEDULE (ELECTRICAL) - BOAT, RIGID, INFLATABLE, 24 FT - APPLICABLE TO HFX, IRO AND PTR CLASSES, FDU`S, CFNOS, CFFS(E) AND NAVAL RESERVE UNITS

A37	C-23-343-000/NY-Z05	NAVAL PREVENTIVE MAINTENANCE SCHEDULE (ELECTRICAL - BOAT, RIGID, INFLATABLE, 24 FT - APPLICABLE TO IRO, PTR AND HFX CLASSES, FDU'S, CFNOS, CFFS(E) AND NAVAL RESERVE UNITS
A38	C-03-010-546/MS-001	AC MOTOR CONTROLLERS
A39	C-23-050-001/PT-001	LAB VOLT STUDENT MANUAL
A40	C-98-002-CAD/FP-003	DND STANDARDS - DND DRAWING STANDARDS & CONVENTIONS
A41	C-38-288-000/MS-001	OPERATION AND MAINTENANCE MANUAL - FREQUENCY CONVERTER PART NUMBER 1003720300 - TRANSFORMER/CONTACTOR ASSEMBLY PART NUMBER 1003720200 (T/CA)PN 1003790200 APPLICABLE TO HALIFAX CLASS
A42	C-28-463-000/MS-001	HANDBOOK WITH PARTS LIST SLIDING PADEYE RECEIVING UNITS BULKHEAD MOUNTED (MODEL SB-9C) AND RETRACTABLE (MODEL SR-12C) APPLICABLE TO HALIFAX CLASS
A43	C-23-045-000/AG-001	DAMAGE CONTROL (SEA) VOLUME 1 OF 4 SHIPBOARD DAMAGE CONTROL
A44	C-23-045-000/AG-002	DAMAGE CONTROL (SEA)VOLUME 2 OF 4 SHIPBOARD CHEMICAL BIOLOGICAL RADIOLOGICAL NUCLEAR (CBRN) DEFENCE
A45	C-23-045-000/AG-003	DAMAGE CONTROL (SEA) VOLUME 3 OF 3 SHIP STABILITY
A46	C-23-045-000/AG-004	DAMAGE CONTROL (SEA) VOLUME 4 OF 4 - SUBMARINE DAMAGE CONTROL
A47	C-03-005-012/AM-001	NAMMS POLICY AND PROCEDURES - NAVAL MATERIEL MANAGEMENT SYSTEM MANUAL
A48	C-28-010-329/MS-001	WATERTIGHT CLOSURES - INSPECTION, MAINTENANCE AND MINOR REPAIRS FOR SHIPS STAFF - APPL TO ALL SURFACE SHIPS
A49	C-03-011-007/MS-002	HANDBOOK - INSULATION MANUAL FOR HMC SHIPS
A50	D-23-003-005/SF-002	SPECIFICATION FOR MAINTENANCE PAINTING OF HMC SHIPS
A51	D-28-155-000/TB-003	TECHNICAL BULLETIN (MARINE) DECK COVERING MATERIAL FOR HMC SHIPS AND SUBMARINES
A52	D-49-003-003/SF-001	DESCRIPTION AND MAINTENANCE INSTRUCTIONS - WELDING SPECIFICATION FOR HMC SHIPS AND AUXILIARIES
A53	D-03-001-025/SF-001	SPECIFICATION FOR REPAIR AND OVERHAUL OF MARINE VALVES
A54	C-03-005-033/AA-000	DIRECTIVE NAVAL ENGINEERING MANUAL NEM DIRECTION AND GUIDANCE
A55	D-23-003-005/SF-002	SPECIFICATION FOR MAINTENANCE PAINTING OF HMC SHIPS
A56	C-03-010-104/ME-000	INSTRUCTION BOOK INSTALLATION AND MAINTENANCE OF VALVES AND FITTINGS
A57	C-25-007-001/TB-001	TECHNICAL BULLETIN OPERATING INSTRUCTIONS VELON STEAM TRAPS
A58	C-97-325-000/NY-Z01	FIRE FIGHTING SYSTEM TWIN AGENT UNIT (TAU) APPLICABLE TO HALIFAX CLASS
A59	C-97-040-001/TB-001	CARTRIDGE AND FIREFIGHTING AGENT DRY CHEMICAL FIRE EXTINGUISHERS

A60	C-19-A94-000/MS-001	HANDBOOK (COMPREHENSIVE MAINTENANCE MANUAL) AND PARTS LIST AIR CONDITIONING/HEATING UNIT - MODEL ACE-804-H-CUP NSN 4120-01-508-1404
A61	C-28-010-024/MS-001	HANDBOOK - ANCHORS, CHAIN CABLES AND ASSOCIATED ARTICLES
A62	C-29-050-004/PT-001	MARINE HEATING VENTILATION AND AIR CONDITIONING THEORY MANUAL
A63	C-29-355-000/MS-001	Heating Ventilation and Air Conditioning Manual
A64	D-03-003-016/SF-005	SPECIFICATION FOR THE CONSTRUCTION AND OUTFITTING OF SHIPS
A65	C-28-020-001/TB-001	TECHNICAL BULLETIN (MARINE) IN-SERVICE CERTIFICATION REQUIREMENTS FOR SHIPBOARD LIFTING EQUIPMENT APPLICABLE TO ALL CLASSES OF SHIPS
A66	CFCD-105	Seamanship manual (Rigging)
A67	D-03-010-001/SF-001	SPECIFICATION FOR FURNISHING AND FURNISHING MATERIALS FOR HMC SHIPS
A68	C-27-010-016/TB-001	FLEET TECHNICAL BULLETIN - DECONTAMINATION OF DOMESTIC FRESH WATER TANKS
A69	D-03-002-004/SG-007	SHIPS STANDARDS - NON STANDARD PIPEWORK COMPONENETS VOL 17
A70	CFAO 30-2	FIRE REPORTING AND INVESTIGATION
A71	QR&Os	Queen's Regulations and Orders
A72	C-02-015-001/AG-000	POLICY PROCEDURES AND GUIDLINES UNSATISFACTORY CONDITION REPORTING
A73	A-AD-D30-001/JS-001	DND ADMINISTRATIVE AND STAFF PROCEDURES MANUAL, ADMINISTRATIVE PROCEDURES FOR NDHQ
A74	A-AE-000-001/AG-E00	GUIDE TO THE DIVISIONAL SYSTEM (MARITIME COMMAND)
A75	C-03-005-012/AM-007	NAVAL MAINTENANCE MANAGEMENT SYSTEM MANUAL VOLUME 7 (DRAFT) GUIDELINES FOR THE REFIT OF MAJOR WARSHIPS
A76	C-77-010-008/MB-001	STANDARD PRESENTATION DAMAGE CONTROL INCIDENT AND STATUS BOARDS (BILINGUAL)
A77	D-23-002-005/SG-001	STANDARD FOR METRIC DRAUGHT MARKS (HMC SURFACE SHIPS AND CNAV VESSELS)
A78	C-28-199-000/MS-000	HELICOPTER HANGER DOOR IROQUOIS (IRO) CLASS SHIPS
A79	C-97-238-000/MS-001	RANGE GUARD SYSTEM, LIQUID KARBALOY SYSTEM
A80	C-29-354-000/MS-001	Main Refrigeration Applicable to HFX Class
A81	C-28-679-000/MS-001	Installation Operation Manual CV15-1142 Vertical Capstan Handling Winch
A82		Outboard - Maintenance Manual
A83		Small Engine Manual (WAJAX?)
A84	C-02-040-009/AG-001	DND General Safety Standards
A85	D-03-002-006/SG-Z01	Inspection and Run-in Tests - Shipboard Electric Motors

A86	C-24-541-000/MF-001	GEK 50501 LM 2500 Gas Turbine System
A87	C-24-545-A00/MS-002	CPF Mainshaft Flexible Coupling
A88	C-24-545-A00/MS-001	Manesafe Seal Type MA MKII Stern Shaft Seal and Manesafe Type ND Bulkhead Seal
A89	C-24-559-000/MS-001	Controllable Pitch Propellers and Propeller Shafting
A90	C-27-778-000/MS-001	Steering Gear Control Systems (Books 1 - 8)
A91	C-97--325-000/NY-001	Firefighting System Twin Agent Unit (TAU)
A92	C-97-325-000/NK-001	Firefighting System Twin Agent Unit (TAU)
A93	A-LM-007-100 AG-001	Supply Administration Manual
A94	C-27-886-00/MS-001	Hydrovane 188-CW-08-002 Air Compressor
A95	C-27-889-000/MS-001	Comprehensive Manual for HP Air Compressor Model 5436N
A96	C-27-B67-000/MS-001	Hydromem Bilge Water Treatment System Model BWTS100
A97	C-27-B75-000/MS-001	Operation and Maintenance Instructions with Part Identification Steam Generator
A98	C-28-270-000/MS-000	Instruction Bool Probe and Receiver Fuelling System
A99	C-25-547-000/MS-001	Lube Oil Separator
A100		Alfa Laval MAB 104B-14/24 Separator Manual
A101	C-28-270-000/NY-000	Naval Preventative Maintenance Schedule (Engineering) Receiver Assembly (Fuel Probe)
A102	C-77-304-000/ NY-001	Naval preventative maintenance schedule CBRN filtration units
A103		CFCD 132 Damage Control Manual
A104		DRMIS
A105	A-GG-040-063-AG-001	General Safety Training Program Vol. 3
A106	SEMS	
A107	C-02-010-002/AB-000	Electric Motor Repair
A108	C-02-040-009/AG-001	DND General Safety Standards
A109	D-03-02-06/SG-000	Repair and Overhaul and Post- Repair inspection and test procedures for electric motors
A110	NETE	Bearing Course Notes
A111	C-23-050-002/PT-001	Lab-Volt Trainee Manual
A112	C-03-010-546/MS-001	AC Motors and Controllers
A113	C-28-679-000/MS-001	Ventical Capstan/ Line Handling Winch
C1	ISBN 9780827341463	DIRECT CURRENT FUNDAMENTALS 4TH EDITION
C2	ISBN 1418065803	DELMARS STANDARD TEXTBOOK OF ELECTRICITY 4TH EDITION
C3	ISBN 0321374177	INTRODUCTION TO TECHNICAL MATHEMATICS 5TH EDITION
C4		ALTERNATING CURRENT FUNDAMENTALS 4TH EDITION
C5		PRINCIPLES OF FLUID MECHANICS, KREIDER
C6	ISBN 0-02-342-700-0	TECHNICAL DRAWING (SIXTH EDITION)
C7	ISBN 0-87006-577-7	MODERN WOODWORKING
C8	ISBN 0-13-345471-1	FURNITURE AND CABINET CONSTRUCTION

C9	ISBN 0-87345-104-X	METALWORK TECHNOLOGY AND PRACTICE
C10	ISBN 0-87006-429-0	BLUEPRINT READING
C11	ISBN 0-87006-279-4	MODERN WELDING
C12	ISBN 0-07-548922-8	ENGINEERING DRAWING AND DESIGN (THIRD EDITION)
C13		THERMODYNAMICS FOR ENGINEERING TECHNOLOGY, PERRY ROBERT L.
C14		THERMODYNAMICS FOR MARINE ENGINEERING TRADESMAN
C15	ISBN 0-13-898762-2	TECHNICALLY WRITE
C16	ISBN 0-87006-509-2	MODERN METAL WORKING
C17	ISBN 0-07-082462-2	GENERAL METALS
C18	ISBN 87002-307-1	WOOD MATERIALS AND PROCESSES
C19	ISBN 0-471-80043-0	MATHEMATICS FOR THE TRADES
C20	DX-600N	HILTI GUN MANUFACTURERS INSTRUCTIONS
C21		ARCAIR SLICE CUTTING SYSTEM INSTRUCTION MANUAL, THERMADYNE ARCAIR COMPANY
C22		APPLIED MECHANICS FOR ENGINEERS - VOL 2, THOMAS REED
C23		ENGINEERING THERMODYNAMICS WITH APPLICATIOINS, BURGHARDT, DAVID M.
C24	ISBN 1-56637-300-X	Modern Refrigeration and Air Conditioning, Althouse/Turnquist/Bracciano
C25	ISBN 7735288-8	Volvo Penta Workshop Manual
C26		Daniels Manufacturing Corporation - Tooling Guide for Aerospace Wiring
C27		Daniels Manufacturing Corporation - Tensile Test Systems
C28	ISBN 9781408175965	Reeds Vol 8 General Engineering Knowledge for Marine Engineers, Paul Anthony Russell, 1st Edition
C29	ISBN 0-113-177691-6	Electrical Machines, Drives, and Power Systems 6th Edition.
D1	ICS 19-2002	National Electric Manufacturers Association (NEMA) Standards Publication

ANNEX B – TASK LIST (TRAIN)

Task No	Task Statement	PO
AFT0340	Prepare unsatisfactory condition reports (UCR)	001
AFT0350	Prepare (initiate) pre-installation failure (PIF) reports	001
AFT0310	Prepare marine engineering equipment for shipment or storage	001
AZT0360	Test fuel pump	002
AZT0070	Inspect fuel pump	002
AKT2180	Repair positive displacement pump	002
AJT2320	Repair rotary vane pump	002
AKT2310	Repair rotary piston pump	002
AZT0180	Repair fuel pump	002
CT0190	Investigate operation of replenishment at sea systems	002
AHT0760	Repair replenishment at sea fuel receiver (support of RAS)	002
AKT0130	Assemble / disassemble emergency breakaway fuel coupling	002
AKT0140	Assemble / disassemble fuel probe	002
AKT1610	Repair emergency breakaway fuel coupling	002
AHT0510	Assemble / disassemble fuel oil system components	002
AHT0520	Assemble / disassemble main lube oil filling and transfer system components	002
AKT2820	Troubleshoot centrifuge	002
AIT0510	Assemble / disassemble fuel oil filling/transfer system components	002
AYT0250	Repair oil pollution abatement treatment system membranes	002
AYT0220	Repair oil pollution abatement treatment system batch tank	002
AYT0030	Assemble / disassemble oil pollution abatement collection system components	002
AYT0040	Assemble / disassemble oil pollution abatement treatment system components	002
UT0250	Operate oil pollution abatement system locally	002
ANT1460	Test twin agent units	003
FT0440	Operate twin agent unit	003
ANT1110	Repair twin agent unit system	003
ANT1740	Troubleshoot twin agent unit	003
ANT0110	Assemble / disassemble twin agent unit components	003
ANT0590	Inspect twin agent units	003
ANT1180	Survey twin agent units	003
ATT0570	Inspect valves	003
ANT0620	Inspect visually, emergency escape breathing device (EEBD).	003
ANT1450	Test thermal Imaging camera	003
ANT1100	Repair thermal imaging camera	003
ANT1000	Repair portable damage control equipment	003
ANT0390	Inspect emergency escape breathing device (EEBD).	003
AJT0180	Inspect portable rescue equipment.	003
AJT0370	Repair portable rescue equipment.	003
AJT0010	Adjust portable rescue equipment	003

AJT0620	Troubleshoot portable rescue equipment.	003
ANT1730	Troubleshoot self-contained breathing apparatus (SCBA)	003
AYT0430	Troubleshoot respirators	003
ANT0840	Repair emergency escape breathing device (EEBD).	003
AJT0520	Test portable rescue equipment.	003
AMT0100	Assemble / disassemble shock mount arrangements	004
AMT0670	Repair plastic parts and components	004
AMT0070	Assemble / disassemble plastic parts and components	004
AMT0300	Fabricate plastic shapes	004
AMT0290	Fabricate plastic regalia / momento	004
AMT0250	Fabricate metal parts and components	004
AMT0720	Repair wood parts and components	004
AMT0370	Fabricate wood parts and components	004
AMT1010	Install wood parts and components	004
AMT0400	Inspect plastic parts and components	004
BBT0430	Investigate ship's fluid tanks	004
BBT0330	Inspect visually, ship's fluid tanks	004
BBT0320	Inspect visually, ship's appendages	004
BBT0340	Inspect visually, ship's minor structure	004
BBT0350	Inspect visually, ship's structure	004
ANT0200	Configure casualty power cable route	005
ANT0340	Inspect casualty power riser terminal cables	005
AKT0110	Assemble / disassemble combustion engine acoustic enclosure cooling system components	006
AKT1500	Repair combustion engine acoustic enclosure	006
AKT1470	Repair combustion engine acoustic enclosure cooling system	006
AKT0570	Inspect combustion engine acoustic enclosure	006
AIT0530	Assemble / disassemble HP air compressor	006
AIT0550	Assemble / disassemble LP air compressor	006
AIT0590	Install auxiliary systems components	006
AHT0690	Repair compressed air automatic lubrication system	006
AIT0540	Assemble/ disassemble HP air system components	006
AIT0560	Assemble/ disassemble LP air system components	006
AIT0810	Repair low pressure compressed air control station	006
ATT0140	Assemble/ disassemble pneumatic valve	006
AHT0550	Clean auxiliary steam generator tubes	007
AHT0780	Repair steam generator fuel igniter	007
AIT0670	Repair auxiliary steam generator fuel nozzle	007
AHT0700	Repair feedwater chemical injection system	007
AIT0650	Repair auxiliary steam generator air damper	007
AIT0660	Repair auxiliary steam generator forced air supply fan	007
AT0090	Conduct salinity test	007
ET0120	Operate auxiliary steam generator remotely	007
AHT0800	Test aux steam generator electronic safety control	007
AHT0810	Test aux steam generator high fire electrode	007

ET0110	Operate auxiliary steam generator locally	008
ATT1410	Troubleshoot fluid piping system (steam, water, aux circ, chilled water, fuel, air)	008
AOT0190	Assemble / disassemble reverse osmosis desalination (ROD) unit	008
AOT0310	Clean reverse osmosis desalination (ROD) membrane	008
JT0480	Test potable water quality	008
AOT1010	Preserve reverse osmosis desalination (ROD) membrane	008
AOT0040	Adjust chiller units	008
AWT0120	Repair double seated refrigeration service valve	008
AT0060	Conduct pressure test	008
AT0160	Conduct vacuum test (not for sub)	008
AT0280	Test pilot operated thermostatic control valve	008
AT0270	Test mechanical thermostatic control valve	008
AOT0770	Investigate chilled water and hydronic system components	008
AOT0780	Investigate HVAC/NBCD air distribution system	008
AKT1820	Repair HVAC air flow switch	008
AOT0030	Adjust chilled water and hydronic system components	008
AVT0280	Assemble / disassemble diesel engine combustion air system components	009
AVT0880	Repair diesel engine crankcase explosion door	009
AVT1130	Repair main diesel engine (MDE) lube oil system self-cleaning filter	009
AVT0830	Repair diesel engine air start lubrication system.	009
AZT0100	Inspect spark plug gaps	009
AZT0120	Investigate operation of small gasoline engine	009
AZT0020	Adjust small boat fuel pump	009
AZT0200	Repair small boat engine fuel pump	009
AZT0390	Troubleshoot small gasoline engine	009
AZT0290	Repair small gasoline engine	009
AZT0240	Repair small boat outboard engine	009
VT0020	Operate small gasoline engine	009
AZT0190	Repair rigid inflatable boat electrical systems	009
AZT0370	Test rigid inflatable boat electrical systems	009
AZT0400	Inspect small boat propeller	009
AZT0260	Repair small boat shaftline shear coupling	009
AZT0210	Repair small boat hull	009
AZT0270	Repair small boat throttle assembly	009
AZT0300	Repair steering system of a small boat	009
AZT0250	Repair small saltwater impeller	009
AVT1430	Troubleshoot gas turbine water wash system	010
AVT0390	Assemble / disassemble gas turbine water wash system components	010
AHT0580	Install ancilliary systems components	010
AVT1120	Repair gas turbine water wash tank	010
AKT0700	Inspect mechanical interlocks	010
XT0020	Investigate operation of inflatable seal	010
AUT1150	Inspect flexible coupling	010
BT0190	Repair main shaftline bulkhead seal	010
BT0010	Adjust stern tube gland	010

BAT0010	Adjust rudder angle indicator	010
BAT0250	Repair mechanical rudder locking arrangement	010
BAT0300	Test auto steering unit	010
AUT2210	Repair electrical connections	011
AUT2870	Repair uninterruptible power supplies wiring harness	011
AUT2070	Repair cannon plugs	011
AMT0270	Fabricate pin connectors	011
ALT1230	Repair pin connectors	011
AUT3080	Test cannon plugs	011
AUT3470	Test limit switches	011
AUT3200	Test differential switches	011
AUT1420	Inspect rheostat	011
AUT0960	Inspect differential switches	011
AUT1450	Inspect solenoid	011
AUT1540	Inspect temperature switches	011
AUT2330	Repair field devices	011
AUT1390	Inspect Pressure sensors	011
AUT1530	Inspect temperature sensors	011
AUT1590	Inspect thermostats	011
AKT2530	Repair vacuum sensors	011
AUT2830	Repair thermocouples	011
ANT1350	Test heat detector	011
AUT2960	Test Thermocouples	011
AKT2420	Repair solenoid	011
AUT2640	Repair rheostat	011
AUT2470	Repair micro switch	011
AUT2670	Repair selector switches	011
AUT2620	Repair push buttons	011
AUT2440	Repair limit switches	011
ATT0260	Assemble/ disassemble temperature switches	011
AUT2770	Repair temperature switches	011
AKT0530	Inspect bearings	012
AKT0690	Inspect magnetic brake	012
AUT1280	Inspect motor impeller	012
AUT1290	Inspect motor seals	012
AKT1410	Repair bearings	012
AUT2260	Repair electrical motor	012
AKT2910	Adjust brake pads	012
AUT0670	Clean electrical motor components	012
AUT2450	Repair low voltage fans (24V DC)	012
AUT3290	Test electrical motor	012
AUT3170	Test DC motor	012
AUT0980	Inspect electrical coil	012
AUT1710	Inspect visually internal wiring	012
AUT1340	Inspect overload heater elements	012

AUT1350	Inspect overload relay	012
AUT1010	Inspect electrical contacts	012
AUT1600	Inspect timing relay	012
AUT0270	Adjust timing relay	012
AUT2190	Repair electrical coil	012
AUT0660	Clean electrical controller components	012
AUT0350	Assemble/ disassemble controller	012
AUT2390	Repair internal wiring	012
AUT2540	Repair overload heater elements	012
AUT2550	Repair overload relay	012
AUT2230	Repair electrical contacts	012
AKT0470	Conduct current test	012
AUT3230	Test electrical coil	012
AUT3240	Test electrical contractor	012
AZT0350	Test electrical motor starter	012
AUT1660	Inspect vanaxial fan	012
AUT1100	Inspect enclosure fans	012
AUT2320	Repair enclosure fans	012

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 01 May 2017 – 26 May 2017

Location: WP1020 (Haida –Venture)

Board Members:

Surname	Initials	Rank	MOSID	Unit
Walters (Chair)	R.	Lt(N)	00345	HMCS REGINA
Robinson (TDO)	T.	Lt(N)	00211	NTDC(P)
Sprang (TDO)	L.A.	Lt(N)	00211	NTDC(P)
Moazzezi Moghaddam (OPI)	A.	Lt(N)	00345	NTDC(P)
Gagne (NTG Rep)	Y.S.	CPO2	00124	NTG HQ
Semple (Secretary)	T.W.	A/SLt	00207	NFS(P)
Hockey (SME)	M.L.	CPO2	00225	F4 ENG
Moody (SME)	M.E.	CPO2	00125	NTDC(P)
Lambert (SME)	S.J.	CPO2	00367	NTDC(P)
Robinson (SME)	M.J.	CPO2	00124	NTDC(P)
Penney (SME)	K.G.	PO1	00125	NTDC(P)
Lester (SME)	D.I.	PO1	00367	NTDC(P)
Medal (SME)	R.W.	MS	00124	NTDC(P)
Kale (SME)	C.	LS	00367	HMCS VANCOUVER
Osborne (SME)	C.G.	MS	00125	NTDC(P)

Board Objective:

The purpose of this writing board is to develop the QSP for the RQS-AB Machinery Maintainer USQ in the overall context of the Marine Technician trade development MIP. This is the second QSPWB to be convened in the process.

Specification Task – No Train Decisions:

Decisions are captured as Comments in the JTAR where an NT occurred. NT tasks were usually decided on redundancy or they were covered at another level such as NETP.

Discussions/Decisions:

1 May 17

Introduction Brief was conducted by TDO to familiarize everyone with the QSPWB expectations and the outline for how the board will run. Many of the board members had also participated in the QS RQ QSPWB, and continue its development as we are conducting this board. The Board Chair also provided some background on his career, how disputes will be handled, and what the timings will be. After intro brief, the Review Document brief was given. Sorting the PO clusters on the whiteboard commenced. This was done IOT organize all POs so that the board could then analyze each one and decide which components are to be considered No Train (NT). Initial groupings of POs were established by the East coast, and adjusted as required. After lunch, board members continued to decide together if a component of the POs were NT (Refer to JTAR to examine which is NT and for what reason).

HELO Crash Rescue content was removed from AB USQ because needs to be its own standalone USQ. The reasoning is that not every MAR TECH will be required/able to take the training due to physical/medical limitations. Furthermore, the board was given information that Boatswains posted to the interim tanker may require HCRFF training.

Attack Team Leader and Flood Team Leader requirements were drawn from Hull Tech QL5.

2 May 17

Board members discussed what they believe the basic expectations of an AB Maintainer/ Roundsperson should be. This included: responding to ENG emergencies, have more academic knowledge, respond to damage control, improve hand skills and knowledge, being able to work with less supervision, and interpret reading.

The board discussed the requirements to train Breathing Air Systems as a core training requirement vice class specific. It was agreed by all, including the MSE Div Cdr, to train as a core requirement as these systems will be available on all current and future classes.

TDO discussed the importance of referencing others work and paraphrasing information IOT avoid plagiarism

The board discussed and agreed to include the increased academic requirements for Math, Physics, Chemistry, and Thermodynamics requirements under the PO for Maintain Aux Equipment and systems.

Keep PO numbers the same. They can be used as a future reference point if they need to be further discussed or changed.

Admin EO can change name to Perform Maintenance Administration

ATL and FLT should be taught last on this course so they transfer right into HCRFF. Shop stuff in the middle.

When POs and EOs have been defined and typed out the secretary will go around and save them into one document for the board to review

Board members broke off into teams and began scalaring IOT finalize the POs and the EOs. At the same time, the groups also removed anymore components within the POs which they decided to be NT. Board commenced with typing out Chapter 2 of the QSB and getting the POs outlined.

Chapter 1 of the QSB was reviewed and amended together.

3 May 17 – Board recommends that minimum entry requirement for MAR TECHs should be grade 12 due to the challenges with math and sciences, any age or gender, positive attitude and mechanical aptitude, conscience of safety, and critical thinking because they have to be able to solve and analyze problems. Furthermore, the board recommend that future MAR TECHs have a certain number of prerequisite sea days before they are eligible for USQ-AB for a consolidation of knowledge. The board discussed about how many sea days are appropriate for ABs coming into the course A spectrum of attitude, whether positive or negative, is a common problem. Once qualified, ABs are sent ashore to make room for non-qualified members. Furthermore, the discussion of an OJT package was brought up.

MSE Div Cmd addressed the board and discussed about the flexibility to accommodate for PLARs, civilian training, and more.

Board Chair discussed how chemistry, physics, and thermodynamics can become their own separate EOs. They do not need to be placed amongst others IOT facilitate modularization. Scaler is to be built from the bottom to the top. Furthermore, previously taught materials in RQ-OS stay as place holders, and as reminders for more advanced topics covered in the USQ-AB.

The board discussed making sure that all minor details should be placed in teaching points. Doing this means that more labels will have to be added to the whiteboards. However, the TDO mentioned that this will aid in the writing of the master lesson plan at a later date. At some point detail must be added, it can be done now while the board is happening IOT ensure that there are no vague points being taught for future students and that a specific standard can be created.

TAU Maintenance – The Board has decided to leave TAU maintenance training at the AB level and not move it to the OS/LS level as a Class specific USQ. The reasoning behind the decision is that the majority of MAR TECHs will be employed in HFX Class ships. Future AOPS, JSS and CSC classes are expected to have Flight OPS capabilities and will have TAUs of some form as fitted equipment; and

RHIB Maintainer – The Board has decided to incorporate the RHIB Maintainer Course into Small Boat Maintenance within the AB USQ. This training is required for MAR TECHs who will be employed as a Port Security Unit RHIB Maintainer or Upper Deck Equipment Maintainer (former UDER) on a ship, at the AB to MS level.

Some members of the board further discussed certain teaching points and decided whether they should be considered NT.

Board members broke off into teams and commenced scalaring EOs within each PO.
4 May 17- The board debated if certain teaching points placed on the “Parking Lot” were Train or NT.

There is to be a standard that needs to be followed to clean up the EOs and teaching points. This should be a program of Inspect, Repair, and Test/Operate. This will be done IOT take away confusion in the EOs and allow for further teaching points to fall under a certain category. Repair will include: Troubleshot, Assemble/Dis, Adjust, and clean. Test will include: conduct, perform, and operate. Finally, Inspect will include investigation. These sections will also have the task numbers of specific teaching points IOT to further organize the process.

JTAR corrections needed to be made. Some teaching points that were considered NT because of the RQ-OS were not listed in the RQ-OS section of the JTAR. This issue needed to be rectified IOT to figure out which teaching points would remain NT and which would be switched back to train and fitted back into their appropriate POs and EOs. Through this discussion, it came to the boards' attention that certain teaching points were not taught at the RQ-OS, however, the general knowledge of the same task was taught in the RQ-OS, it was just taught under another name or similar name. This is significant because the board discussed how certain topics can be spread out over a long time of training simply due to different paraphrasing of tasks.

5 May 17- The Board attempted to decide where the teaching point "Fluid Coupling" would appropriately fit in. It was decided that it would be NT because it is covered under flex coupling. The board began to write down beside each PO how they see the students demonstrating and applying the PO based on all of the content included within the PO. This will help with the planning of the assessment plan when writing commenced. The TDO mentioned that there needed to be congruency between the title of the PO and the PC of the specific PO. This needs to be done IOT see the connection between what the students will learn and what they will be tested on. Furthermore, the PC needs to focus on the elements that are most important such as safety and security.

The Board decided to make Thermography as a NT due to the fact that it is briefly taught at the OS level. It was decided that while the Thermography teaching point is more in depth, it can be handled at the OS level without issue.

The EO 002.04 *Ships Structure and Hull* was moved to the PO *Maintain Shipboard Components*. The wad due to the fact that the *Ships Structure and Hull* was previously placed in the Admin EO and there is no admin involved in inspecting the hull, the major teaching point.

The board broke off into teams and commenced the writing of the EOs. This was done IOT expedite the process.
8th May 17- under the repair section of the EOs, the word "install" was added for future teaching points. Paragraphs 1-3 of EOs reviewed to ensure common format and layout.

9th May 17 – EO 014.03 titled *Maintain Bearings* was removed and integrated into 014.02 IOT have better flow within the EOs.

PO 007 was put into the parking lot because it needed to be decided whether the entire PO would solely cover the topic of diesel engines or if it would include other types of engine systems.
10th May 17- The board decided that PO 007 would be changed to "Maintain Diesel Engines" because its main focus was directed toward diesel engines.

It was agreed that "Trialing" would be added into the "repair" task.

EO 007.03 would be blended into 007.02 titled *Maintain sub system engine components.*

11th May 17- The board discussed how there should one piece of practical learning equipment for each student on the course. This should be done IOT for all students to learn and test at once. Furthermore, if this situation could be possible this would lead to a reduction in total course time. Additionally, with each student having their own practical training material lead to the discussion of where individual invigilators would be pulled from if this scenario was to take place.

Total course time will be calculated based on 1hr per student. Therefore, if there are 16 students on the course, then that will be equivalent to 16 hours of training.

Teaching points continued to be written and reviewed by the board.

12 May 17- Manual DC plotting will be incorporated into PO 006s 3 EOs IOT maintain a specific skill set. Furthermore, it was decided that TAU maintenance would be moved into USQ AB.

Purifier is used in conjunction with the centrifuge due to the fact that they are interchangeable with one another.

SMEs agreed that the “repair” task will also include the term “replace” if required. This was done because some tasks and teaching points that say “repair” are completed by replacing the specific material.

15 May 17 – The board continued to write the teaching points out for each EO.

The board went through completed para 10s for PO 001 and 014 IOT establish the standard for each para 10s would look like. Furthermore, it gave the board the chance to bring up any questions or concerns regarding formatting or issues within the teaching points.

Secretary began to put together Annex A with board chair.

If there is specific instructions or considerations for a particular training point it can be added in the para. 9 “remarks”.

16 May 17- The board decided that Critical ECs for academics will get a percentage mark where everything else will be a Pass or Fail.

It was discussed that recommendations can be made by the Snr. Instructor on the trainee course reports that can reflect what training stream (electrical or mechanical) the trainee can go into based on their aptitude throughout the course.

“RAS Trainer” was added into the Training Aids Portion in EO 002.02 para 6.

Teaching Points must match the headings that are written in Para 3.

EO 007.01 titled *Maintain enclosure components* was to be listed as NT because the trainees will learn that through experience. The teaching point A1 *Inspecting vanaxial fan* (AUT1660), A2 *Inspecting enclosure fans* (AUT1100), and B3 *Repairing enclosure fans* (AUT2320) are all being moved to EO 013.02.

17 May 17 – Members of the board had a meeting with the civilian academic advisor to discuss the math, chemistry, physics, thermodynamics, and rotating machinery and electronic circuit analysis portions of the training. The civilian adviser said that he would look over the teaching points and provide feedback to the board by early next week.

The board decided that there needed to be two tables made for course timings. The first table will display course timing with the current resources. The second table will be created with ideal conditions. This will be done IOT display the effects of course timings based on resources.

PO 003, 005, and 006 para 10s were reviewed by the board.

18 May 17- The board decided that PO 007 will be covered in PO 011 *Small Boat Maintenance.*

EO 002.02, 013, and 009 were reviewed by the board IOT examine the teaching points and make sure the points are appropriate for the respective EO.

It was emphasised that timings need to be put in minutes so that there can be an idea of how long the course will take to complete.

19 May 17- The board continued to work on para 10s for their respective EOs.

24 May 17- It was decided that the max load for the AB course will be 12, the same as the optimum. This was decided because The RQ OS course follows civilian standard for safety practices for the practical phases (ratio of 1:12). Therefore, resource implications would need to be considered if running the course above max load.

There is no available equipment at the school to perform practical learning from PO 008. EOs will have to be theoretical unless the necessary equipment is purchased. Furthermore, tours, will have to be conducted on any available HMCS with functioning equipment.

25th May 17- An unforeseen issue with internet connection for two days (24th and 25th) posed an issue for the board as external communication was unavailable making it difficult for work to be passed on or reviewed.

PO 007.04 was NT because it was previously covered in the RQ-OS.

Module 5 was separated into two MODS to accommodate reservists.

26th May 17- CECs are required for academic EOs to ensure that the required knowledge is gained for critical thinking.

Delta training will be conducted after the board.

Due to availability of personnel, HCRFF will be conducted after the board.

PO 001 Perform Tech Admin is now to be included in MOD 6.

29th May 17- MSE Div Cmd and the board reviewed the Delta training and Mods for the AB course. It was stressed that the USQ AB needs to be thoroughly looked through IOT determine what can be cut out if it is deemed not necessary and what items, if any, needs to be added to the document.

30th May 17- Victoria Day

31st May 17- The board reviewed the how many instructors and invigilators were needed for each EO within each module. This was done IOT ensure that the proper number of instructors and invigilators would be ready to teach and test future trainees.

HCRFF PO continued to be discussed and created between members of the board.

EO 012.02 was changed from 700 minutes to 650 minutes total.

1st June 17:

Lessons Learned:

See above comments

Guidance to Instructional Staff:

See applicable EOs in respective POs.

At this time, no OJPR is planned for this QSP. However, it was acknowledged that the required number, type, and design of the requisite trainers may not be available at the Fleet Schools. As such, it is recommended to use the ships themselves as trainers where necessary, or for the procurement of even partial trainers as a high priority for the schools, rather than shunting large amounts of material to a QSP. This will reduce the training burden on the fleet as well as prepare trainees for the subsequent class package training.

Timings for testing are indicated in time per student. What that means is that a given EC or PC, if it is noted that 'you have 50minutes for this test' that means that is the time allocated for one student to complete the test. This was to accommodate the difference in training facilities. The scheduling authority for each school will be required to accommodate the individual facilities. To continue the example, in the case of a 50 minutes/single period test, if a school has five lab benches the test will require 50 minutes for five students, 100 minutes for ten students, etc.

5 June 2018

Reviewed EO 003.03, Maintain Portable Rescue Equipment. Removed reference to Gas Powered Rescue Saw from para 3 a. (2), b. (2), c. (2) as well as listed item of training aids. Also removed from TP1, TP2 and TP3 all reference to the Gas Powered Rescue Saw. Reference item is no longer required kit held onboard and has been removed from the Sea Training Guide and required equipment as part of the air readiness inspection.
PO1 Shawn Mann NTDC CCD analyst.

4 July 2018

Removed item from EO 004.02 TP1 b (4) minor bulkheads. This information is already covered in EO 004.02 TP1 a (7).
PO1 Shawn Mann NTDC CCD analyst.

17 Jan 2019

After the review process from the NFS (pre-pilot) was completed, many changes where applied to the QSP, such as:

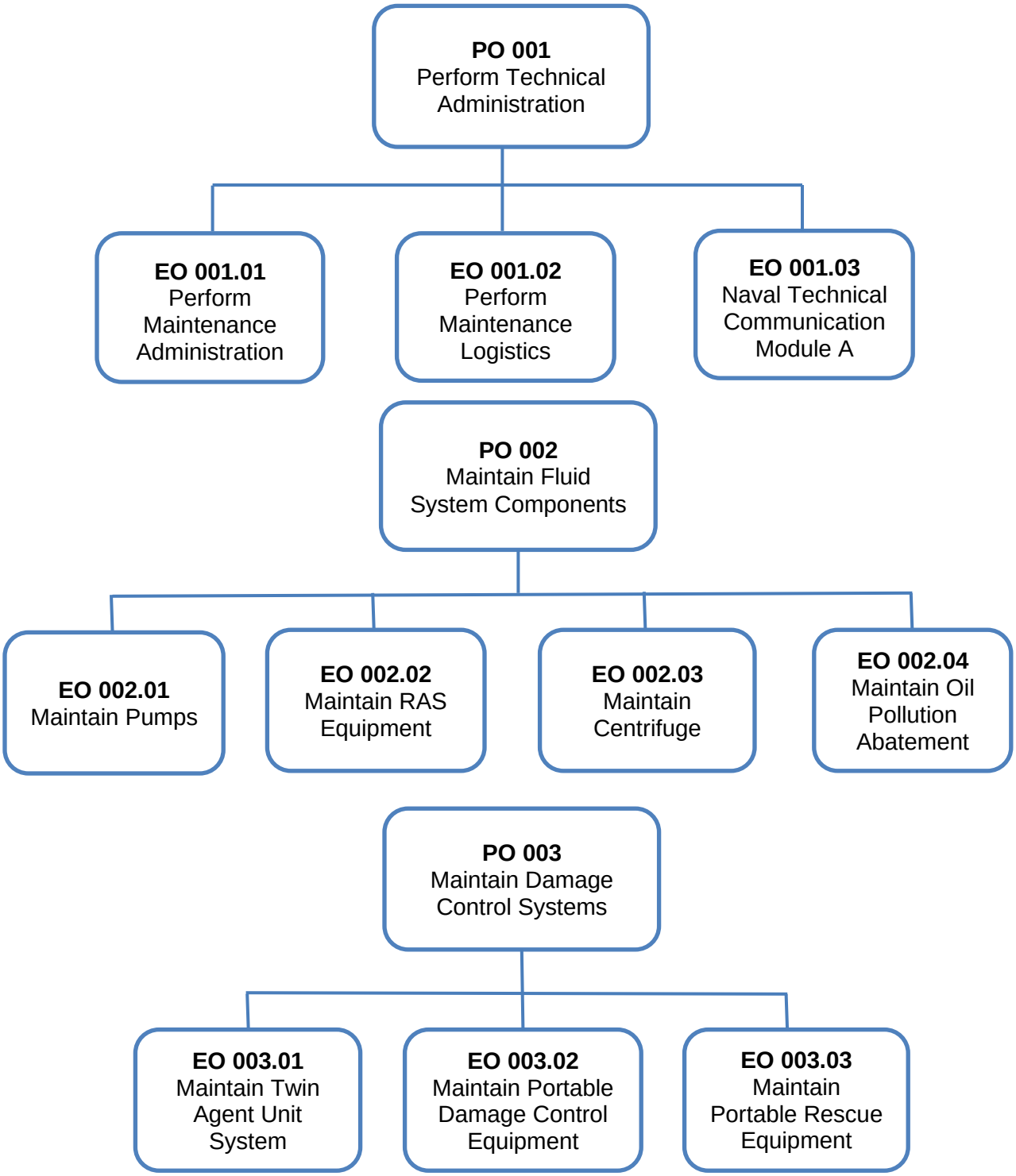
a. Re-formatted all EO/PO's for consistency;
b. Changed the EC's within PO 003;
c. Reversed the order for EO 004.01 and EO 004.02;
d. Removed PO 006 and placed content within the new EO 005.03 – Lead a flood control team.
e. Added EO 005.04 – Lead a Casualty Power team.
f. Re-worked PO 009 Perform small boat maintenance, adding an additional EO.
g. Changed many TP within the 6 academic EO's, resulting in larger timings.

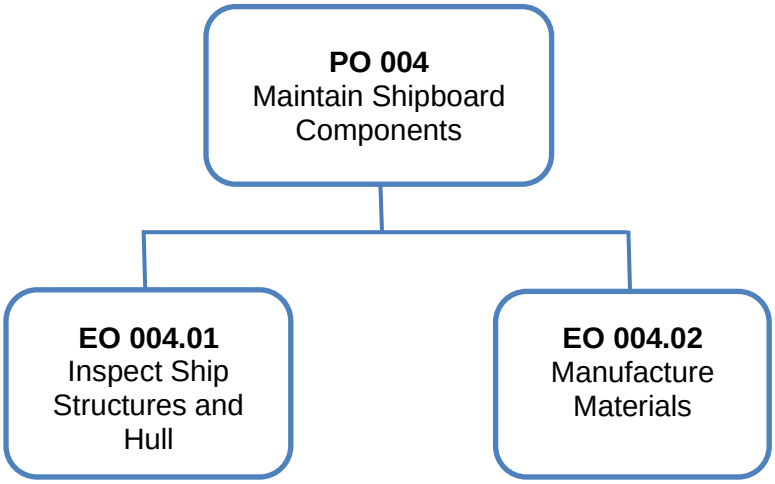
Follow Up Action:

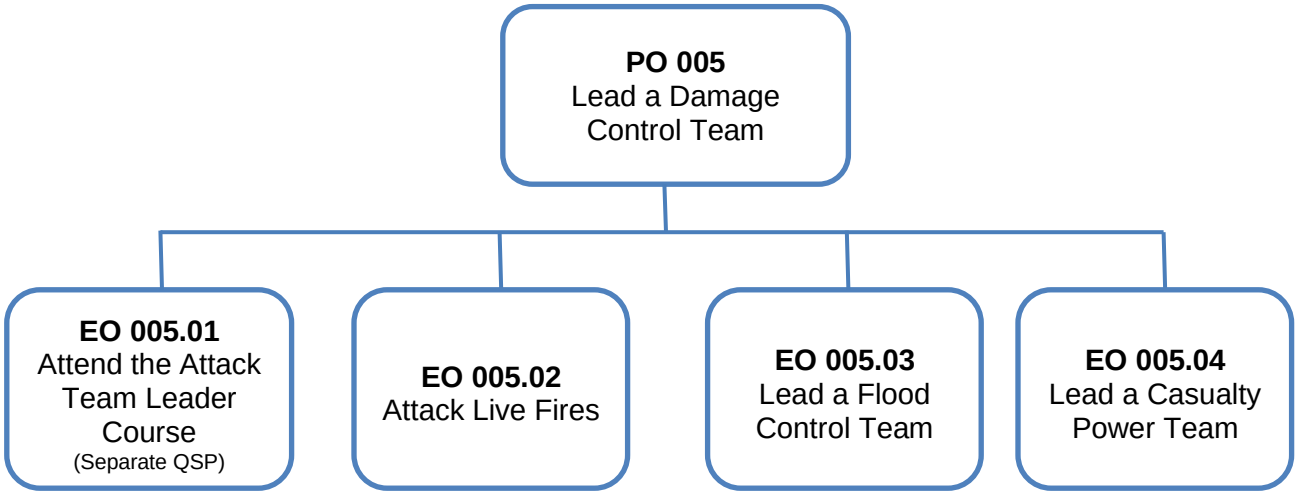
Item	Action	OPI	Expected Date of Completion
1			
2			

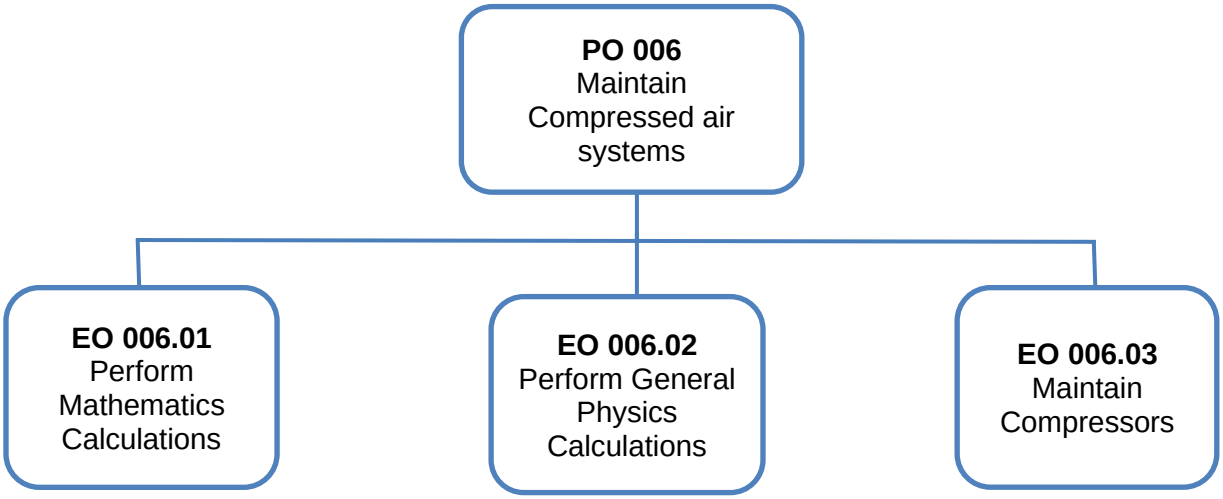
Other Considerations (if applicable):

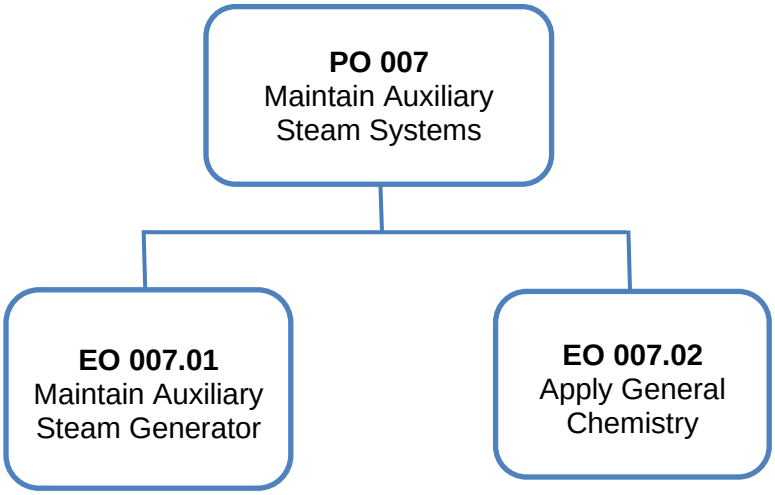
ANNEX D - SCALARS

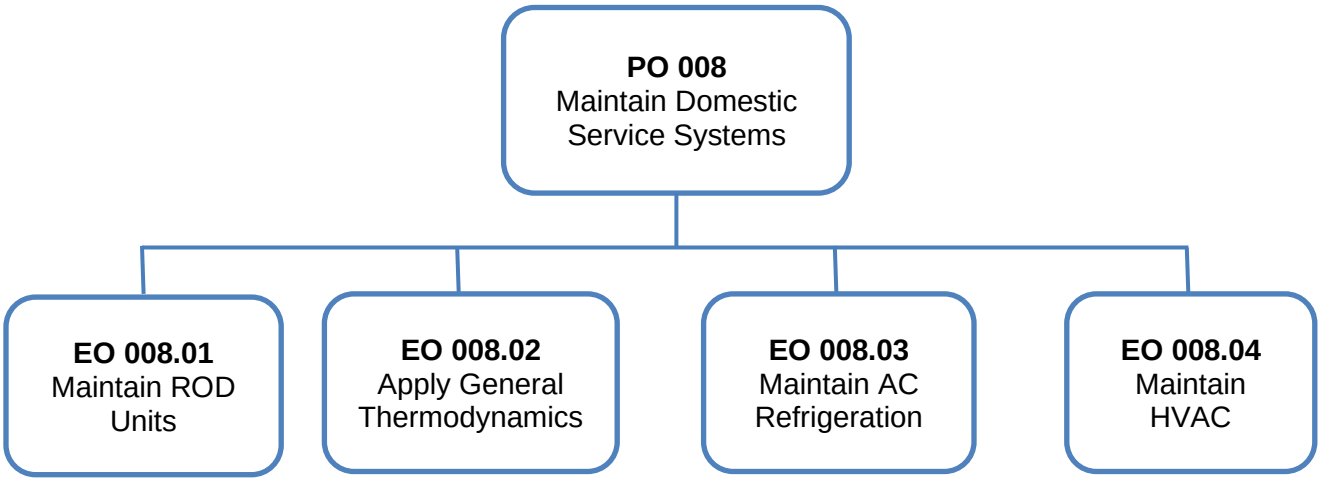


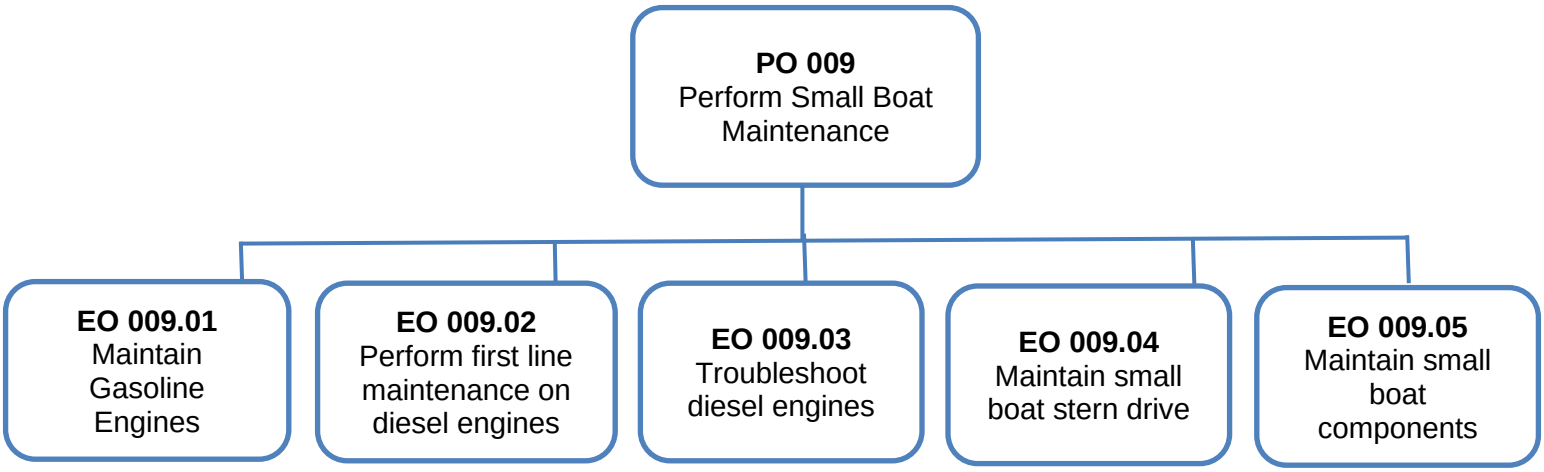

PO 004
Maintain Shipboard Components
EO 004.01
Inspect Ship Structures and Hull
EO 004.02
Manufacture Materials

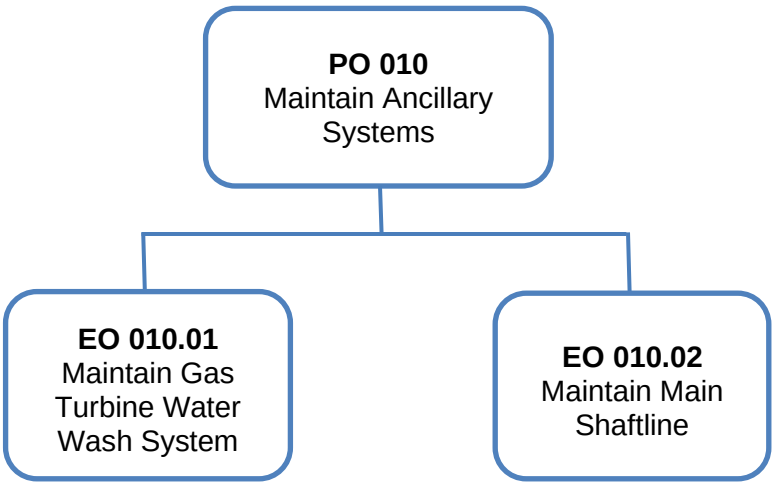

PO 005
Lead a Damage Control Team
EO 005.01
Attend the Attack Team Leader Course
(Separate QSP)
EO 005.02
Attack Live Fires
EO 005.03
Lead a Flood Control Team
EO 005.04
Lead a Casualty Power Team

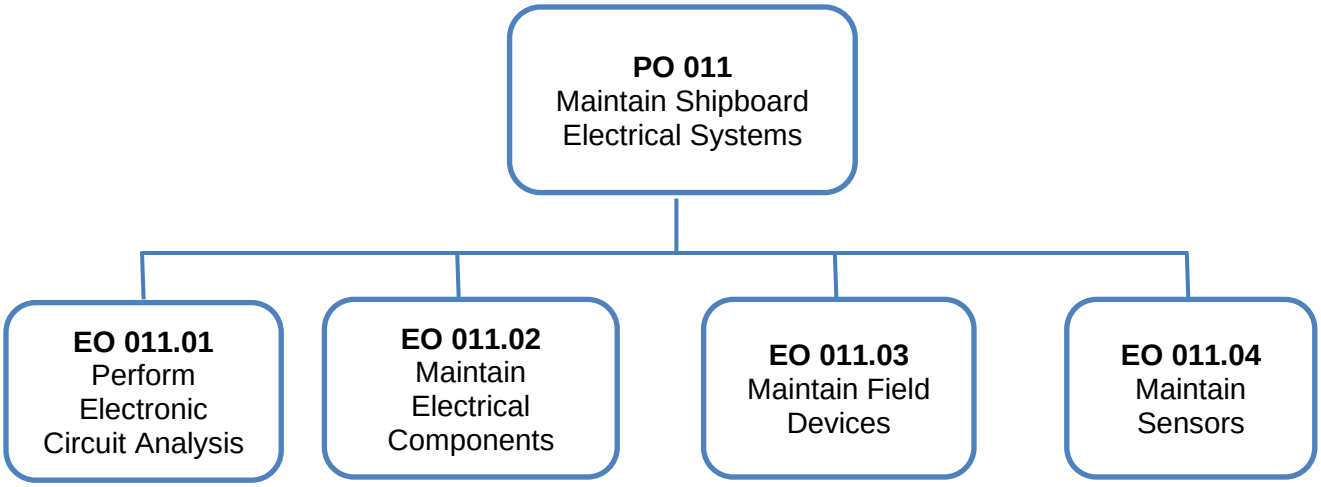

PO 006
Maintain Compressed air systems
EO 006.01
Perform Mathematics Calculations
EO 006.02
Perform General Physics Calculations
EO 006.03
Maintain Compressors

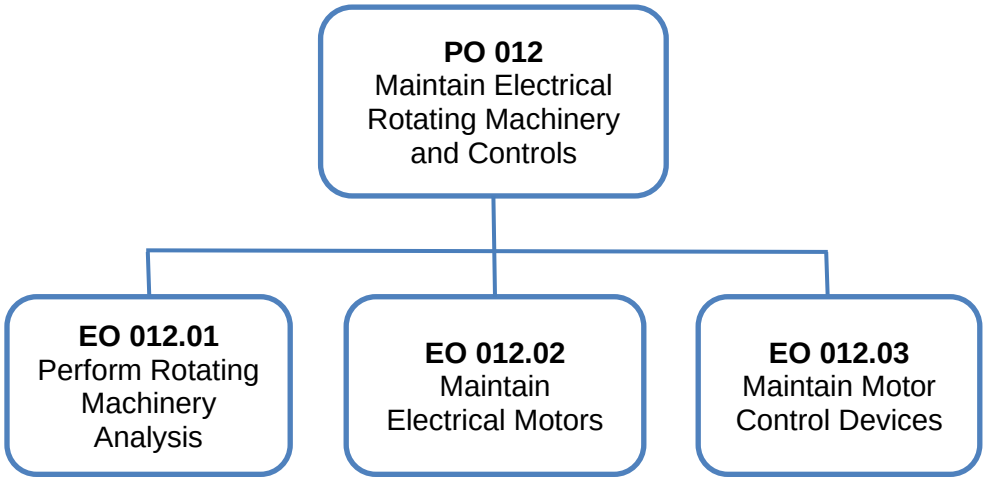

PO 007
Maintain Auxiliary Steam Systems
EO 007.01
Maintain Auxiliary Steam Generator
EO 007.02
Apply General Chemistry


PO 008
Maintain Domestic Service Systems
EO 008.01
Maintain ROD Units
EO 008.02
Apply General Thermodynamics
EO 008.03
Maintain AC Refrigeration
EO 008.04
Maintain HVAC


PO 009
Perform Small Boat Maintenance
EO 009.01
Maintain Gasoline Engines
EO 009.02
Perform first line maintenance on diesel engines
EO 009.03
Troubleshoot diesel engines
EO 009.04
Maintain small boat stern drive
EO 009.05
Maintain small boat components


PO 010
Maintain Ancillary Systems
EO 010.01
Maintain Gas Turbine Water Wash System
EO 010.02
Maintain Main Shaftline


PO 011
Maintain Shipboard Electrical Systems
EO 011.01
Perform Electronic Circuit Analysis
EO 011.02
Maintain Electrical Components
EO 011.03
Maintain Field Devices
EO 011.04
Maintain Sensors


PO 012
Maintain Electrical Rotating Machinery and Controls
EO 012.01
Perform Rotating Machinery Analysis
EO 012.02
Maintain Electrical Motors
EO 012.03
Maintain Motor Control Devices

ANNEX E – ON JOB PERFORMANCE RECORD (OJPR)

1. There is no formal OJPR associated to the USQ-AB Marine Technician Machinery Maintainer (ALMI) (MITE ID 121388) qualification.

ANNEX F – ACRONYMS AND ABBREVIATIONS

Acronym and Abbreviation	Term or Definition
AC	Air Conditioner
AC or I_{AC}	Alternating Current
ACOS	Automatic Change Over Switch
Admin	Administration
AFFF	Aqueous Film Forming Foam
ACB	Air Insulated Circuit Breaker
BAM	Bus Access Module
BDCS	Battle Damage Control System
BW	Black Water
CBRN	Chemical Biological Radiological Nuclear
CIP	Clean In Place
DA	Diesel Alternator
DC	Damage Control
DC or I_{DC}	Direct Current
DG	Diesel Generator
DMP	Designated Maintenance Period
DP	Demo-Performance
DRMIS	Defence Resource Management Information System
DWP	Docking Work Period
EC	Engineering Change
ECWP	Engineering Change Work Period
EEBD	Emergency Escape Breathing Device
EOTIs	Engineering Officer’s Technical Instructions
EWP	Extended Work Period
FW	Fresh Water
FA	First Aid
FFE	Fire Fighting Ensemble
FOS	Fuel Oil Services
FOT	Fuel Oil Transfer
FT	Field Trip
FTOIs	Fleet Technical Officer’s Instructions
GD	Guided Discussion
GT	Gas Turbine
GTOIs	Group Technical Officer’s Instructions
GW	Grey Water

HMI	Human Machine Interface
HP	High Pressure
HVAC	Heating, Ventilation, and Air Conditioning
HPR	High Priority Requisition
HZ	Hertz or Cycles
LCPA	Local Cash Procurement Account
LOGREQ	Logistics Request
LP	Low Pressure
LPO	Local Purchase Order
IPMS	Integrated Platform Management System
MA	Motor Alternator
IL	Interactive Lecture
MAF	Maintenance Action Form
MCB	Moulded Circuit Breaker
MCOS	Manual Change Over Switch
MLO	Main Lube Oil
MWM	Mann Diesel Engine
ODS	Ozone Depleting Substance
OPDEF	Operational Deficiency
NATO	North Atlantic Treaty Organization
NSN	NATO Stock Number
PDE	Pielstick Diesel Engine
PG&D	Power Generation & Distribution
PIF	Pre-installation Failure
PLC	Programmable Logic Controller
PP	Pump
PPE	Personal Protective Equipment
RAS	Replenishment At Sea
ROD	Reverse Osmosis Desalinator
RTU	Remote Terminal Unit
SA	Study Assignment
SFC	Static Frequency Converter
SI	Summary Investigation
SITREP	Situation Report
SLoc	Storage Location (Former SCA/MAST)
SS	Self-Study
SSS	Synchronous Self Shifting
SW	Sea Water
SWB	Saltwater Ballast
SWBD	Switchboard
SWP	Short Work Period

TAU	Twin Agent Unit
TI	Technical Investigation
TIC	Thermal Imaging Camera
TRANREQ	Transfer Requisition
UCR	Unsatisfactory Condition Report
UPS	Uninterruptable Power Supply
WEAF	Work Estimate Action Form
WTD	Water Tight Door
VAC	Voltage, Alternating

ANNEX G - SUGGESTED SEQUENCES AND TIMINGS

1. The USQ AB course is subject to limiting factors that may change the overall length of the course. These limiting factors are as follows:

   a. Number of Trainees;
   b. Number of Instructors; and
   c. Number of Equipment/Facilities.

2. Below is a table based on a ratio of one (1) trainee to one (1) instructor to one (1) piece of equipment/facility. As seen below, when the number of trainee's increase, while the other two (2) factors remain the same, the timings will increase. Optimally, all three (3) factors should increase equally.

3. The suggested sequence for delivering the training is as follows:

PO / EO	PO / EO Title		Multiplier based on number of Trainees												Limiting Factor
			1	2	3	4	5	6	7	8	9	10	11	12	
PO 001 Perform Technical Administration															
EO 001.01	Perform Maintenance Administration	EO	175												
		EC	0												
EO 001.02	Perform Logistical Administration	EO	340												
		EC	0												
EO 001.03	Naval Technical Communication Module A	EO	2150												
		EC	0												
Perform Technical Administration		PC	75												
PO 001 Total:			2740												
PO 002 Maintain Fluid System Components															
EO 002.01	Maintain Pumps	EO	980												
		EC	15	30	45	60	75	90	105	120	135	150	165	180	Inst / Pumps
EO 002.02	Maintain Replenishment at Sea (RAS) Equipment	EO	500												
		EC	20	40	60	80	100	120	140	160	180	200	220	240	Inst / Equip

EO 002.03	Maintain Centrifuges	EO	1425												
		EC	25	50	75	100	125	150	175	200	225	250	275	300	Inst/ Centrifuge
EO 002.04	Maintain Oil Pollution Abatement Systems	EO	500												
		EC	50	100	150	200	250	300	350	400	450	500	550	600	Inst / Equip
Maintain Fluid System Components		PC	0												
PO 002 Total:			3515	3625	3735	3845	3955	4065	4175	4285	4395	4505	4615	4725	
PO 003 Maintain Damage Control Systems															
EO 003.01	Maintain Twin Agent Unit System	EO	675												
		EC	50	100	150	200	250	300	350	400	450	500	550	600	Inst / TAU
EO 003.02	Maintain Portable Damage Control Equipment	EO	550												
		EC	30	60	90	120	150	180	210	240	270	300	330	360	Inst / DC Equip
EO 003.03	Maintain Portable Rescue Equipment	EO	650												
		EC	30	60	90	120	150	180	210	240	270	300	330	360	Inst / Rescue Equip
Maintain Damage Control Systems		PC	0												
PO 003 Total:			1985	2095	2205	2315	2425	2535	2645	2755	2865	2975	3085	3195	
PO 004 Maintain Shipboard Components															
EO 004.01	Inspect Ships Structures and Hull	EO	600												
		EC	50												
EO 004.02	Manufacture Materials	EO	3000												
		EC1	50												
		EC2	50												
		EC3	800												
		EC4	200												
		EC5	250												
Maintain Shipboard Components		PC1	700												
		PC2	350												
		PC3	200												
PO 004 Total:			6250												
PO 005 Lead a Damage Control Team															
EO 005.01	Attack Team Leader (ATL) Course	EO	800												
		EC	0												
EO 005.02	Attack Live Fires	EO	400												

		EC	0												
EO 005.03	Lead a Flood Control Team	EO	875												
		EC	50	100	150	200	250	300	350	400	450	500	550	600	Inst / Facility
EO 005.04	Lead a Casualty Power Team	EO	700												
		EC	50	100	150	200	250	300	350	400	450	500	550	600	Inst / Facility
Lead a Damage Control Team		PC	0												
PO 005 Total:			2875	2975	3075	3175	3275	3375	3475	3575	3675	3775	3875	3975	
PO 006 Maintain Compressed Air Systems															
EO 006.01	Apply General Mathematics	EO	2875												
		ECR	50												
		EC	100												
EO 006.02	Perform General Applied Physics	EO	2845												
		ECR	200												
		EC	100												
EO 006.03	Maintain Compressed Air System Components	EO	2200												
		EC	75	150	225	300	375	450	525	600	675	750	825	900	Inst / Compressors
Maintain Compressed Air Systems		PC	0												
PO 006 Total:			8445	8520	8595	8670	8745	8820	8895	8970	9045	9120	9195	9270	
PO 007 Maintain Auxiliary Steam Systems															
EO 007.01	Maintain Auxiliary Steam Generator	EO	1025												
		EC	15	30	45	60	75	90	105	120	135	150	165	180	Inst / Steam Generator
		CEC	50	100	150	200	250	300	350	400	450	500	550	600	
EO 007.02	Perform General Chemistry	EO	1685												
		ECR	50												
		EC	50												
Maintain Auxiliary Steam Systems		PC	0												
PO 007 Total:			2875	2940	3005	3070	3135	3200	3265	3330	3395	3460	3525	3590	
PO 008 Maintain Domestic Service Systems															
EO 008.01	Maintain Reverse Osmosis Desalination	EO	565												
		EC	0												
EO 008.02	Apply General Thermodynamics	EO	2955												
		ECR	100												
		EC	100												

EO 008.03	Maintain Refrigeration Components	EO	545												
		EC	0												
EO 008.04	Maintain Heating, Ventilation and Air Conditioning	EO	625												
		EC	50												
Maintain Domestic Service Systems		PC	0												
PO 009 Total:			4940												
PO 009 Perform Small Boat Maintenance															
EO 009.01	Maintain Small Boat Gasoline Engines	EO	340												
		EC	25	50	75	100	125	150	175	200	225	250	275	300	Inst / Gasoline Eng
EO 009.02	Perform First Line Maintenance on Small Boat Diesel Engines	EO	200												
		EC1	50												
		EC2	100	200	300	400	500	600	700	800	900	1000	1100	1200	Inst / Diesel
EO 009.03	Troubleshoot Small Boat Diesel Engines	EO	250												
		EC	20	40	60	80	100	120	140	160	180	200	220	240	Inst / Diesel
EO 009.04	Maintain Small Boat Stern Drive	EO	300												
		EC1	50												
		EC2	30	60	90	120	150	180	210	240	270	300	330	360	Inst / Stern Drive
EO 009.05	Maintain Small Boat Components	EO	415												
		EC1	50												
		EC2	100	200	300	400	500	600	700	800	900	1000	1100	1200	Inst / Equip
Perform Small Boat Maintenance		PC	50	100	150	200	250	300	350	400	450	500	550	600	Inst / Equip
PO 009 Total:			1980	2305	2630	2955	3280	3605	3930	4255	4580	4905	5230	5555	
PO 010 Maintain Ancillary Systems															
EO 010.01	Maintain Gas Turbine Water Wash System	EO	300												
		EC	0												
EO 010.02	Maintain Main Shaftline	EO	480												
		EC	75	150	225	300	375	450	525	600	675	750	825	900	Inst / Equip
Maintain Ancillary Systems		PC	0												
PO 010 Total:			855	930	1005	1080	1155	1230	1305	1380	1455	1530	1605	1680	
PO 011 Maintain Shipboard Electrical Systems															
EO 011.01	Perform Electrical Circuit Analysis	EO	2450												
		ECR	50												

		EC	100												
EO 011.02	Maintain Electrical Components	EO	650												
		EC	50	100	150	200	250	300	350	400	450	500	550	600	Inst
EO 011.03	Maintain Field Devices	EO	600												
		EC	50	100	150	200	250	300	350	400	450	500	550	600	Inst / Equip
EO 011.04	Maintain Sensors	EO	900												
		EC	50												
Maintain Shipboard Electrical Systems		PC	0												
PO 011 Total:			4900	5000	5100	5200	5300	5400	5500	5600	5700	5800	5900	6000	
PO 012 Maintain Electrical Rotating Machinery and Control															
EO 012.01	Perform Rotating Machinery Analysis	EO	3700												
		ECR	50												
		EC	100												
EO 012.02	Maintain Electric Motors	EO	825												
		EC	100	200	300	400	500	600	700	800	900	1000	1100	1200	Inst / Equip
EO 012.03	Maintain Motor Control Devices	EO	1250												
		EC	50												
Maintain Electrical Rotating Machinery and Control		PC	100	200	300	400	500	600	700	800	900	1000	1100	1200	Inst / Equip
PO 012 Total:			6175	6375	6575	6775	6975	7175	7375	7575	7775	7975	8175	8375	
Overall Total:			47535	48695	49855	51015	52175	53335	54495	55655	56815	57975	59135	60295	

ANNEX H - TRAINING RESOURCES

1. The following is a list of the resources identified as being required to conduct the Marine Technician Machinery Maintainer Course. Some of the resources listed currently exist at the Naval Fleet Schools; however, there are new resources that will be required to maximize the efficiency of training delivery and testing. Purchasing the new items will increase the quality of the student graduating from the training, and significantly reduce the number of required training days ultimately saving time and money over the long-term.

2. The following is a list of classroom requirements:

   a. One large classroom to accommodate maximum number of candidates which include:

      (1) Individual candidate's desks c/w chair;
      (2) Smart Board;
      (3) Dry erase board;
      (4) Projector (power point capable) & screen;
      (5) BluRay Player & TV;
      (6) Podium;
      (7) DWAN / LAN connections;
      (8) Laptops/PCs;
      (9) Calculators;
      (10) PPS; and
      (11) Student handouts and references.

   b. Diesel Classroom to accommodate maximum number of trainees, to include:

      (1) XXXXXX

3. The following workshops will be required to support the practical aspects of the training:

   a. One electrical workshop and one lab volt lab, to include:

      (1) PPE;
      (2) Appropriate work bench and seating
      (3) Bearing pullers;
      (4) Heat guns;
      (5) Lab-Volt Trainers.
      (6) Motors for bearing change
      (7) Stepped machined shafts for bearing changes
      (8) Pin connector kits
      (9) Pin insertion Tools
      (10) Crimping tool
      (11) Canon plugs
      (12) Field devices for students to include but not limited too;
         (a) RTD
         (b) Bimetallic strip

(c) Thermometer
(d) Thermocouple
(e) Thermistors
(f) flow devices
(g) Pressure Sensors
(h) Level Sensors
(i) Position Sensors
(j) Magnetic Pick-ups

(13) Field devices for students to include but not limited too;
(a) Rheostats
(b) Macro switches
(c) Differential switches
(d) Solenoids
(e) Limit switches

(14) ROXTEC Kits
(15) Various tools

b. One woodworking shop, to include:

(1) PPE;
(2) Woodworking benches power equipped to accommodate maximum number of trainees fitted with a woodworking vice;
(3) Storage arrangement for building materials;
(4) band-saws;
(5) drill-presses;
(6) Pedestal grinders;
(7) Pedestal sanders;
(8) Stowage area for wood;
(9) Radial arm saws;
(10) Table saws;
(11) Thickness planer;
(12) Jointer;
(13) Shaper;
(14) Mortise & tendon machine;
(15) Routers & tables;
(16) Collection system;
(17) Honing grinders; and
(18) Sander.

c. One Oxyacetylene shop to include:

(1) PPE;
(2) Welding benches to safely accommodate maximum number of trainees fitted with a metal vice and a weld metal support bracket;
(3) complete oxyacetylene welding sets to accommodate maximum number of trainees;
(4) Cylinders of both oxygen & acetylene to accommodate maximum number of trainees;
(5) Bench grinders;
(6) Power hacksaw;
(7) Guided bend test machine shared with Arc Welding;

(8) Stowage area for metal;
(9) exhaust system for fumes;
(10) Cutting benches;
(11) complete oxyacetylene cutting sets to accommodate maximum number of trainees;
(12) rota-belt sanders;
(13) Oxyacetylene cutting guide (radiograph) cutters; and
(14) One tonne electric hoist.

d. One metalwork shop to include:

(1) PPE;
(2) Metalwork benches to accommodate maximum number of trainees fitted with one metal vice each;
(3) rota-belt sander;
(4) drill-presses;
(5) Power hacksaw;
(6) Metal worker;
(7) Furnace;
(8) Anvils;
(9) Bench grinders;
(10) Exhaust system for fumes; and
(11) Bench fitted with metal working vices.

e. One paint shop to accommodate maximum number of trainees, to include:

(1) PPE;
(2) Work benches;
(3) Ventilated paint mixing area;
(4) Paint mixing machine;
(5) Spray booths;
(6) Paint sprayers;
(7) Exhaust system for fumes; and
(8) Classroom to accommodate maximum number of trainees, for theory portion.

f. Refrigeration shop to accommodate maximum number of trainees, to include:

(1) PPE;
(2) Suitable workbench;
(3) Sufficient Refrigeration Trainers;
(4) complete acetylene brazing sets to accommodate maximum number of trainees
(5) 2x R134a cylinders and storage;
(6) 2x Nitrogen cylinders and storage;
(7) 2x Acetylene cylinders and storage

g. Diesel shop to accommodate maximum number of trainees, to include:

(1) PPE;
(2) Suitable work benches;
(3) Hazmat Spill/Clean-up materials and appropriate receptacles;

(4) HAZMAT storage for consumable Oils and Lubricants; and
(5) Suitable storage for consumable parts.

h. Fitted Fire Fighting Systems shop to accommodate maximum number of trainees to perform PM training on the Twin Agent Unit.

i. Pump training classroom or workshop to accommodate maximum number of trainees, to include:

(1) PPE;
(2) Suitable workbenches with pump for assembly/disassembly;
(3) Pump trainee platform;
(4) Working Reverse Osmosis Distillation (ROD) plant for assembly/disassembly;
(5) Working steam generator for assembly/disassembly; and
(6) Toolbox with appropriate tools.

j. Workshop for oil treatment and processing to accommodate maximum number of trainees, to include:

(1) PPE;
(2) Centrifuges for assembly/disassembly;
(3) Hydromem for assembly/disassembly; and
(4) Toolbox with appropriate tools.

4. Library facilities to include references and accountability tracking, to accommodate sufficient textbooks for two USQ AB serials running concurrently (30x each textbook) for the following subjects, to include:

a. Math;

b. Physics;

c. Chemistry;

d. Thermodynamics;

e. Electric Circuit Analysis; and

f. Electric Field Device/Controls.

5. Access to outside resources will be required for the following, to include:

a. Damage Control Training Facilities (Kootenay & Galiano), in support of EO 005.01 and PO 006;

b. Civilian Fire Fighting School, in support of Live Fire requirements in EO 005.02;

c. Static RAS Trainer in support of EO 002.02; and

d. RHIBs located at Boatshed in support of PO 01

ANNEX I – COSTING

Course:	AB USQ					
TD	No. Of Students/ Instructors	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Not Applicable						
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)	Initial TE Set-up	Each successive course	5 Year total
Classroom/Stationary Supplies						
- DWAN Connected Laptops	12	$1,800.00	1	$21,600.00		$21,600.00
- Pens/Pencils/Erasers/Paper/etc.	1	$400.00	1	$400.00	$400.00	$2,000.00
			Sub-Total:	$22,000.00	$400.00	$23,600.00
Instructor Stationary Supplies						
- DWAN Connected Laptops	2	$1,800.00	1	$3,600.00		$3,600.00
- Photo copier	1	$1,968.00		$1,968.00		$1,968.00
- Pens/Pencils/Erasers/Paper/etc.	1	$150.00	1	$150.00	$150.00	$750.00
- Printer for 11' x 17' paper	1	$2,500.00	1	$2,500.00		$2,500.00
- Paper 11x17 per pkg of 500	2	$15.00	1	$30.00	$30.00	$150.00
			Sub-Total:	$8,248.00	$180.00	$8,968.00
References						
- Math Textbooks	30	$150.00		$4,500.00		$4,500.00
- Physics Textbooks	30	$150.00		$4,500.00		$4,500.00
- Chemistry Textbooks	30	$150.00		$4,500.00		$4,500.00
- Thermodynamics Textbooks	30	$150.00		$4,500.00		$4,500.00
- Electric Circuit Analysis Textbooks	30	$200.00		$6,000.00		$6,000.00
- Field Device/Controls Textbooks	30	$200.00		$6,000.00		$6,000.00
- Miscellaneous References	30	$100.00		$3,000.00		$3,000.00
			Sub-Total:	$33,000.00	$0.00	$33,000.00
Diesel Classroom Supplies						

- No additional requirements				$0.00		0
			Sub-Total:	$0.00	$0.00	$0.00
Electrical Workshop						
- Bearings	24	$4.99	1	$119.76	$119.76	$598.80
- Bearing pullers	12	$26.99	1	$323.88		$323.88
- Heat guns	6	$29.99	1	$179.94		$179.94
- ROXTEX Gland CF 24 Kit	12	$217.21	1	$2,606.52	$2,606.52	$13,032.60
- Multi core wire 12 gauge (100 ft.)	1	$195.00	1	$195.00	$195.00	$975.00
- Additional LabVolt Trainers	12					
- Lab Volt supplies and Maintenance	5	$100.00	1	$500.00	$500.00	$2,500.00
- Pin connectors	12	$40.69	1	$488.28	$488.28	$2,441.40
			Sub-Total:	$4,413.38	$3,909.56	$20,051.62
Woodworking Shop Supplies						
- Softwood (2"x4"x8')	114	$3.99	1	$454.86	$454.86	$2,274.30
- Softwood (1"x3"x8')	4	$2.53		$10.12	$10.12	$50.60
- Wood Screws, various sizes	1	$150.00	1	$150.00	$150.00	$750.00
- Plywood (4'x8'x0.5")	13	$24.22		$314.86	$314.86	$1,574.30
			Sub-Total:	$929.84	$929.84	$4,649.20
Oxyacetylene Shop Supplies						
- Oxygen	1	$100.00		$100.00	$100.00	$500.00
- Acetylene	1	$100.00		$100.00	$100.00	$500.00
			Sub-Total:	$200.00	$100.00	$1,000.00
Metalwork Shop Supplies						
- Aluminum (6061) (0.125" x 1.0" x 30")	1	$20.40		$20.40	$20.40	$102.00
- Aluminum (6061) (0.25" x 1.0" x 30")	1	$39.34		$39.34	$39.34	$196.70
- Sheet, Cold Roll (0.125" x 1.0" x 30")	1	$3.99		$3.99	$3.99	$19.95
- Sheet, Cold Roll (0.25" x 1.0" x 30")	1	$55.19		$55.19	$55.19	$275.95
			Sub-Total:	$118.92	$118.92	$594.60
Refrigeration Shop Supplies						
- Refrigeration Trainer	2					
- ACR copper tubing ¼", 3/8", ½", 5/8"	12	$25.00		$300.00	$300.00	$1,500.00
- Brazing rods	12	$16.00		$192.00	$192.00	$960.00
- silver solder	12	$8.00		$96.00	$96.00	$480.00
- rosin core solder	12	$4.00		$48.00	$48.00	$240.00

- brazing flux	12	$5.00		$60.00	$60.00	$300.00
- assorted copper fittings	12	$40.00		$480.00	$480.00	$2,400.00
- assorted brass fittings	12	$40.00		$480.00	$480.00	$2,400.00
- vacuum pump oil	12	$8.00		$96.00	$96.00	$480.00
- polyol ester compressor oil	12	$5.00		$60.00	$60.00	$300.00
- snoop leak detect solution	12	$5.00		$60.00	$60.00	$300.00
- R134a refrigerant, <2lbs	12	$1.00		$12.00	$12.00	$60.00
- electrical components/wiring	12	$16.00		$192.00	$192.00	$960.00
- nitrogen, 1-2 cylinders large	1	$100.00		$100.00	$100.00	$500.00
- acetylene, 2-3 cylinders half size-pony	1	$100.00		$100.00	$100.00	$500.00
			Sub-Total:	$2,276.00	$2,276.00	$11,380.00
Diesel Shop Supplies						
- Racor Filter (4330-01-188-3158)	6	$8.00		$48.00	$48.00	$240.00
- Oil Filter (2940-01-201-6794)	6	$13.00		$78.00	$78.00	$390.00
- Air Filter (2940-01-319-7600)	6	$30.00		$180.00	$180.00	$900.00
- Fuel Filter (2910-00-238-0033)	6	$6.00		$36.00	$36.00	$180.00
- 15W40 Oil (9150-21-883-2944)	6	$2.00		$12.00	$12.00	$60.00
- Oil soak up pads	12	$8.00		$96.00	$96.00	$480.00
- Small Gasoline Engine	1					
- Outboard Motor (6 East/4 West TE)	6					
			Sub-Total:	$450.00	$450.00	$2,250.00
Boiler Shop Supplies						
- Boiler Assembly Trainer(includes feedwater system)	1	$5,000.00		$5,000.00		$5,000.00
- Boiler Burner Assembly	6	$1,000.00		$6,000.00		$6,000.00
			Sub-Total:	$11,000.00	$0.00	$11,000.00
General Requirements (Not currently associated to a shop)						
- Lube Oil Centrifuge (7 East/11 West TE)	11					
- GT Waterwash assembly	1					
- HP Air Compressors	2					
- LP Air Compressors	2					
- Magenhelic Gauge Assembelies	12					
			Sub-Total:	$0.00	$0.00	$0.00
Fitted DC Systems Shop						

- TAU	2					
			Sub-Total:	$0.00	$0.00	$0.00
Miscellaneous Replacements						
	1	$2,500.00		$2,500.00		$2,500.00
			Sub-Total:	$85,136.14	$8,364.32	$118,993.42
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Live Fire Training, East Coast						
- Training Cost	12	$250.00		$3,000.00	$3,000.00	$15,000.00
- Meals	12	$11.91		$142.92	$142.92	$714.60
- Transportation	12	$97.78		$1,173.36	$1,173.36	$5,866.80
Live Fire Training, West Coast						
- Training Cost	12	$310.00		$3,720.00	$3,720.00	$18,600.00
- Meals	12	$12.21		$146.52	$146.52	$732.60
- Transportation	12	$97.78		$1,173.36	$1,173.36	$5,866.80
			Sub-Total:	$5,039.88	$5,039.88	$25,199.40
Total Estimated Costs				$90,176.02	$13,404.20	$144,192.82

ANNEX J- COURSE HISTORY

The Marine Technician Machinery Maintainer Sailor Second Class (S2) USQ is an amalgamation of elements of legacy Marine Engineer QL 5, Electrical Technician QL 4 and 5, MESO QL 3, and Hull Technician QL 3 and 5. This course is a follow-on from the Marine Technician Basic Maintainer RQ-OS course, and is a prerequisite for all the Marine Technician RQ – LS courses (watch-keeping, mechanical, and electrical technician). This course was developed with input from representatives from each of the four aforementioned legacy MOSIDs.

Sept 22 Change made to have additional MITE codes granted upon completion of course (Fire Fighter Refresher – MITE ID 109369, Flood Refresher – MITE ID 109370). Supported by Training Manager. Updated in Chapter 1, MITE, Para 8.

14 June 24 – This course was Modded. Mod A Electrical and Mod B Mechanical with a "wrapper" MITE ID(for administrative purposes). This was done in advance of the S2 Elec and Mech Split to provide an easier transition period. To provide the right training at the right level, immediately for the trade to expedite Mar-Tech 2.0 delivery.

October 24 – Cleaned up Prerequisites to include new CSQS2 course, as well as included NQUALs.

07 November 2024 – QSP reverted to unmodularized version. Considerations for NFSA not made when planning course. POs remains the same. No TT increase/decrease. No change in delivery strategy.

ANNEX K – ASSESSMENT PLAN

Mod	EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria	Impact
1	CEC 006.01	EO 006.01	To assess the trainee's ability to apply general mathematics.	Written Test	60%	First CEC Fail results in a retest Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
	CEC 006.02	EO 006.02	To assess the trainee's ability to perform general applied physics.	Written Test	60%	First CEC Fail results in a retest Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
	CEC 007.02	EO 007.02	To assess the trainee's ability to perform general chemistry.	Written Test	60%	First CEC Fail results in a retest Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
	CEC 008.02	EO 008.02	To assess the trainee's ability to apply general thermodynamics.	Written Test	60%	First CEC Fail results in a retest Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
2	CEC 011.01	EO 011.01	To assess the trainee's ability to perform electrical circuit analysis.	Written Test	60%	First CEC Fail results in a retest Second CEC failure results in a TRB IAW NPTORD. Remedial

						training/counselling as required.
	CEC 012.01	EO 012.01	To assess the trainee's ability to perform rotational machinery analysis.	Written Test	60%	First CEC Fail results in a retest Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
3	EC 003.01	EO 003.01	To assess the trainee's ability to maintain twin agent unit system.	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 003.02	EO 003.02	To assess the trainee's ability to maintain portable damage control equipment.	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 003.03	EO 003.03	To assess the trainee's ability to maintain portable rescue equipment.	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	PC 003	PO 003	To assess the trainee's ability to maintain dc systems.	Developmental Observations	Nil.	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
4	EC 005.01	EO 005.01	To assess the trainee's ability to lead an attack team.	Practical Assessment.	AS per ATL QSP.	As per DCS P/A SOP.
	EC 005.02	EO 005.02	To assess the trainee's ability to attack live fires.	Practical Assessment.	As per civilian standard.	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction

	EC 005.03	EO 005.03	To assess the trainee's ability to lead a flood team	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction
	EC 005.04	EO 005.04	To assess the trainee's ability to lead a casualty power team	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction
	PC 005	PO 005	To assess the trainee's ability to lead a damage control team.	Developmental Observations	Nil.	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction
5	EC 004.01	EO 004.01	To assess the trainee's ability to inspect ship's structures and hull.	Written assessment	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 004.02 Part A	EO 004.02	To assess the trainee's ability to manufacture materials.	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 004.02 Part B			Practical Assessment.	Pass/Fail IAW EC Checklist	
	EC 004.02 Part C			Practical Assessment.	Pass/Fail IAW EC Checklist	
	EC 004.02 Part D			Practical Assessment.	Pass/Fail IAW EC Checklist	
	EC 004.02 Part E			Practical Assessment.	Pass/Fail IAW EC Checklist	
	PC 004 Part A	EO 004.02	To assess the trainee's ability to maintain	Practical Assessment.	Pass/Fail IAW PC Checklist	First PC Fail results in a retest

	PC 004 Part B		shipboard components	Practical Assessment	Pass/Fail IAW PC Checklist	Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
	PC 004 Part C			Practical Assessment	Pass/Fail IAW PC Checklist	
6	EC 001.01	EO 001.01	To assess the trainee's ability to perform maintenance administration.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 001.02	EO 001.02	To assess the trainee's ability to perform logistical administration.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 001.03	EO 001.03	IAW Naval Technical Communication Module A	Developmental Observations	IAW Naval Technical Communication Module A	IAW Naval Technical Communication Module A
	PC 001	PO 001	To assess the trainee's ability to perform technical administration.	Practical Assessment.	Pass/Fail IAW PC Checklist	First PC Fail results in a retest Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
7	EC 011.02	EO 011.02	To assess the trainee's ability to maintain electrical components.	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 011.03	EO 011.03	To assess the trainee's ability to maintain field devices.	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

	EC 011.04	EO 011.04	To assess the trainee's ability to maintain sensors.	Written assessment	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	PC 011	PO 011	To assess the trainee's ability to maintain shipboard electrical systems	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 012.02	EO 012.02	To assess the trainee's ability to maintain electrical motors.	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 012.03	EO 012.03	To assess the trainee's ability to maintain motor control devices.	Written Test	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	PC 012	PO 012	To assess the trainee's ability to electrical rotating machinery and control.	Practical Assessment.	Pass/Fail	First PC Fail results in a retest Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
8	CEC 002.03	EO 002.03	To assess the trainee's ability to maintain centrifuges.	Practical Assessment.	Pass/Fail IAW CEC Checklist	First CEC Fail results in a retest Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
	EC 002.04	EO 002.04	To assess the trainee's ability to maintain oil pollution	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided

			abatement systems.			additional training/instruction.
	PC 002	PO 002	To assess the trainee's ability to maintain fluid system components	Developmental Observations	Nil.	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 010.01	EO 010.01	To assess the trainee's ability to maintain gas turbine water wash system.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 010.02	EO 010.02	To assess the trainee's ability to maintain main shaftline.	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	PC 010	PO 010	To assess the trainee's ability to maintain ancillary systems	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 002.01	EO 002.01	To assess the trainee's ability to maintain fluid system components.	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
9	CEC 002.02	EO 002.02	To assess the trainee's ability to maintain RAS equipment.	Practical Assessment.	Pass/Fail IAW CEC Checklist	First CEC Fail results in a retest Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.

EC 006.03	EO 006.03	To assess the trainee's ability to maintain compressed air system components	Practical Assessment	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
PC 006	PO 006	To assess the trainee's ability to maintain compressed air systems	Developmental Observations	Nil.	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 007.01	EO 007.01	To assess the trainee's ability to maintain auxiliary steam generator.	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
CEC 007.01	EO 007.01			Pass/Fail IAW CEC Checklist	First CEC Fail results in a retest Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
PC 007	PO 007	To assess the trainee's ability to maintain auxiliary steam systems.	Developmental Observations	Nil.	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction
EC 008.01	EO 008.01	To assess the trainee's ability to maintain reverse osmosis desalination.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 008.03	EO 008.03	To assess the trainee's ability to maintain refrigeration components.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided

						additional training/instruction.
	EC 008.04	EO 008.04	To assess the trainee's ability to maintain HVAC systems	Written Test	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction
	PC 008	PO 008	To assess the trainee's ability to maintain domestic service systems	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction
10	EC 009.01	EO 009.01	To assess the trainee's ability to maintain small boat gasoline engines	Practical Assessment.	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction
	EC 009.02	EO 009.02	To assess the trainee's ability to perform first line maintenance on small boat diesel engines	Written Test	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
				Practical Assessment.	Pass/Fail IAW EC Checklist	
	EC 009.03	EO 009.03	To assess the trainee's ability to troubleshoot small boat diesel engines	Practical Assessment	Pass/Fail IAW EC Checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
	EC 009.04	EO 009.04	To assess the trainee's ability to maintain small boat stern drives	Written Test	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
				Practical Assessment	Pass/Fail IAW EC Checklist	
	EC 009.05	EO 009.05	To assess the trainee's ability to	Written Test	60%	Trainee issued Trainee

			maintain small boat components	Practical Assessment	Pass/Fail IAW EC Checklist	Experiencing Difficulty (TED) and provided additional training/instruction.
	PC 009	PO 009	To assess the trainee's ability to perform small boat maintenance	Practical Assessment	Pass/Fail IAW PC Checklist	First PC Fail results in a retest Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.

ANNEX L – COURSE MODULARIZATION

S2 USQ

Required Course Staffing							
MOD	EO	Ratio		Impact		TEL Requirements	
		East	West	East	West	East	West
1	006.01	12:1		NIL		This course can be developed in the future to be a partial online course.	
	006.02						
	007.02						
	008.02						
2	011.01	2:1		Test is CEC	Test is CEC	LABVOLT 12 BAYS	LAB VOLT 12 BAYS
	011.01			Test is CEC	Test is CEC	LABVOLT 12 BAYS	LAB VOLT 12 BAYS
3	003.01	1:1		Test is EC	Test is EC	Require TAU to test and 2 test tools (NSN 4210-01-535-7128)	Require TAU to test and 2 test tools (NSN 4210-01-535-7128)
	003.02	1:1		Test is EC	Test is EC	Requires DC Lockers	Requires DC Lockers
	003.03	1:1		Test is EC	Test is EC	Require DC Rescue Equipment	Require DC Rescue Equipment
4	005.01	12:1		Taught by DC Div/Civ live fire trainer		Live fire trainer	
	005.02	12:1					
	005.03	12:1					
	005.04	12:1					
5	004.01	12:1		nil	nil	nil	nil
	004.02	12:1		nil	nil	nil	nil
6	001.01	12:1		NIL		DRMIS Access	
	001.02						
	001.03			Taught by Civ		Taught by Civ	
6	009.01	5:1		Can only test 10 trainees(2 per engine) at a time		1 Small Gas Engine 6 outboard engines	1 small gas engine 4 outboard engines
	009.02	12:1		Small diesel engine		Small diesel engine	
	009.03	12:1		For this EO only 6 small boats are needed (2 trainees per boat)		NIL	
	009.04	12:1					
	009.05	12:1					

7	011.02	1:1	All trainees can be tested at the same time	36 X PIN connector; 12 X Canon plug; 12 X Crimper 12 X PIN insertion/extraction tool. 12 X Wire crimp pull tester 12 x roxtec kits	36 X PIN connector; 12 X Canon plug; 12 X Crimper 12 X PIN insertion/extraction tool. 12 X Wire crimp pull tester 12 x roxtec kits
	011.03	1:1	Written EC	12 X various field devices for students to include but not limited too; • Rheostats • Macro switches • Differential switches • Solenoids • Limit switches 12 X Lab Volt Bays	12 X various field devices for students to include but not limited too; • Rheostats • Macro switches • Differential switches • Solenoids • Limit switches 12 X Lab Volt Bays
	011.04	1:1	Written EC	12 X various field devices for students to include but not limited too; • RTD • Bimetallic strip • Thermometer • Thermocouple • Thermistors • flow devices • Pressure Sensors • Level Sensors	12 X various field devices for students to include but not limited too; • RTD • Bimetallic strip • Thermometer • Thermocouple • Thermistors • flow devices • Pressure Sensors • Level Sensors

					• Position Sensors • Magnetic Pick-ups	• Position Sensors Magnetic Pick-ups
	012.02	1:1		All trainees can be tested at the same time	12 X Bearing training stepped shafts Motor (IE NSN 12-3234785) 24 X Bearings 6 x Bearing Heater	12 X Bearing training stepped shafts Motor (IE NSN 12-3234785) 24 X Bearings 6 x Bearing Heater
	012.03	1:1		Ideally one invigilator for 2 trainees; therefore would need 6 invalidators. Testing time would increase if this has to be reduced (FOR PC)	12 X Lab Volt Bays 12 Motor Control Schematics (Various, Full Size) Motor Control Devices (various Ships Fitted Motor Controller (For Explanation)	12 X Lab Volt Bays 12 Motor Control Schematics (Various, Full Size) Motor Control Devices (various Ships Fitted Motor Controller (For Explanation)
8	002.03	5:1	1:1	Can only test five trainees at a time	Can only test one trainee at a time	7 centrifuges	11 centrifuges
	002.04	1:1		Can only test one trainee at a time	Can only test one trainee at a time	The hydromem would be too large of an asset to purchase to have one on one testing	The hydromem would be too large of an asset to purchase to have one on one testing
	010.01	0		Cannot test	Cannot test	Functional GT water wash system	Functional water wash system
	010.02	5:1		Due to only having one of each system, only five can be tested at a time	Due to only having one of each system, only five can be tested at a time	Drive belt training aid	Drive belt training aid
9	002.01	1:1		Can only test one trainee at a time	Can only test one trainee at a time	6 pump trainers needed	6 pump trainers needed
	002.02	4:1		Can only train four	Can only train four trainees at	1 fuel probes, 2 fuel receivers, 2	1 fuel probes, 2 fuel receivers, 2

				trainees at a time due to having only one RAS trainer	a time due to having only one RAS trainer	emergency breakaway coupling and 6 7” fuel hoses with clamps.	emergency breakaway coupling and 6 7” fuel hoses with clamps.
	007.01	1:1	0	Can only test one trainee at a time	Cannot test	6 burner assemblies, and 6 ignitor assemblies	1 functional steam generator, 6 burner assemblies, and 6 ignitor assemblies.
	008.01	1:1		Can only test one trainee at a time	Can only test one trainee at a time	Due to the size of the ROD unit, only one should be at the TE for training purposes.	Due to the size of the ROD unit, only one should be at the TE for training purposes.
	008.03	4:1		Can only test four at a time	Can only test four at a time	2 fridge trainers from Lab Volt	2 fridge trainers from Lab Volt
10	006.03	2:1		Can only train two trainees at a time (one per comp)	Can only train two trainees at a time (one per comp)	2 HP Air Comp and 2 LP air Comp	2 HP Air Comp and 2 LP air Comp
	008.04	12:1		Nil		Nil	